

- If TRANS = 'N' and $M < N$, rows 1 to N of B contain the minimum norm solution vectors of the underdetermined system $AX = B$;
- If TRANS = 'T' and $M \geq N$, rows 1 to M of B contain the minimum norm solution vectors of the underdetermined system $A^T X = B$;
- If TRANS = 'T' and $M < N$, rows 1 to M of B contain the solution vectors of the LLS problem $\min_{\mathbf{x}} \|\mathbf{b} - A^T \mathbf{x}\|_2$; the residual sum of squares for the solution in each column is given by the sum of the squares of elements $M+1$ to N in that column.

WORK: If INFO = 0, WORK(1) returns the optimal LWORK.

INFO: INFO = 0 if successful exit. If INFO = $-j$, the j th input argument had an illegal value.

Consider an LLS problem (75.2) with a 6-by-5 matrix A and a 6-by-1 matrix $\mathbf{b}$:

$$A = \begin{bmatrix} -74 & 80 & 18 & -11 & -4 \\ 14 & -69 & 21 & 28 & 0 \\ 66 & -72 & -5 & 7 & 1 \\ -12 & 66 & -30 & -23 & 3 \\ 3 & 8 & -7 & -4 & 1 \\ 4 & -12 & 4 & 4 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 51 \\ -61 \\ -56 \\ 69 \\ 10 \\ -12 \end{bmatrix}.$$

The exact solution of the LLS problem is $\mathbf{x} = [1 \quad 2 \quad -1 \quad 3 \quad -4]^T$ with residual $\|\mathbf{b} - A\mathbf{x}\|_2 = 0$. Upon calling SGELS, the first 5 elements of B are overwritten by the solution vector $\mathbf{x}$:

$$\mathbf{x} = \begin{bmatrix} 1.0000176 \\ 2.0000196 \\ -0.9999972 \\ 3.0000386 \\ -4.0000405 \end{bmatrix},$$

while the sixth element of B contains the residual sum of squares 0.0000021. With $M = 6$, $N = 5$, NRHS = 1, LWORK has been set to 11. For such a small matrix, the minimal workspace is also the optimal workspace.

75.4 The Linear Equality-Constrained Least Squares Problem

Definitions:

The linear equality-constrained least squares (LSE) problem is

$$\min_{\mathbf{x}} \|\mathbf{c} - A\mathbf{x}\|_2 \quad \text{subject to} \quad B\mathbf{x} = \mathbf{d}, \quad (75.3)$$

where A is an m -by- n matrix and B is a p -by- n matrix, $\mathbf{c}$ is an m -vector, and $\mathbf{d}$ is a p -vector, with $p \leq n \leq m + p$.

Backgrounds:

Under the assumptions that B has full row rank p and the matrix $\begin{bmatrix} A \\ B \end{bmatrix}$ has full column rank n , the LSE problem (75.3) has a unique solution $\mathbf{x}$.

Driver Routines:

The driver routine for solving the LSE is `xGGLSE`, which uses a generalized QR factorization of the matrices A and B .

Examples:

Let us show how to use the driver routine `SGGLSE` to solve the LSE problem (75.3). `SGGLSE` first computes a generalized QR decomposition of A and B , and then computes the solution by back substitution. `SGGLSE` has the following calling sequence:

```
CALL SGGLSE( M, N, P, A, LDA, B, LDB, C, D, X, WORK, LWORK, INFO )
```

Input to `SGGLSE`:

M , P : The numbers of rows of the matrices A and B , respectively. $M \geq 0$ and $P \geq 0$.

N : The number of columns of the matrices A and B . $N \geq 0$. Note that $0 \leq P \leq N \leq M+P$.

A , LDA : The M -by- N matrix A and the leading dimension of the array A . $LDA \geq \max(1, M)$.

B , LDB : The P -by- N matrix B and the leading dimension of the array B . $LDB \geq \max(1, P)$.

C , D : The right-hand side vectors for the least squares part, and the constrained equation part of the LSE, respectively.

$WORK$, $LWORK$: The workspace array and its dimension. $LWORK \geq \max(1, M + N + P)$.

If $LWORK = -1$, then a workspace query is assumed; the routine only calculates the optimal size of the $WORK$ array, and returns this value as the first entry of the $WORK$ array.

Output from `SGGLSE`:

C : The residual sum of squares for the solution is given by the sum of squares of elements $N-P+1$ to M of vector C .

X : The solution of the LSE problem.

$WORK$: If $INFO = 0$, $WORK(1)$ returns the optimal $LWORK$.

$INFO$: $= 0$ if successful exit. If $INFO = -j$, the j th argument had an illegal value.

Let us demonstrate the use of `SGGLSE` to solve the LSE problem (75.3), where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}.$$

The unique exact solution is $\mathbf{x} = \frac{1}{8}[46 \quad -2 \quad 12]^T$. Upon calling `SGGLSE` with this input data and $M = 4, N = 3, P = 2, LWORK = 9$, an approximate solution of the LSE problem is returned in X :

$$X = [5.7500000 \quad -0.2500001 \quad 1.4999994]^T.$$

The array C is overwritten by the residual sum of squares for the solution:

$$C = [4.2426405 \quad 8.9999981 \quad 2.1064947 \quad 0.2503501]^T.$$

75.5 A General Linear Model Problem

Definitions:

The general linear model (GLM) problem is

$$\min_{\mathbf{x}, \mathbf{y}} \|\mathbf{y}\|_2 \quad \text{subject to} \quad \mathbf{d} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y}, \quad (75.4)$$

where $\mathbf{A}$ is an n -by- m matrix, $\mathbf{B}$ is an n -by- p matrix, and $\mathbf{d}$ is a n -vector, with $m \leq n \leq m + p$.

Backgrounds:

When $\mathbf{B} = \mathbf{I}$, the problem reduces to an ordinary linear least squares problem (75.2). When $\mathbf{B}$ is square and nonsingular, the GLM problem is equivalent to the *weighted linear least squares problem*:

$$\min_{\mathbf{x}} \|\mathbf{B}^{-1}(\mathbf{d} - \mathbf{A}\mathbf{x})\|_2.$$

Note that the GLM is equivalent to the LSE problem

$$\min_{\mathbf{x}, \mathbf{y}} \left\| \begin{bmatrix} 0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \right\|_2 \quad \text{subject to} \quad \begin{bmatrix} \mathbf{A} & \mathbf{B} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \mathbf{d}.$$

Therefore, the GLM problem has a unique solution of the matrix $\begin{bmatrix} 0 & \mathbf{I} \\ \mathbf{A} & \mathbf{B} \end{bmatrix}$ and has full column rank $m + p$.

Driver Routines:

The driver routine for solving the GLM problem (75.4) is xGGGLM, which uses a generalized QR factorization of the matrices $\mathbf{A}$ and $\mathbf{B}$.

Examples:

Let us show how to use the driver routine SGGGLM to solve the GLM problem (75.4). SGGGLM computes a generalized QR decomposition of the matrices $\mathbf{A}$ and $\mathbf{B}$, and then computes the solution by back substitution. SGGGLM has the following calling sequence:

```
CALL SGGGLM( N, M, P, A, LDA, B, LDB, D, X, Y, WORK, LWORK, INFO )
```

Input to SGGGLM:

N: The number of rows of the matrices $\mathbf{A}$ and $\mathbf{B}$. $N \geq 0$.

M, P: The number of columns of the matrices $\mathbf{A}$ and $\mathbf{B}$, respectively. $0 \leq M \leq N$ and $P \geq N - M$.

A, LDA: The N -by- M matrix $\mathbf{A}$ and the leading dimension of the array $\mathbf{A}$. $LDA \geq \max(1, N)$.

B, LDB: The N -by- P matrix $\mathbf{B}$ and the leading dimension of the array $\mathbf{B}$. $LDB \geq \max(1, N)$.

D: The left-hand side of the GLM equation.

WORK, LWORK: The workspace array and its dimension. $LWORK \geq \max(1, N + M + P)$.

If $LWORK = -1$, then a workspace query is assumed; the routine only calculates the optimal size of the **WORK** array, and returns this value as the first entry of the **WORK** array.

Output from SGGGLM:

X, Y : Solution vectors.

WORK: If $INFO = 0$, $WORK(1)$ returns the optimal LWORK.

INFO: $INFO = 0$ if successful exit. If $INFO = -j$, the j th argument had an illegal value.

Let us demonstrate the use of SGGGLM for solving the GLM problem (75.4), where

$$A = \begin{bmatrix} 1 & 2 & 1 & 4 \\ -1 & 1 & 1 & 1 \\ -1 & -2 & -1 & 1 \\ -1 & 2 & -1 & -1 \\ 1 & 1 & 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & -2 \\ 3 & 1 & 6 \\ 2 & -2 & 4 \\ 1 & -1 & 2 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 7.99 \\ 0.98 \\ -2.98 \\ 3.04 \\ 4.02 \end{bmatrix}.$$

Upon calling SGGGLM with this input data, $N = 5, M = 4, P = 3, LWORK = 12$, the program successfully exits and returns the following solution vectors:

$$X = [1.002950 \quad 2.001435 \quad -0.987797 \quad 0.009080]^T$$

and

$$Y = [0.003435 \quad -0.004417 \quad 0.006871]^T.$$

75.6 Symmetric Eigenproblems

Definitions:

The symmetric eigenvalue problem (SEP) is to find the eigenvalues, λ , and corresponding eigenvectors, $\mathbf{x} \neq 0$, such that

$$A\mathbf{x} = \lambda\mathbf{x}, \quad (75.5)$$

where A is real and symmetric. If A is complex and Hermitian, i.e., $A^* = A$, then it is referred to as the *Hermitian eigenvalue problem*.

Backgrounds:

When all eigenvalues and eigenvectors have been computed, we write

$$A = X\Lambda X^*, \quad (75.6)$$

where Λ is a diagonal matrix whose diagonal elements are real and are the eigenvalues, and X is an orthogonal (or unitary) matrix whose columns are the eigenvectors. This is the classical *spectral decomposition* of A . The theoretical and algorithmic backgrounds of the solution of the symmetric eigenvalue problem are discussed in Chapter 42.

Driver Routines:

There are four types of driver routines for solving the SEP (75.5) and each has its own variations that take advantage of the special structure or storage of the matrix A , as summarized in the following table.

Types of Matrix (Storage Scheme)	Routine Names			
	Simple Driver	Divide-and-Conquer	Expert Driver	RRR Driver
General symmetric	xSYEV	xSYEVD	xSYEVX	xSYEVR
General symmetric (packed storage)	xSPEV	xSPEVD	xSPEVX	–
Band matrix	xSBEV	xSBEVD	xSBEVX	–
Tridiagonal matrix	xSTEV	xSTEVD	xSTEVX	xSTEVR

The simple driver computes all eigenvalues and (optionally) eigenvectors. The expert driver computes all or a selected subset of the eigenvalues and (optionally) eigenvectors. The divide-and-conquer driver has the same functionality as, yet outperforms, the simple driver, but it requires more workspace. The relative robust representation (RRR) driver computes all or a subset of the eigenvalues and (optionally) the eigenvectors. The last one is generally faster than any other types of driver routines and uses the least amount of workspace.

Examples:

Let us show how to use the simple driver SSYEV to solve the SEP (75.5) by computing the spectral decomposition (75.6). SSYEV first reduces A to a tridiagonal form, and then uses the implicit QL or QR algorithm to compute eigenvalues and optionally eigenvectors. SSYEV has the following calling sequence:

```
CALL SSYEV( JOBZ, UPLO, N, A, LDA, W, WORK, LWORK, INFO )
```

Input to SSYEV:

JOBZ: = 'N', compute eigenvalues only;

= 'V', compute eigenvalues and eigenvectors.

UPLO: = 'U', the upper triangle of A is stored in the array A ; if UPLO = 'L', the lower triangle of A is stored.

N: The order of the matrix A . $N \geq 0$.

A, LDA: The symmetric matrix A and the leading dimension of the array A . $LDA \geq \max(1, N)$.

WORK, LWORK: The workspace array and its dimension. $LWORK \geq \max(1, 3 * N - 1)$.

If LWORK = -1, then a workspace query is assumed; the routine only calculates the optimal size of the WORK array, and returns this value as the first entry of the WORK array.

Output from SSYEV:

A: The orthonormal eigenvectors X , if JOBZ = 'V'.

W: The eigenvalues λ in ascending order.

WORK: If INFO = 0, WORK(1) returns the optimal LWORK.

INFO: = 0 if successful exit. If INFO = - j , the j th input argument had an illegal value. If INFO = j , the j off-diagonal elements of an intermediate tridiagonal form did not converge to zero.

Let us demonstrate the use of SSYEV to solve the SEP (75.5), where

$$A = \begin{bmatrix} 5 & 4 & 1 & 1 \\ 4 & 5 & 1 & 1 \\ 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 \end{bmatrix}.$$

The exact eigenvalues are 1, 2, 5, and 10. Upon calling SSYEV with the matrix A and $N = 4$, $LWORK = 3 * N - 1 = 11$, A is overwritten by its orthonormal eigenvectors X .

$$X = \begin{bmatrix} 0.7071068 & -0.0000003 & 0.3162279 & 0.6324555 \\ -0.7071068 & 0.0000001 & 0.3162278 & 0.6324555 \\ 0.0000002 & 0.7071069 & -0.6324553 & 0.3162278 \\ -0.0000001 & -0.7071066 & -0.6324556 & 0.3162278 \end{bmatrix}.$$

The eigenvalues that correspond to the eigenvectors in each column of X are returned in W :

$$W = [0.9999996 \quad 1.9999999 \quad 4.9999995 \quad 10.0000000].$$

75.7 Nonsymmetric Eigenproblems

As is customary in numerical linear algebra, in this section the term *left eigenvector* of A means a (column) vector y such that $y^* A = \lambda y^*$. This is contrary to the definition in Section 4.3, under which y^* would be called a left eigenvector.

Definitions:

The nonsymmetric eigenvalue problem (NEP) is to find the eigenvalues, λ , and corresponding (right) eigenvectors, $x \neq 0$, such that

$$Ax = \lambda x \tag{75.7}$$

and, perhaps, the left eigenvectors, $y \neq 0$, satisfying

$$y^* A = \lambda y^*. \tag{75.8}$$

Backgrounds:

The problem is solved by computing the *Schur decomposition* of A , defined in the real case as

$$A = ZTZ^T,$$

where Z is an orthogonal matrix and T is an upper quasi-triangular matrix with 1-by-1 and 2-by-2 diagonal blocks, the 2-by-2 blocks corresponding to complex conjugate pairs of eigenvalues of A . In the complex case, the Schur decomposition is

$$A = ZTZ^*,$$

where Z is unitary and T is a complex upper triangular matrix.

The columns of Z are called the *Schur vectors*. For each k ($1 \leq k \leq n$), the first k columns of Z form an orthonormal basis for the invariant subspace corresponding to the first k eigenvalues on the diagonal of T . It is possible to order the Schur factorization so that any desired set of k eigenvalues occupies the k leading positions on the diagonal of T . The theoretical and algorithmic background of the solution of the nonsymmetric eigenvalue problem is discussed in Chapter 43.

Driver Routines:

Both the simple driver xGEEV and expert driver xGEEVX are provided. The simple driver computes all the eigenvalues of A and (optionally) the right or left eigenvectors (or both). The expert driver performs the same task as the simple driver plus the additional feature that it balances the matrix to try to improve the conditioning of the eigenvalues and eigenvectors, and it computes the condition numbers for the eigenvalues or eigenvectors (or both).

Examples:

Let us show how to use the simple driver SGEEV to solve the NEP (75.7). SGEEV first reduces A to an upper Hessenberg form (a Hessenberg matrix is a matrix where all entries below the first lower subdiagonal are

zeros), and then uses the implicit QR algorithm to compute the Schur decomposition, and finally computes eigenvectors of the upper quasi-triangular matrix. SGEEV has the following calling sequence:

```
CALL SGEEV( JOBLV, JOBVR, N, A, LDA, WR, WI, VL, LDVL, VR, LDVR, WORK,
LWORK, INFO )
```

Input to SGEEV:

JOBLV, JOBVR: = 'V', the left and/or right eigenvectors are computed;
= 'N', the left and/or right eigenvectors are not computed.

N: The order of the matrix A . $N \geq 0$.

A, LDA: The matrix A and the leading dimension of the array A . $LDA \geq \max(1, N)$.

LDVL, LDVR: The leading dimensions of the arrays VL and VR if the left and right eigenvectors are computed. $LDVL, LDVR \geq N$.

WORK, LWORK: The workspace array and its dimension. $LWORK \geq \max(1, 3 * N)$. If eigenvectors are computed, $LWORK \geq 4 * N$. For good performance, $LWORK$ must generally be larger.

If $LWORK = -1$, then a workspace query is assumed; the routine only calculates the optimal size of the $WORK$ array, and returns this value as the first entry of the $WORK$ array.

Output from SGEEV:

WR, WI: The real and imaginary parts of the computed eigenvalues. Complex conjugate pairs of eigenvalues appear consecutively with the eigenvalue having the positive imaginary part first.

VL: If the j th eigenvalue λ_j is real, then the j th left eigenvector $\mathbf{y}_j$ is stored in $VL(:, j)$. If the j th and $(j + 1)$ -st eigenvalues λ_j and λ_{j+1} form a complex conjugate pair, then $VL(:, j) + i \cdot VL(:, j + 1)$ and $VL(:, j) - i \cdot VL(:, j + 1)$ are the corresponding left eigenvectors $\mathbf{y}_j$ and $\mathbf{y}_{j+1}$.

VR: If the j th eigenvalue λ_j is real, then the j th right eigenvector $\mathbf{x}_j$ is stored in $VR(:, j)$. If the j th and $(j + 1)$ -st eigenvalues λ_j and λ_{j+1} form a complex conjugate pair, then $VR(:, j) + i \cdot VR(:, j + 1)$ and $VR(:, j) - i \cdot VR(:, j + 1)$ are the corresponding right eigenvectors $\mathbf{x}_j$ and $\mathbf{x}_{j+1}$.

WORK: If $INFO = 0$, $WORK(1)$ returns the optimal $LWORK$.

INFO: = 0 if successful exit. If $INFO = -j$, the j th argument had an illegal value. If $INFO = j$, the QR algorithm failed to compute all the eigenvalues, and no eigenvectors have been computed; elements $j + 1 : N$ of WR and WI contain eigenvalues, which have converged.

Let us demonstrate the use of SGEEV for solving the NEP (75.7), where

$$A = \begin{bmatrix} 4 & -5 & 0 & 3 \\ 0 & 4 & -3 & -5 \\ 5 & -3 & 4 & 0 \\ 3 & 0 & 5 & 4 \end{bmatrix}.$$

The exact eigenvalues are 12 , $1 + i \cdot 5$, $1 - i \cdot 5$, and 2 . Upon calling SGEEV with this matrix and $N = 4$, $LWORK = 4 * N = 16$, each eigenvalue, λ_j , is retrieved by combining the j th entry in WR and WI .

such that $\lambda_j = \text{WR}(j) + i \cdot \text{WI}(j)$. If $\text{WI}(j)$ is 0, then the j th eigenvalue is real. For this example, we have

$$\begin{aligned}\lambda_1 &= 12.0000000 \\ \lambda_2 &= 1.000000 + i \cdot 5.0000005 \\ \lambda_3 &= 1.000000 - i \cdot 5.0000005 \\ \lambda_4 &= 1.9999999.\end{aligned}$$

The left eigenvectors are stored in VL. Since the first and fourth eigenvalues are real, their eigenvectors are the corresponding columns in VL, that is, $\mathbf{y}_1 = \text{VL}(:, 1)$ and $\mathbf{y}_4 = \text{VL}(:, 4)$. Since the second and third eigenvalues form a complex conjugate pair, the second eigenvector, $\mathbf{y}_2 = \text{VL}(:, 2) + i \cdot \text{VL}(:, 3)$ and the third eigenvector, $\mathbf{y}_3 = \text{VL}(:, 2) - i \cdot \text{VL}(:, 3)$. If we place all the eigenvectors in a matrix Y where $Y = [\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4]$, we have

$$Y = \begin{bmatrix} -0.5000001 & 0.0000003 - i \cdot 0.4999999 & 0.0000003 + i \cdot 0.4999999 & 0.5000000 \\ 0.4999999 & -0.5000002 & -0.5000002 & 0.5000001 \\ -0.5000000 & -0.5000000 - i \cdot 0.0000002 & -0.5000000 + i \cdot 0.0000002 & -0.4999999 \\ -0.5000001 & -0.0000003 + i \cdot 0.5000000 & -0.0000003 - i \cdot 0.5000000 & 0.5000001 \end{bmatrix}.$$

The right eigenvectors $\mathbf{x}_j$ can be recovered from VR in the way similar to the left eigenvectors. The right eigenvector matrix X is

$$X = \begin{bmatrix} -0.5000000 & 0.5000002 & 0.5000002 & 0.5000001 \\ 0.4999999 & -0.0000001 - i \cdot 0.5000000 & -0.0000001 + i \cdot 0.5000000 & 0.5000000 \\ -0.5000000 & -0.0000001 - i \cdot 0.4999999 & -0.0000001 + i \cdot 0.4999999 & -0.5000000 \\ -0.5000001 & -0.5000001 & -0.5000001 & 0.5000000 \end{bmatrix}.$$

75.8 Singular Value Decomposition

Definitions:

The singular value decomposition (SVD) of an m -by- n matrix A is

$$A = U \Sigma V^T \quad (A = U \Sigma V^* \quad \text{in the complex case}), \quad (75.9)$$

where U and V are orthogonal (unitary) and Σ is an m -by- n diagonal matrix with real diagonal elements, σ_j , such that

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min(m,n)} \geq 0.$$

The σ_j are the singular values of A and the first $\min(m, n)$ columns of U and V are the left and right singular vectors of A .

Backgrounds:

The singular values σ_j and the corresponding left singular vectors $\mathbf{u}_j$ and right singular vectors $\mathbf{v}_j$ satisfy

$$A \mathbf{v}_j = \sigma_j \mathbf{u}_j \quad \text{and} \quad A^T \mathbf{u}_j = \sigma_j \mathbf{v}_j \quad (\text{or} \quad A^* \mathbf{u}_j = \sigma_j \mathbf{v}_j \quad \text{in complex case}),$$

where $\mathbf{u}_j$ and $\mathbf{v}_j$ are the j th columns of U and V , respectively. (See Chapter 17 and Chapter 45 for more information on singular value decompositions.)

Driver Routines:

Two types of driver routines are provided in LAPACK. The simple driver xGESVD computes, all the singular values and (optionally) left and/or right singular vectors. The divide and conquer driver xGESDD has the same functionality as the simple driver except that it is much faster for larger matrices, but uses more workspace.

Examples:

Let us show how to use the simple driver SGESVD to compute the SVD (75.9). SGESVD first reduces A to a bidiagonal form, and then uses an implicit QR-type algorithm to compute singular values and optionally singular vectors. SGESVD has the following calling sequence:

```
CALL SGESVD( JOBU, JOBVT, M, N, A, LDA, S, U, LDU, VT, LDVT, WORK,
             LWORK, INFO )
```

Input to SGESVD:

JOBU: Specifies options for computing all or part of the left singular vectors U :

- = 'A', all M columns of U are returned in the array U ;
- = 'S', the first $\min(M, N)$ columns of U are returned;
- = 'O', the first $\min(M, N)$ columns of U are overwritten on the array A ;
- = 'N', no left singular vectors are computed. *Note that JOBVT and JOBU cannot both be 'O'.*

JOBVT: Specifies options for computing all or part of the right singular vectors V^T :

- = 'A', all N rows of V^T are returned in the array VT ;
- = 'S', the first $\min(M, N)$ rows of V^T are returned;
- = 'O', the first $\min(M, N)$ rows of V^T are overwritten on the array A ;
- = 'N', no right singular vectors are computed.

M, N : The number of rows and columns of the matrix A . $M, N \geq 0$.

A, LDA : The M -by- N matrix A and the leading dimension of the array A . $LDA \geq \max(1, M)$.

$LDU, LDVT$: The leading dimension of the arrays U and VT . $LDU, LDVT \geq 1$;

If $JOBU = 'S'$ or $'A'$, $LDU \geq M$.

If $JOBVT = 'A'$, $LDVT \geq N$; If $JOBVT = 'S'$, $LDVT \geq \min(M, N)$.

$WORK, LWORK$: The workspace array and its dimension. $LWORK \geq \max(3 \min(M, N) + \max(M, N), 5 \min(M, N))$.

If $LWORK = -1$, then a workspace query is assumed; the routine only calculates the optimal size of the $WORK$ array and returns this value as the first entry of the $WORK$ array.

Output from SGESVD:

A : If $JOBU = 'O'$, A is overwritten with the first $\min(M, N)$ columns of U (the left singular vectors, stored columnwise);

If $JOBVT = 'O'$, A is overwritten with the first $\min(M, N)$ rows of V^T (the right singular vectors, stored rowwise);

S : Singular values of A , sorted so that $S(i) \geq S(i + 1)$.

U : If $JOBU = 'A'$, U contains M -by- M orthogonal matrix U . If $JOBU = 'S'$, U contains the first $\min(M, N)$ columns of U . If $JOBU = 'N'$ or $'O'$, U is not referenced.

VT: If JOBV = 'A', VT contains right N-by-N orthogonal matrix V^T . If JOBV = 'S', VT contains the first $\min(M, N)$ rows of V^T (the right singular vectors stored rowwise). If JOBV = 'N' or 'O', VT is not referenced.

WORK: If INFO = 0, WORK(1) returns the optimal LWORK.

INFO: = 0 if successful exit. If INFO = -j, the jth argument had an illegal value. If INFO > 0, the QR-type algorithm (subroutine SBDQR) did not converge. INFO specifies how many superdiagonals of an intermediate bidiagonal form B did not converge to zero. WORK(2:min(M, N)) contains the unconverged superdiagonal elements of an upper bidiagonal matrix B whose diagonal is in S (not necessarily sorted). B satisfies $A = UB^T$, so it has the same singular values as A , and singular vectors related by U and V^T .

Let us show the numerical results of SGESVD in computing the SVD by an 8-by-5 matrix A as follows:

$$A = \begin{bmatrix} 22 & 10 & 2 & 3 & 7 \\ 14 & 7 & 10 & 0 & 8 \\ -1 & 13 & -1 & -11 & 3 \\ -3 & -2 & 13 & -2 & 4 \\ 9 & 8 & 1 & -2 & 4 \\ 9 & 1 & -7 & 5 & -1 \\ 2 & -6 & 6 & 5 & 1 \\ 4 & 5 & 0 & -2 & 2 \end{bmatrix}.$$

The exact singular values are $\sqrt{1248}$, 20, $\sqrt{384}$, 0, 0. The rank of the matrix A is 3. Upon calling SGESVD with $M = 8$, $N = 5$, LWORK = 25, the computed singular values of A are returned in S:

$$S = [35.3270454 \quad 20.0000038 \quad 19.5959187 \quad 0.0000007 \quad 0.0000004].$$

The columns in U contain the left singular vectors U of A :

$$U = \begin{bmatrix} -7.0711e-001 & 1.5812e-001 & -1.7678e-001 & 2.4818e-001 & -4.0289e-001 & -3.2305e-001 & -3.3272e-001 & -6.9129e-002 \\ -5.3033e-001 & 1.5811e-001 & 3.5355e-001 & -6.2416e-001 & 2.5591e-001 & -3.9178e-002 & 3.0548e-001 & -1.3725e-001 \\ -1.7678e-001 & -7.9057e-001 & 1.7677e-001 & 3.0146e-001 & 1.9636e-001 & -3.1852e-001 & 2.3590e-001 & -1.6112e-001 \\ 0 & 1.5811e-001 & 7.0711e-001 & 2.9410e-001 & 3.1907e-001 & 4.7643e-002 & -5.2856e-001 & 7.1055e-002 \\ -3.5355e-001 & -1.5811e-001 & -1.0000e-006 & 2.3966e-001 & -7.8607e-002 & 8.7800e-001 & 1.0987e-001 & -5.8528e-002 \\ -1.7678e-001 & 1.5812e-001 & -5.3033e-001 & 1.7018e-001 & 7.9071e-001 & -7.0484e-003 & -9.0913e-002 & -8.3220e-004 \\ 0 & 4.7434e-001 & 1.7678e-001 & 5.2915e-001 & -1.5210e-002 & -1.3789e-001 & 6.6193e-001 & 7.9763e-002 \\ -1.7678e-001 & -1.5811e-001 & -1.0000e-006 & -7.1202e-002 & 1.3965e-002 & -2.0712e-002 & 4.9676e-002 & 9.6726e-001 \end{bmatrix}.$$

The rows in VT contain the right singular vectors V^T of A :

$$V^T = \begin{bmatrix} -8.0064e-001 & -4.8038e-001 & -1.6013e-001 & 0 & -3.2026e-001 \\ 3.1623e-001 & -6.3246e-001 & 3.1622e-001 & 6.3246e-001 & -1.8000e-006 \\ -2.8867e-001 & -3.9000e-006 & 8.6603e-001 & -2.8867e-001 & 2.8868e-001 \\ -4.0970e-001 & 3.4253e-001 & -1.2426e-001 & 6.0951e-001 & 5.7260e-001 \\ 8.8224e-002 & -5.0190e-001 & -3.3003e-001 & -3.8100e-001 & 6.9730e-001 \end{bmatrix}.$$

75.9 Generalized Symmetric Definite Eigenproblems

Definitions:

The generalized symmetric definite eigenvalue problem (GSEP) is to find the eigenvalues, λ , and corresponding eigenvectors, $\mathbf{x} \neq 0$, such that

$$A\mathbf{x} = \lambda B\mathbf{x} \quad (\text{type 1}) \tag{75.10}$$

or

$$AB\mathbf{x} = \lambda\mathbf{x} \quad (\text{type 2}) \quad (75.11)$$

or

$$BA\mathbf{x} = \lambda\mathbf{x} \quad (\text{type 3}) \quad (75.12)$$

where A and B are symmetric or Hermitian and B is positive definite.

Backgrounds:

For all these problems the eigenvalues λ are real. The matrix Z of the computed eigenvectors satisfies $Z^*AZ = \Lambda$ (problem types 1 and 3) or $Z^{-1}AZ^{-*} = I$ (problem type 2), where Λ is a diagonal matrix with the eigenvalues on the diagonal. Z also satisfies $Z^*BZ = I$ (problem types 1 and 2) or $Z^*B^{-1}Z = I$ (problem type 3). These results are consequences of spectral theory for symmetric matrices. For example, the GSEP type 1 can be rearranged as

$$B^{-\frac{1}{2}}AB^{-\frac{1}{2}}\mathbf{y} = \lambda\mathbf{y},$$

where $\mathbf{y} = B^{\frac{1}{2}}\mathbf{x}$.

Driver Routines:

There are three types of driver routines for solving the GSEP, and each has variations that take advantage of the special structure or storage of the matrices A and B , as shown in the following table:

Types of Matrix (Storage Scheme)	Routine Names		
	Simple Driver	Divide-and-Conquer	Expert Driver
General dense	xSYGV/xHEGV	xSYGVD/xHEGVD	xSYGVX/xHEGVX
General dense (packed storage)	xSPGV/xHPGV	xSPGVD/xHPGVD	xSPGVX/xHPGVX
Band matrix	xSBGV/xHBGV	xSBBVD/xHBGVD	xSBGVX/xHBGVX

The simple driver computes all the eigenvalues and (optionally) the eigenvectors. The expert driver computes all or a selected subset of the eigenvalues and (optionally) eigenvectors. The divide-and-conquer driver solves the same problem as the simple driver. It is much faster than the simple driver, but uses more workspace.

Examples:

Let us show how to use the simple driver SSYGV to compute the GSEPs (75.10), (75.11), and (75.12). SSYGV first reduces each of these problems to a standard symmetric eigenvalue problem, using a Cholesky decomposition of B , and then computes eigenvalues and eigenvectors of the standard symmetric eigenvalue problem by an implicit QR-type algorithm. SSYGV has the following calling sequence:

```
CALL SSYGV( ITYPE, JOBZ, UPLO, N, A, LDA, B, LDB, W, WORK, LWORK, INFO )
```

Input to SSYGV:

ITYPE: Specifies the problem type to be solved:

JOBZ: = 'N', compute eigenvalues only;

= 'V', compute eigenvalues and eigenvectors.

UPLO: = 'U', the upper triangles of A and B are stored;

= 'L', the lower triangles of A and B are stored.

N: The order of the matrices A and B . $N \geq 0$.

A, LDA: The symmetric matrix A and the leading dimension of the array A. $\text{LDA} \geq \max(1, N)$.

B: The symmetric positive definite matrix B and the leading dimension of the array B. $\text{LDB} \geq \max(1, N)$.

WORK, LWORK: The workspace array and its length. $\text{LWORK} \geq \max(1, 3 * N - 1)$.

If $\text{LWORK} = -1$, then a workspace query is assumed; the routine only calculates the optimal size of the WORK array, and returns this value as the first entry of the WORK array.

Output from SSYGV:

A: Contains the normalized eigenvector matrix Z if requested.

B: If $\text{INFO} \leq N$, the part of B containing the matrix is overwritten by the triangular factor U or L from the Cholesky factorization $B = U^T U$ or $B = L L^T$.

W: The eigenvalues in ascending order.

WORK: If $\text{INFO} = 0$, $\text{WORK}(1)$ returns the optimal LWORK.

$\text{INFO} = 0$, then successful exit. If $\text{INFO} = -j$, then the j th argument had an illegal value.

If $\text{INFO} > 0$, then SSYGV returned an error code:

- $\text{INFO} \leq N$: if $\text{INFO} = j$, the algorithm failed to converge;
- $\text{INFO} > N$: if $\text{INFO} = N + j$, for $1 \leq j \leq N$, then the leading minor of order j of B is not positive definite. The factorization of B could not be completed and no eigenvalues or eigenvectors were computed.

Let us show the use of SSYGV to solve the type 1 GSEP (75.10) for the following 5-by-5 matrices A and B :

$$A = \begin{bmatrix} 10 & 2 & 3 & 1 & 1 \\ 2 & 12 & 1 & 2 & 1 \\ 3 & 1 & 11 & 1 & -1 \\ 1 & 2 & 1 & 9 & 1 \\ 1 & 1 & -1 & 1 & 15 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 12 & 1 & -1 & 2 & 1 \\ 1 & 14 & 1 & -1 & 1 \\ -1 & 1 & 16 & -1 & 1 \\ 2 & -1 & -1 & 12 & -1 \\ 1 & 1 & 1 & -1 & 11 \end{bmatrix}.$$

Upon calling SSYGV with $N = 5$, $\text{LWORK} = 3 * N - 1 = 14$, A is overwritten by the eigenvector matrix Z :

$$Z = \begin{bmatrix} -0.1345906 & 0.0829197 & -0.1917100 & 0.1420120 & -0.0763867 \\ 0.0612948 & 0.1531484 & 0.1589912 & 0.1424200 & 0.0170980 \\ 0.1579026 & -0.1186037 & -0.0748390 & 0.1209976 & -0.0666645 \\ -0.1094658 & -0.1828130 & 0.1374690 & 0.1255310 & 0.0860480 \\ 0.0414730 & 0.0035617 & -0.0889779 & 0.0076922 & 0.2894334 \end{bmatrix}.$$

The corresponding eigenvalues are returned in W :

$$W = \begin{bmatrix} 0.4327872 & 0.6636626 & 0.9438588 & 1.1092844 & 1.4923532 \end{bmatrix}.$$

75.10 Generalized Nonsymmetric Eigenproblems

Definitions:

The generalized nonsymmetric eigenvalue problem (GNEP) is to find the eigenvalues, λ , and corresponding (right) eigenvectors, $\mathbf{x} \neq 0$, such that

$$A\mathbf{x} = \lambda B\mathbf{x} \tag{75.13}$$

and optionally, the corresponding left eigenvectors $\mathbf{y} \neq 0$, such that

$$\mathbf{y}^* A = \lambda \mathbf{y}^* B, \quad (75.14)$$

where A and B are n -by- n matrices. In this section the terms *right eigenvector* and *left eigenvector* are used as just defined.

Backgrounds:

Sometimes an equivalent notation is used to refer to the GNEP of the pair (A, B) . The GNEP can be solved via the generalized Schur decomposition of the pair (A, B) , defined in the real case as

$$A = QSZ^T, \quad B = QTZ^T,$$

where Q and Z are orthogonal matrices, T is upper triangular, and S is an upper quasi-triangular matrix with 1-by-1 and 2-by-2 diagonal blocks, the 2-by-2 blocks corresponding to complex conjugate pairs of eigenvalues. In the complex case, the generalized Schur decomposition is

$$A = QSZ^*, \quad B = QTZ^*,$$

where Q and Z are unitary and S and T are both upper triangular. The columns of Q and Z are called left and right generalized Schur vectors and span pairs of deflating subspaces of A and B . Deflating subspaces are a generalization of invariant subspaces: For each k , $1 \leq k \leq n$, the first k columns of Z span a right deflating subspace mapped by both A and B into a left deflating subspace spanned by the first k columns of Q . It is possible to order the generalized Schur form so that any desired subset of k eigenvalues occupies the k leading position on the diagonal of (S, T) . (See Chapter 43 and Chapter 15 for more information on generalized eigenvalue problems.)

Driver Routines:

Both the simple and expert drivers are provided in LAPACK. The simple driver xGGEV computes all eigenvalues of the pair (A, B) , and optionally the left and/or right eigenvectors. The expert driver xGGEVX performs the same task as the simple driver routines; in addition, it also balances the matrix pair to try to improve the conditioning of the eigenvalues and eigenvectors, and computes the condition numbers for the eigenvalues and/or left and right eigenvectors.

Examples:

Let us show how to use the simple driver SGGEV to solve the GNEPs (75.13) and (75.14). SGGEV first reduces the pair (A, B) to generalized upper Hessenberg form (H, R) , where H is upper Hessenberg (zero below the first lower subdiagonal) and R is upper triangular. Then SGGEV computes the generalized Schur form (S, T) of the generalized upper Hessenberg form (H, R) , using an QZ algorithm. The eigenvalues are computed from the diagonals of (S, T) . Finally, SGGEV computes left and/or right eigenvectors if requested. SGGEV has the following calling sequence:

```
CALL SGGEV( JOBVL, JOBVR, N, A, LDA, B, LDB, ALPHAR, ALPHAI, BETA, VL,
LDVL, VR, LDVR, WORK, LWORK, INFO )
```

Input to SGGEV:

JOBVL, JOBVR: = 'N', do not compute the left and/or right eigenvectors;
= 'V', compute the left and/or right eigenvectors.

N: The order of the matrices A and B . $N \geq 0$.

A, LDA: The matrix A and the leading dimension of the array A. $LDA \geq \max(1, N)$.

B, LDB: The matrix B and the leading dimension of the array B. $LDB \geq \max(1, N)$.

LDVL, LDVR: The leading dimensions of the eigenvector matrices VL and VR. $\text{LDVL}, \text{LDVR} \geq 1$.
If eigenvectors are required, then $\text{LDVL}, \text{LDVR} \geq N$.

WORK, LWORK: The workspace array and its length. $\text{LWORK} \geq \max(1, 8 * N)$. For good performance, LWORK must generally be larger.

If $\text{LWORK} = -1$, then a workspace query is assumed; the routine only calculates the optimal size of WORK, and returns this value in $\text{WORK}(1)$ on return.

Output from SGGEV:

ALPHAR, ALPHAI, BETA: $(\text{ALPHAR}(j) + i \cdot \text{ALPHAI}(j)) / \text{BETA}(j)$ for $j = 1, 2, \dots, N$, are the generalized eigenvalues. If $\text{ALPHAI}(j)$ is zero, then the j th eigenvalue is real; if positive, then the j th and $(j + 1)$ -st eigenvalues are a complex conjugate pair, with $\text{ALPHAI}(j + 1)$ negative.

VL: If $\text{JOBVL} = 'V'$, the left eigenvectors $\mathbf{y}_j$ are stored in the columns of VL, in the same order as their corresponding eigenvalues. If the j th eigenvalue is real, then $\mathbf{y}_j = \text{VL}(:, j)$, the j th column of VL. If the j th and $(j + 1)$ th eigenvalues form a complex conjugate pair, then $\mathbf{y}_j = \text{VL}(:, j) + i \cdot \text{VL}(:, j + 1)$ and $\mathbf{y}_{j+1} = \text{VL}(:, j) - i \cdot \text{VL}(:, j + 1)$.

VR: If $\text{JOBVR} = 'V'$, the right eigenvectors $\mathbf{x}_j$ are stored one after another in the columns of VR, in the same order as their eigenvalues. If the j th eigenvalue is real, then $\mathbf{x}_j = \text{VR}(:, j)$, the j th column of VR. If the j th and $(j + 1)$ th eigenvalues form a complex conjugate pair, then $\mathbf{x}_j = \text{VR}(:, j) + i \cdot \text{VR}(:, j + 1)$ and $\mathbf{x}_{j+1} = \text{VR}(:, j) - i \cdot \text{VR}(:, j + 1)$.

WORK: If $\text{INFO} = 0$, $\text{WORK}(1)$ returns the optimal LWORK.

INFO: $\text{INFO} = 0$ if successful exit. If $\text{INFO} = -j$, the j th argument had an illegal value. If $\text{INFO} = 1, \dots, N$, then the QZ iteration failed. No eigenvectors have been calculated, but $\text{ALPHAR}(j)$, $\text{ALPHAI}(j)$, and $\text{BETA}(j)$ should be correct for $j = \text{INFO} + 1, \dots, N$. If $\text{INFO} = N + 1$, then other than QZ iteration failed in SHGEQZ. If $\text{INFO} = N + 2$, then error return from STGEVC.

Note that the quotients $\text{ALPHAR}(j) / \text{BETA}(j)$ and $\text{ALPHAI}(j) / \text{BETA}(j)$ may easily over- or underflow, and $\text{BETA}(j)$ may even be zero. Thus, the user should avoid naively computing the ratio. However, ALPHAR and ALPHAI will be always less than and usually comparable to $\|A\|$ in magnitude, and BETA always less than and usually comparable to $\|B\|$.

Let us demonstrate the use of SGGEV in solving the GNEP of the following 6-by-6 matrices A and B :

$$A = \begin{bmatrix} 50 & -60 & 50 & -27 & 6 & 6 \\ 38 & -28 & 27 & -17 & 5 & 5 \\ 27 & -17 & 27 & -17 & 5 & 5 \\ 27 & -28 & 38 & -17 & 5 & 5 \\ 27 & -28 & 27 & -17 & 16 & 5 \\ 27 & -28 & 27 & -17 & 5 & 16 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 16 & 5 & 5 & 5 & -6 & 5 \\ 5 & 16 & 5 & 5 & -6 & 5 \\ 5 & 5 & 16 & 5 & -6 & 5 \\ 5 & 5 & 5 & 16 & -6 & 5 \\ 5 & 5 & 5 & 5 & -6 & 16 \\ 6 & 6 & 6 & 6 & -5 & 6 \end{bmatrix}.$$

The exact eigenvalues are $\frac{1}{2} + i \cdot \frac{\sqrt{3}}{2}$, $\frac{1}{2} + i \cdot \frac{\sqrt{3}}{2}$, $\frac{1}{2} - i \cdot \frac{\sqrt{3}}{2}$, $\frac{1}{2} - i \cdot \frac{\sqrt{3}}{2}$, ∞ , ∞ . Upon calling SGGEV with $N = 6$, $\text{LWORK} = 48$, on exit, arrays ALPHAR, ALPHAI, and BETA are

$$\begin{aligned} \text{ALPHAR} &= \begin{bmatrix} -25.7687130 & 6.5193458 & 5.8156629 & 5.8464251 & 5.5058141 & 11.2021322 \end{bmatrix}, \\ \text{ALPHAI} &= \begin{bmatrix} 0.0000000 & 11.2832556 & -10.0653677 & 10.1340599 & -9.5436525 & 0.0000000 \end{bmatrix}, \\ \text{BETA} &= \begin{bmatrix} 0.0000000 & 13.0169611 & 11.6119413 & 11.7124090 & 11.0300474 & 0.0000000 \end{bmatrix}. \end{aligned}$$

Therefore, there are two infinite eigenvalues corresponding to $\text{BETA}(1) = \text{BETA}(6) = 0$ and four finite eigenvalues $\lambda_j = (\text{ALPHAR}(j) + i \cdot \text{ALPHAI}(j))/\text{BETA}(j)$ for $j = 2, 3, 4, 5$.

$$(\text{ALPHAR}(2:5) + i \cdot \text{ALPHAI}(2:5))/\text{BETA}(2:5) = \begin{bmatrix} 0.50083 + i \cdot 0.86681 \\ 0.50083 - i \cdot 0.86681 \\ 0.49917 + i \cdot 0.86524 \\ 0.49917 - i \cdot 0.86524 \end{bmatrix}.$$

The left eigenvectors $\mathbf{y}_j$ are stored in VL . Since $\text{ALPHAI}(1) = \text{ALPHAI}(6) = 0$, $\mathbf{y}_1 = \text{VL}(:, 1)$ and $\mathbf{y}_6 = \text{VL}(:, 6)$. The second and third eigenvalues form a complex conjugate pair, the $\mathbf{y}_2 = \text{VL}(:, 2) + i \cdot \text{VL}(:, 3)$ and $\mathbf{y}_3 = \text{VL}(:, 2) - i \cdot \text{VL}(:, 3)$. Similarly, $\mathbf{y}_4 = \text{VL}(:, 4) + i \cdot \text{VL}(:, 5)$ and $\mathbf{y}_5 = \text{VL}(:, 4) - i \cdot \text{VL}(:, 5)$. If we place all the left eigenvectors in a matrix Y , where $Y = [\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4, \mathbf{y}_5, \mathbf{y}_6]$, we have

$$Y = \begin{bmatrix} -0.1666666 & 0.2632965 + i \cdot 0.3214956 & 0.2632965 - i \cdot 0.3214956 & -0.4613968 + i \cdot 0.1902102 & -0.4613968 - i \cdot 0.1902102 & 0.1666667 \\ -0.1666666 & -0.2834885 - i \cdot 0.7165115 & -0.2834885 + i \cdot 0.7165115 & 0.9231794 - i \cdot 0.0765849 & 0.9231794 + i \cdot 0.0765849 & 0.1666667 \\ -0.1666666 & 0.1623165 + i \cdot 0.7526108 & 0.1623165 - i \cdot 0.7526108 & -0.9240005 - i \cdot 0.0759995 & -0.9240005 + i \cdot 0.0759995 & 0.1666667 \\ -0.1666666 & 0.0396326 - i \cdot 0.4130635 & 0.0396326 + i \cdot 0.4130635 & 0.4619284 + i \cdot 0.1907958 & 0.4619284 - i \cdot 0.1907958 & 0.1666666 \\ -0.1666671 & -0.0605860 + i \cdot 0.0184893 & -0.0605860 - i \cdot 0.0184893 & 0.0000969 - i \cdot 0.0761408 & 0.0000969 + i \cdot 0.0761408 & 0.1666666 \\ 1.0000000 & -0.0605855 + i \cdot 0.0184900 & -0.0605855 - i \cdot 0.0184900 & 0.0000959 - i \cdot 0.0761405 & 0.0000959 + i \cdot 0.0761405 & -1.0000000 \end{bmatrix}.$$

The right eigenvectors can be recovered from VR in a way similar to the left eigenvectors. If we place all the right eigenvectors in a matrix X , where $X = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6]$, we have

$$U = \begin{bmatrix} 0.1666672 & -0.2039835 - i \cdot 0.5848466 & -0.2039835 + i \cdot 0.5848466 & 0.5722237 - i \cdot 0.0237538 & 0.5722237 + i \cdot 0.0237538 & 0.1666672 \\ 0.1666664 & -0.7090308 - i \cdot 0.2908980 & -0.7090308 + i \cdot 0.2908980 & 0.4485306 - i \cdot 0.5514694 & 0.4485306 + i \cdot 0.5514694 & 0.1666664 \\ 0.1666666 & -0.7071815 + i \cdot 0.2928185 & -0.7071815 - i \cdot 0.2928185 & -0.0709520 - i \cdot 0.7082051 & -0.0709520 + i \cdot 0.7082051 & 0.1666666 \\ 0.1666666 & -0.2013957 + i \cdot 0.5829236 & -0.2013957 - i \cdot 0.5829236 & -0.4667411 - i \cdot 0.3361499 & -0.4667411 + i \cdot 0.3361499 & 0.1666666 \\ 1.0000000 & -0.2023994 + i \cdot 0.0000001 & -0.2023994 - i \cdot 0.0000001 & 0.0536732 - i \cdot 0.1799536 & 0.0536732 + i \cdot 0.1799536 & 1.0000000 \\ 0.1666666 & -0.2023991 - i \cdot 0.0000002 & -0.2023991 + i \cdot 0.0000002 & 0.0536734 - i \cdot 0.1799532 & 0.0536734 + i \cdot 0.1799532 & 0.1666664 \end{bmatrix}.$$

75.11 Generalized Singular Value Decomposition

Definitions:

The generalized (or quotient) singular value decomposition (GSVD or QSVD) of an m -by- n matrix A and a p -by- n matrix B is given by the pair of factorizations

$$A = U \Sigma_1 \begin{bmatrix} 0 & R \end{bmatrix} Q^T \quad \text{and} \quad B = V \Sigma_2 \begin{bmatrix} 0 & R \end{bmatrix} Q^T. \quad (75.15)$$

The matrices in these factorizations have the following properties:

- U is m -by- m , V is p -by- p , Q is n -by- n , and all three matrices are orthogonal. If A and B are complex, these matrices are unitary instead of orthogonal, and Q^T should be replaced by Q^* in the pair of factorizations.
- R is r -by- r , upper triangular and nonsingular. $\begin{bmatrix} 0 & R \end{bmatrix}$ is r -by- n (in other words, the 0 is an r -by- $(n-r)$ zero matrix). The integer r is the rank of $\begin{bmatrix} A \\ B \end{bmatrix}$.
- Σ_1 is m -by- r and Σ_2 is p -by- r . Both are real, nonnegative, and diagonal, satisfying $\Sigma_1^T \Sigma_1 + \Sigma_2^T \Sigma_2 = I$. Write $\Sigma_1^T \Sigma_1 = \text{diag}(\alpha_1^2, \dots, \alpha_r^2)$ and $\Sigma_2^T \Sigma_2 = \text{diag}(\beta_1^2, \dots, \beta_r^2)$. The ratios α_j/β_j for $j = 1, 2, \dots, r$ are called the *generalized singular values*.

Σ_1 and Σ_2 have the following detailed structures, depending on whether $m-r \geq 0$ or $m-r < 0$.

- In the first case, when $m - r \geq 0$,

$$\Sigma_1 = \begin{matrix} & k & \ell \\ & \begin{pmatrix} I & 0 \\ 0 & C \\ 0 & 0 \end{pmatrix} \\ \begin{matrix} k \\ \ell \\ m - k - \ell \end{matrix} & \end{matrix} \quad \text{and} \quad \Sigma_2 = \begin{matrix} & k & \ell \\ & \begin{pmatrix} 0 & S \\ 0 & 0 \end{pmatrix} \\ \begin{matrix} \ell \\ p - \ell \end{matrix} & \end{matrix}. \quad (75.16)$$

Here $k + \ell = r$, and ℓ is the rank of B . C and S are diagonal matrices satisfying $C^2 + S^2 = I$, and S is nonsingular. Let c_j and s_j be the diagonal entries of C and S , respectively. Then we have $\alpha_1 = \cdots = \alpha_k = 1$, $\alpha_{k+j} = c_j$ for $j = 1, \dots, \ell$, $\beta_1 = \cdots = \beta_k = 0$, and $\beta_{k+j} = s_j$ for $j = 1, \dots, \ell$. Thus, the first k generalized singular values $\alpha_1/\beta_1, \dots, \alpha_k/\beta_k$ are infinite and the remaining ℓ generalized singular values are finite.

- In the second case, when $m - r < 0$,

$$\Sigma_1 = \begin{matrix} & k & m - k & k + \ell - m \\ & \begin{pmatrix} I & 0 & 0 \\ 0 & C & 0 \end{pmatrix} \\ \begin{matrix} k \\ m - k \end{matrix} & \end{matrix} \quad \text{and} \quad \Sigma_2 = \begin{matrix} & k & m - k & k + \ell - m \\ & \begin{pmatrix} 0 & S & 0 \\ 0 & 0 & I \\ 0 & 0 & 0 \end{pmatrix} \\ \begin{matrix} m - k \\ k + \ell - m \\ p - \ell \end{matrix} & \end{matrix}. \quad (75.17)$$

Again, $k + \ell = r$, and ℓ is the rank of B . C and S are diagonal matrices satisfying $C^2 + S^2 = I$, and S is nonsingular. Let c_j and s_j be the diagonal entries of C and S , respectively. Then we have $\alpha_1 = \cdots = \alpha_k = 1$, $\alpha_{k+j} = c_j$ for $j = 1, \dots, m - k$, $\alpha_{m+1} = \cdots = \alpha_r = 0$, and $\beta_1 = \cdots = \beta_k = 0$, $\beta_{k+j} = s_j$ for $j = 1, \dots, m - k$, $\beta_{m+1} = \cdots = \beta_r = 1$. Thus, the first k generalized singular values $\alpha_1/\beta_1, \dots, \alpha_k/\beta_k$ are infinite, and the remaining ℓ generalized singular values are finite.

Backgrounds:

Here are some important special cases of the QSVD. First, when B is square and nonsingular, then $r = n$ and the QSVD of A and B is equivalent to the SVD of AB^{-1} , where the singular values of AB^{-1} are equal to the generalized singular values of A and B :

$$AB^{-1} = (U \Sigma_1 R Q^T)(V \Sigma_2 R Q^T)^{-1} = U(\Sigma_1 \Sigma_2^{-1})V^T.$$

Second, if the columns of $\begin{bmatrix} A^T & B^T \end{bmatrix}^T$ are orthonormal, then $r = n$, $R = I$, and the QSVD of A and B is equivalent to the CS (Cosine-Sine) decomposition of $\begin{bmatrix} A^T & B^T \end{bmatrix}^T$:

$$\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} U & 0 \\ 0 & V \end{bmatrix} \begin{bmatrix} \Sigma_1 \\ \Sigma_2 \end{bmatrix} Q^T.$$

Third, the generalized eigenvalues and eigenvectors of the pencil $A^T A - \lambda B^T B$ can be expressed in terms of the QSVD of A and B , namely,

$$X^T A^T A X = \begin{bmatrix} 0 & 0 \\ 0 & \Sigma_1^T \Sigma_1 \end{bmatrix} \quad \text{and} \quad X^T B^T B X = \begin{bmatrix} 0 & 0 \\ 0 & \Sigma_2^T \Sigma_2 \end{bmatrix},$$

where $X = Q \begin{bmatrix} I & 0 \\ 0 & R^{-1} \end{bmatrix}$. Therefore, the columns of X are the eigenvectors of $A^T A - \lambda B^T B$, and the “nontrivial” eigenvalues are the squares of the generalized singular values. “Trivial” eigenvalues are those corresponding to the leading $n - r$ columns of X , which span the common null space of $A^T A$ and $B^T B$.

The “trivial eigenvalues” are not well defined.¹ (See Chapter 15 for more information on generalized singular value problems.)

Driver Routines:

The driver routine xGGSVD computes the GSVD (75.15) of A and B .

Examples:

Let us show how to use the driver routine SGGSDV to compute the QSVD (75.15). SGGSDV first reduces the matrices A and B to a pair of triangular matrices, and then use a Jacobi-like method to compute the QSVD of the triangular pair. SGGSDV has the following calling sequence:

```
CALL SGGSDV( JOBU, JOBV, JOBQ, M, N, P, K, L, A, LDA, B, LDB, ALPHA,
BETA, U, LDU, V, LDV, Q, LDQ, WORK, IWORK, INFO )
```

Input to SGGSDV:

JOBU, JOBV, JOBQ: = 'U', orthogonal matrices U , V and Q are computed;
= 'N', these orthogonal matrices are not computed.

M, N, P: The number of rows or columns of the matrices A and B as defined in (15)

A, LDA: The M -by- N matrix A and the leading dimension of the array A. $LDA \geq \max(1, M)$.

B, LDB: The P -by- N matrix B and the leading dimension of the array B. $LDB \geq \max(1, P)$.

LDU, LDV, LDQ: The leading dimension of the arrays U, V, and Q if the orthogonal matrices U , V , and Q are computed, $LDU \geq \max(1, M)$, $LDV \geq \max(1, P)$, $LDQ \geq \max(1, N)$.

WORK: The real workspace array, dimension $\max(3N, M, P) + N$.

IWORK: The integer workspace array, dimension N .

Output from SGGSDV:

K, L: The dimension of the subblocks described in the definition of GSVD. $K + L$ is the effective numerical rank of the matrix $\begin{bmatrix} A^T & B^T \end{bmatrix}^T$.

A: The entire triangular matrix R is stored in $A(1 : K+L, N-K-L+1 : N)$ if $m - r \geq 0$.
Otherwise, the subblock $R(1 : m, 1 : k + \ell)$ of R are stored in $A(1 : M, N-K-L+1 : N)$.

B: The subblock $R(m + 1 : k + \ell, m + 1 : k + \ell)$ of R are stored in $B(M-K+1 : L, N+M-K-L+1 : N)$ if $m - r < 0$.

ALPHA, BETA: The generalized singular value pairs;

$ALPHA(1 : K) = 1$ and $BETA(1 : K) = 0$.

- If $M-K-L \geq 0$, then $ALPHA(K + 1 : K + L) = C$ and $BETA(K + 1 : K + L) = S$.
- If $M-K-L < 0$, then

$ALPHA(K + 1 : M) = C$ and $ALPHA(M + 1 : K + L) = 0$,

$BETA(K + 1 : M) = S$ and $BETA(M + 1 : K + L) = 1$;

And $ALPHA(K+L+1 : N) = 0$, $BETA(K+L+1 : N) = 0$.

U, V, Q: Contains computed orthogonal matrices U , V , and Q if requested.

¹If we tried to compute the trivial eigenvalues in the same way as the nontrivial ones, that is by taking ratios of the leading $n - r$ diagonal entries of $X^T A^T A X$ and $X^T B^T B X$, we would get $0/0$.

INFO: INFO = 0 if successful exit. If INFO = $-j$, then the j th argument had an illegal value. If INFO = 1, the Jacobi-type procedure failed to converge.

Let us demonstrate the use of SGGSD in computing the QSVD of the following 6-by-5 matrices A and B :

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 & 5 \\ 0 & 3 & 2 & 0 & 2 \\ 1 & 0 & 2 & 1 & 0 \\ 0 & 2 & 3 & 0 & -1 \\ 1 & 0 & 2 & 1 & 1 \\ 0 & 2 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -2 & 2 & 1 & 1 \\ 0 & 3 & 0 & 0 & 0 \\ 1 & -2 & 2 & 1 & 1 \\ 0 & 2 & 0 & 0 & 0 \\ 2 & -4 & 4 & 2 & 2 \\ 1 & 3 & 2 & 1 & 1 \end{bmatrix}.$$

Upon calling SGGSD with $M = 6, P = 6, N = 5, LWORK = 20$, we have $K = 2$ and $L = 2$. The QSVD (75.15) of A and B falls in the first case (75.16) since $M - K - L = 6 - 2 - 2 = 2 > 0$. The arrays ALPHA and BETA are

$$\begin{aligned} \text{ALPHA} &= \begin{bmatrix} 1.0000000 & 1.0000000 & 0.1537885 & 0.5788464 & 0.0000000 \end{bmatrix}, \\ \text{BETA} &= \begin{bmatrix} 0.0000000 & 0.0000000 & 0.9881038 & 0.8154366 & 0.0000000 \end{bmatrix}. \end{aligned}$$

Hence, Σ_1 and Σ_2 have the structure as described in (75.16), namely,

$$\Sigma_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0.1537885 & 0 \\ 0 & 0 & 0 & 0.5788464 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \Sigma_2 = \begin{bmatrix} 0 & 0 & 0.9881038 & 0 \\ 0 & 0 & 0 & 0.8154366 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The first two generalized singular values are infinite, $\alpha_1/\beta_1 = \alpha_2/\beta_2 = \infty$, and the remaining two generalized singular values are finite, $\alpha_3/\beta_3 = 0.15564$ and $\alpha_4/\beta_4 = 0.70986$.

Furthermore, the array $A(1:4, 2:5)$ contains the 4-by-4 upper triangular matrix R as defined in (75.15):

$$R = \begin{bmatrix} 3.6016991 & -1.7135643 & -0.2843603 & 1.8104467 \\ 0 & -2.6087811 & -4.2943931 & 5.1107349 \\ 0 & 0 & 6.9692163 & 3.5063875 \\ 0 & 0 & 0 & 7.3144341 \end{bmatrix}.$$

The orthogonal matrices U , V , and Q are returned in the arrays U , V , and Q , respectively:

$$\begin{aligned} U &= \begin{bmatrix} -0.6770154 & -0.4872811 & -0.4034495 & -0.2450049 & -0.2151961 & 0.1873468 \\ -0.0947438 & -0.5723576 & 0.4163284 & 0.1218751 & 0.0785425 & -0.6848933 \\ 0.2098812 & 0.0670342 & 0.2612190 & -0.7393155 & -0.5670457 & -0.1228532 \\ 0.6974092 & -0.5903998 & -0.3678919 & 0.0010751 & -0.0196356 & 0.1712235 \\ 0.0000000 & 0.0000001 & -0.0735656 & -0.6152450 & 0.7822418 & -0.0644937 \\ -0.0473719 & -0.2861788 & 0.6744684 & -0.0019711 & 0.1170180 & 0.6687696 \end{bmatrix}, \\ V &= \begin{bmatrix} -0.3017521 & -0.2581125 & 0.9018297 & -0.0002676 & -0.1695592 & -0.0166328 \\ 0.4354534 & -0.2679386 & 0.1028928 & 0.0704557 & 0.2595005 & -0.8097517 \\ -0.3017520 & -0.2581124 & -0.1784097 & -0.8828155 & -0.0002829 & -0.1764375 \\ 0.2903022 & -0.1786257 & -0.1298870 & -0.0008522 & -0.9259184 & -0.0980879 \\ -0.6035041 & -0.5162248 & -0.3568195 & 0.4625078 & -0.0224125 & -0.1660080 \\ 0.4240036 & -0.7046767 & -0.0097810 & -0.0419325 & 0.2146671 & 0.5250862 \end{bmatrix}, \\ Q &= \begin{bmatrix} -0.7071068 & -0.2073452 & -0.5604916 & -0.0112638 & -0.3777966 \\ 0.0000000 & 0.0000000 & 0.0000000 & 0.9995558 & -0.0298012 \\ 0.0000001 & 0.5853096 & 0.2932303 & -0.0225276 & -0.7555932 \\ 0.7071067 & -0.2073452 & -0.5604916 & -0.0112638 & -0.3777965 \\ -0.0000001 & -0.7559289 & 0.5345224 & -0.0112638 & -0.3777965 \end{bmatrix}. \end{aligned}$$

References

[CLA] <http://www.netlib.org/clapack/>.

[JLA] <http://www.netlib.org/java/f2j/>.

[LUG99] E. Anderson, Z. Bai, C. Bischof, S. Blackford, J. Demmel, J. Dongarra, J. Du Croz, A. Greenbaum, S. Hammarling, A. McKenney, and D. Sorensen, *LAPACK Users' Guide*, 3rd ed., SIAM, Philadelphia, 1999.

[LAP] <http://www.netlib.org/lapack/>.

[LAP95] <http://www.netlib.org/lapack95/>.

[LA+] <http://www.netlib.org/lapack++/>.

[Sca] <http://www.netlib.org/scalapack/>.

76

Use of ARPACK and EIGS

D. C. Sorensen
Rice University

76.1	The ARPACK Software	76-1
76.2	Reverse Communication	76-2
76.3	Directory Structure and Contents	76-3
76.4	Naming Conventions, Precisions, and Types	76-3
76.5	Getting Started	76-4
76.6	Setting Up the Problem	76-5
76.7	General Use of ARPACK	76-7
76.8	Using the Computational Modes	76-8
76.9	MATLAB's® EIGS	76-9
	References	76-11

ARPACK is a library of Fortran77 subroutines designed to compute a selected subset of the eigenvalues of a large matrix. It is based upon a limited storage scheme called the implicitly restarted Arnoldi method (IRAM) [Sor92]. This software can solve largescale non-Hermitian or Hermitian (standard and generalized) eigenvalue problems. The IRA method is described in Chapter 44, Implicitly Restarted Arnoldi Method.

This chapter describes the design and performance features of the eigenvalue software ARPACK and gives a brief discussion of usage. More detailed descriptions are available in the papers [Sor92] and [Sor02] and in the ARPACK Users' Guide [LSY98].

The design goals were robustness, efficiency, and portability. Two very important principles that have helped to achieve these goals are modularity and independence from specific vendor supplied communication and performance libraries.

In this chapter, multiplication of a vector $\mathbf{x}$ by a scalar λ is denoted by $\mathbf{x}\lambda$ so that the eigenvector-eigenvalue relation is $A\mathbf{x} = \mathbf{x}\lambda$. This convention provides for direct generalizations to the more general invariant subspace relations $AX = XH$, where X is an $n \times k$ matrix and H is a $k \times k$ matrix with $k < n$.

76.1 The ARPACK Software

The ARPACK software has been used on a wide range of applications. P_ARPACK is a parallel extension to the ARPACK library and is targeted for distributed memory message passing systems. Both packages are freely available and can be downloaded at <http://www.caam.rice.edu/software/ARPACK/>.

Features (of ARPACK and P_ARPACK):

1. A reverse communication interface.
2. Computes k eigenvalues that satisfy a user-specified criterion, such as largest real part, largest absolute value, etc.
3. A fixed predetermined storage requirement of $n \cdot \mathcal{O}(k) + \mathcal{O}(k^2)$ words.
4. Driver routines are included as templates for implementing various spectral transformations to enhance convergence and to solve the generalized eigenvalue problem, or the SVD problem.
5. Special consideration is given to the generalized problem $A\mathbf{x} = M\mathbf{x}\lambda$ for singular or ill-conditioned symmetric positive semidefinite M .
6. A Schur basis of dimension k that is numerically orthogonal to working precision is always computed. These are also eigenvectors in the Hermitian case. In the non-Hermitian case eigenvectors are available on request. Eigenvalues are computed to a user specified accuracy.

76.2 Reverse Communication

Reverse communication is a means to overcome certain restrictions in the Fortran language; with reverse communication, control is returned to the calling program when interaction with the matrix is required. This is a convenient interface for experienced users. However, it may be more challenging for inexperienced users. It has proven to be extremely useful for interfacing with large application codes.

This interface avoids having to express a matrix-vector product through a subroutine with a fixed calling sequence or to provide a sparse matrix with a specific data structure. The user is free to choose any convenient data structure for the matrix representation.

Examples:

1. A typical use of this interface is illustrated as follows:

```

10  continue
    call snaupd (ido, bmat, n, which,..., workd,..., info)
    if (ido .eq. newprod) then
        call matvec ('A', n, workd(ipntr(1)), workd(ipntr(2)))
    else
        return
    endif
    go to 10
%
```

This shows a code segment of the routine the user must write to set up the reverse communication call to the top level ARPACK routine `snaupd` to solve a nonsymmetric eigenvalue problem in single precision. With reverse communication, control is returned to the calling program when interaction with the matrix A is required. The action requested of the calling program is specified by the reverse communication parameter `ido`, which is set in the call to `snaupd`. In this case, there are two possible requests indicated by `ido`. One action is to multiply the vector held in the array `workd` beginning at location `ipntr(1)` by A and then place the result in the array `workd` beginning at location `ipntr(2)`. The other action is to halt the iteration due to successful convergence or due to an error.

When the parameter `ido` indicates a new matrix vector product is required, a call is made to a subroutine `matvec` in this example. However, it is only necessary to supply the action of the matrix on the specified vector and put the result in the designated location. No specified data structure is imposed on A and if a subroutine is used, no particular calling sequence is specified. Because of this, reverse communication is very flexible and even provides a convenient way to use ARPACK interfaced with code written in another language, such as C or C++.

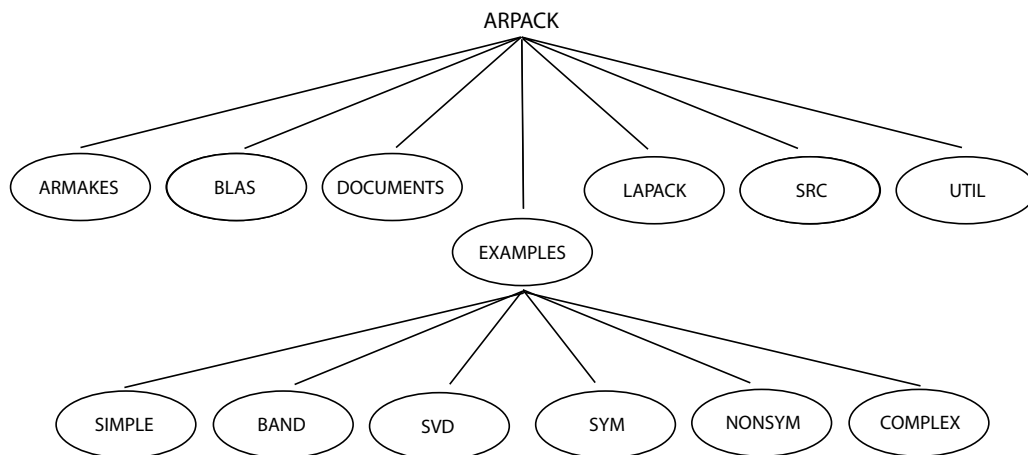


FIGURE 76.1 The ARPACK directory structure.

76.3 Directory Structure and Contents

Once the ARPACK software has been downloaded and unbundled, a directory structure will have been created as pictured in Figure 76.1.

Subdirectories:

1. The `ARMAKES` subdirectory contains sample files with machine specific information needed during the building of the ARPACK library.
2. The `BLAS` and `LAPACK` subdirectories contain the necessary codes from those libraries.
3. The `DOCUMENTS` subdirectory contains files that have example templates showing how to invoke the different computational modes offered by ARPACK.
4. Example driver programs illustrating all of the computational modes, data types, and precisions may be found in the `EXAMPLES` directory.
5. Programs for banded, complex, nonsymmetric, symmetric eigenvalue problems, and singular value decomposition may be found in the directories `BAND`, `COMPLEX`, `NONSYM`, `SYM`, `SVD`.
6. The `README` files in each subdirectory provide further information.
7. The `SRC` subdirectory contains all the ARPACK source codes.
8. The `UTIL` subdirectory contains the various utility routines needed for printing results and timing the execution of the ARPACK subroutines.

76.4 Naming Conventions, Precisions, and Types

1. ARPACK has two interface routines that must be invoked by the user. They are `__aupd` that implements the IRAM and `__eupd` to post process the results of `__aupd`.
2. The user may request an orthogonal basis for a selected invariant subspace or eigenvectors corresponding to selected eigenvalues with `__eupd`. If a spectral transformation is used, `__eupd` automatically transforms the computed eigenvalues of the shift-invert operator to those of the original problem.
3. Both `__aupd` and `__eupd` are available for several combinations of problem type (symmetric and nonsymmetric), data type (real, complex), and precision (single, double). The first letter (*s*, *d*, *c*, *z*) denotes precision and data type. The second letter denotes whether the problem is symmetric (*s*) or nonsymmetric (*n*).

4. Thus, `dnaupd` is the routine to use if the problem is a double precision nonsymmetric (standard or generalized) problem and `dneupd` is the post-processing routine to use in conjunction with `dnaupd` to recover eigenvalues and eigenvectors of the original problem upon convergence. For complex matrices, one should use `_naupd` and `_neupd` with the first letter either `c` or `z` regardless of whether the problem is Hermitian or non-Hermitian.

76.5 Getting Started

Perhaps the easiest way to rapidly become acquainted with the possible uses of ARPACK is to run the example driver routines that have been supplied for each of the computational modes. These may be used as templates and adapted to solve specific problems. To get started, it is recommended that the user execute driver routines from the `SIMPLE` subdirectory.

The `dssimp` driver implements the reverse communication interface to the routine `dsaupd` that will compute a few eigenvalues and eigenvectors of a symmetric matrix. It illustrates the simplest case and has exactly the same structure as shown previously except that the top level routine is `dsaupd` instead of `snaupd`. The full call issued by `dssimp` is as follows.

```
call dsaupd ( ido, bmat, n, which, nev, tol, resid,
&           ncv, v, ldv, iparam, ipntr, workd,
&           workl, lworkl, info )
```

This `dssimp` driver is intended to serve as template to enable a user to create a program to use `dsaupd` on a specific problem in the simplest computational mode. All of the driver programs in the various `EXAMPLES` subdirectories are intended to be used as templates. They all follow the same principle, but the usage is slightly more complicated.

The only thing that must be supplied in order to use this routine on your problem is to change the array dimensions and to supply a means to compute the matrix-vector product

$$\mathbf{w} \leftarrow \mathbf{A}\mathbf{v}$$

on request from `dsaupd`. The selection of which eigenvalues to compute may be altered by changing the parameter `which`.

Once usage of `dsaupd` in the simplest mode is understood, it will be easier to explore the other available options such as solving generalized eigenvalue problems using a shift-invert computational mode.

If the computation is successful, `dsaupd` indicates that convergence has taken place through the parameter `ido`. Then various steps may be taken to recover the results in a useful form. This is done through the subroutine `dseupd` as illustrated below.

```
call dseupd(rvec, howmny, select, d, v, ldv, sigma, bmat,
&          n, which, nev, tol, resid, ncv, v, ldv,
&          iparam, ipntr, workd, workl, lworkl, ierr)
```

Eigenvalues are returned in the first column of the two-dimensional array `d` and the corresponding eigenvectors are returned in the first `NCONV` (`=IPARAM(5)`) columns of the two-dimensional array `v` if requested. Otherwise, an orthogonal basis for the invariant subspace corresponding to the eigenvalues in `d` is returned in `v`.

The input parameters that must be specified are

- The logical variable `rvec` = `.true.` if eigenvectors are requested,
`.false.` if only eigenvalues are desired.
- The character*1 parameter `howmny` that specifies how many eigenvectors are desired.
`howmny` = 'A': compute `nev` eigenvectors;
`howmny` = 'S': compute some of the eigenvectors,
specified by the logical array `select`.

- `sigma` should contain the value of the shift used if `iparam(7) = 3, 4, 5`. It is not referenced if `iparam(7) = 1` or `2`.

When requested, the eigenvectors returned by `dseupd` are normalized to have unit length with respect to the M semi-inner product that was used. Thus, if $M = I$, they will have unit length in the standard 2-norm. In general, a computed eigenvector $\mathbf{x}$ will satisfy $1 = \mathbf{x}^T M \mathbf{x}$.

76.6 Setting Up the Problem

To set up the problem, the user needs to specify the number of eigenvalues to compute which eigenvalues are of interest, the number of basis vectors to use, and whether or not the problem is standard or generalized. These items are controlled by the following parameters.

Parameters for the top-level ARPACK routines:

`ido` — Reverse communication flag.

`nev` — The number of requested eigenvalues to compute.

`ncv` — The number of Arnoldi basis vectors to use through the course of the computation.

`bmat` — Indicates whether the problem is standard `bmat = 'I'` or generalized (`bmat = 'G'`).

`which` — Specifies which eigenvalues of A are to be computed.

`tol` — Specifies the relative accuracy to which eigenvalues are to be computed.

`iparam` — Specifies the computational mode, number of IRAM iterations, the implicit shift strategy, and outputs various informational parameters upon completion of IRAM.

The value of `ncv` must be at least `nev + 1`. The options available for `which` include 'LA' and 'SA' for the algebraically largest and smallest eigenvalues, 'LM' and 'SM' for the eigenvalues of largest or smallest magnitude, and 'BE' for the simultaneous computation of the eigenvalues at both ends of the spectrum. For a given problem, some of these options may converge more rapidly than others due to the approximation properties of the IRAM as well as the distribution of the eigenvalues of A . Convergence behavior can be quite different for various settings of the `which` parameter. For example, if the matrix is indefinite then setting `which = 'SM'` will require interior eigenvalues to be computed and the Arnoldi/Lanczos process may require many steps before these are resolved.

For a given `ncv`, the computational work required is proportional to $n \cdot ncv^2$ FLOPS. Setting `nev` and `ncv` for optimal performance is very much problem dependent. If possible, it is best to avoid setting `nev` in a way that will split clusters of eigenvalues. For example, if the five smallest eigenvalues are positive and on the order of 10^{-4} and the sixth smallest eigenvalue is on the order of 10^{-1} , then it is probably better to ask for `nev = 5` than for `nev = 3`, even if the three smallest are the only ones of interest.

Setting the optimal value of `ncv` relative to `nev` is not completely understood. As with the choice of `which`, it depends upon the underlying approximation properties of the IRAM as well as the distribution of the eigenvalues of A . As a rule of thumb, $ncv \geq 2 \cdot nev$ is reasonable. There are tradeoffs due to the cost of the user supplied matrix-vector products and the cost of the implicit restart mechanism and the cost of maintaining the orthogonality of the Arnoldi vectors. If the user supplied matrix-vector product is relatively cheap, then a smaller value of `ncv` may lead to more user matrix-vector products, but an overall decrease in computation time.

Storage Declarations:

The program is set up so that the setting of the three parameters `maxn`, `maxnev`, `maxncv` will automatically declare all of the work space needed to run `dsaupd` on a given problem.

The declarations allow a problem size of $N \leq \text{maxn}$, computation of $nev \leq \text{maxnev}$ eigenvalues, and using at most $ncv \leq \text{maxncv}$ Arnoldi basis vectors during the IRAM. The user may override the


```

Double precision
&          v(ldv,maxncv), workl(maxncv*(maxncv+8)),
&          workd(3*maxn), d(maxncv,2), resid(maxn),

```

FIGURE 76.2 Storage declarations needed for ARPACK subroutine `dsaupd`.

default settings used for the example problem by modifying `maxn`, `maxnev`, and `maxncv` in the following parameter statement in the `dssimp` code.

```

integer      maxn, maxnev, maxncv, ldv
parameter    (maxn=256, maxnev=10, maxncv=25, ldv=maxn)

```

These parameters are used in the code segment listed in Figure 76.2 for declaring all of the output and work arrays needed by the ARPACK subroutines `dsaupd` and `dseupd`. These will set the storage values in ARPACK arrays.

Stopping Criterion:

The stopping criterion is determined by the user through specification of the parameter `tol`. The default value for `tol` is machine precision ϵ_M . There are several things to consider when setting this parameter. In absence of all other considerations, one should expect a computed eigenvalue λ_c to roughly satisfy

$$|\lambda_c - \lambda_t| \leq \text{tol} \|A\|_2,$$

where λ_t is the eigenvalue of A nearest to λ_c . Typically, decreasing the value of `tol` will increase the work required to satisfy the stopping criterion. However, setting `tol` too large may cause eigenvalues to be missed when they are multiple or very tightly clustered. Typically, a fairly small setting of `tol` and a reasonably large setting of `ncv` is required to avoid missing multiple eigenvalues. However, some care must be taken. It is possible to set `tol` so small that convergence never occurs. There may be additional complications when the matrix A is nonnormal or when the eigenvalues of interest are clustered near the origin.

Initial Parameter Settings:

The reverse communication flag is denoted by `ido`. This parameter must be initially set to 0 to signify the first call to `dsaupd`. Various algorithmic modes may be selected through the settings of the entries in the integer array `iparam`. The most important of these is the value of `iparam(7)`, which specifies the computational mode to use.

Setting the Starting Vector:

The parameter `info` should be set to 0 on the initial call to `dsaupd` unless the user wants to supply the starting vector that initializes the IRAM. Normally, this default is a reasonable choice. However, if this eigenvalue calculation is one of a sequence of closely related problems, then convergence may be accelerated if a suitable starting vector is specified. Typical choices in this situation might be to use the final value of the starting vector from the previous eigenvalue calculation (that vector will already be in the first column of ∇) or to construct a starting vector by taking a linear combination of the computed eigenvectors from the previously converged eigenvalue calculation. If the starting vector is to be supplied, then it should be placed in the array `resid` and `info` should be set to 1 on entry to `dsaupd`. On completion, the parameter `info` may contain the value 0 indicating the iteration was successful or it may contain a nonzero value indicating an error or a warning condition. The meaning of a nonzero value returned in `info` may be found in the header comments of the subroutine `dsaupd`.

Trace Debugging Capability:

ARPACK provides a means to trace the progress of the computation as it proceeds. Various levels of output may be specified from no output (`level = 0`) to voluminous (`level = 3`). A detailed description of trace debugging may be found in [LSY98].

76.7 General Use of ARPACK

The Shift and Invert Spectral Transformation Mode:

The most general problem that may be solved with ARPACK is to compute a few selected eigenvalues and corresponding eigenvectors for

$$A\mathbf{x} = M\mathbf{x}\lambda,$$

where A and M are real or complex $n \times n$ matrices.

The shift and invert spectral transformation is used to enhance convergence to a desired portion of the spectrum. If $(\mathbf{x}, \lambda)$ is an eigen-pair for (A, M) and $\sigma \neq \lambda$, then

$$(A - \sigma M)^{-1} M\mathbf{x} = \mathbf{x}\nu \quad \text{where} \quad \nu = \frac{1}{\lambda - \sigma},$$

where we are requiring that $A - \sigma M$ is nonsingular. Here it is possible for A or M to be singular, but they cannot have a nonzero null vector in common. This transformation is effective for finding eigenvalues near σ since the new eigenvalues ν_j of $C \equiv (A - \sigma M)^{-1} M$ that are largest in magnitude correspond to the new eigenvalues λ_j of the original problem that are nearest to the shift σ in absolute value. As discussed in Chapter 44, these transformed eigenvalues of largest magnitude are precisely the eigenvalues that are easy to compute with a Krylov method. Once they are found, they may be transformed back to eigenvalues of the original problem.

 M Is Hermitian Positive Definite:

If M is Hermitian positive definite and well conditioned ($\|M\| \cdot \|M^{-1}\|$ is of modest size), then computing the Cholesky factorization $M = LL^*$ and converting $A\mathbf{x} = M\mathbf{x}\lambda$ into

$$(L^{-1}AL^{-*})\mathbf{y} = \mathbf{y}\lambda, \quad \text{where} \quad L^*\mathbf{x} = \mathbf{y}$$

provides a transformation to a standard eigenvalue problem. In this case, a request for a matrix vector product would be satisfied with the following three steps:

1. Solve $L^*\mathbf{z} = \mathbf{v}$ for $\mathbf{z}$.
2. Matrix-vector multiply $\mathbf{z} \leftarrow A\mathbf{z}$.
3. Solve $L\mathbf{w} = \mathbf{z}$ for $\mathbf{w}$.

Upon convergence, a computed eigenvector $\mathbf{y}$ for $(L^{-1}AL^{-*})$ is converted to an eigenvector $\mathbf{x}$ of the original problem by solving the triangular system $L^*\mathbf{x} = \mathbf{y}$. This transformation is most appropriate when A is Hermitian, M is Hermitian positive definite, and extremal eigenvalues are sought. This is because $L^{-1}AL^{-*}$ will be Hermitian when A is the same.

If A is Hermitian positive definite and the smallest eigenvalues are sought, then it would be best to reverse the roles of A and M in the above description and ask for the largest algebraic eigenvalues or those of largest magnitude. Upon convergence, a computed eigenvalue $\hat{\lambda}$ would then be converted to an eigenvalue of the original problem by the relation $\lambda \leftarrow 1/\hat{\lambda}$.

 M Is NOT Hermitian Positive Semidefinite:

If neither A nor M is Hermitian positive semidefinite, then a direct transformation to standard form is required. One simple way to obtain a direct transformation to a standard eigenvalue problem $C\mathbf{x} = \mathbf{x}\lambda$

is to multiply on the left by M^{-1} , which results in $C = M^{-1}A$. Of course, one should not perform this transformation explicitly since it will most likely convert a sparse problem into a dense one. If possible, one should obtain a direct factorization of M and when a matrix-vector product involving C is called for, it may be accomplished with the following two steps:

1. Matrix-vector multiply $\mathbf{z} \leftarrow A\mathbf{v}$.
2. Solve: $M\mathbf{w} = \mathbf{z}$.

Several problem dependent issues may modify this strategy. If M is singular or if one is interested in eigenvalues near a point σ , then a user may choose to work with $C \equiv (A - \sigma M)^{-1}M$ but without using the M -inner products discussed previously. In this case the user will have to transform the converged eigenvalues v_j of C to eigenvalues λ_j of the original problem.

76.8 Using the Computational Modes

An extensive set of computational modes has been constructed to implement all of the shift-invert options mentioned previously. The problem set up is similar for all of the available computational modes. A detailed description of the reverse communication loop for the various modes (Shift-Invert for a Real Nonsymmetric Generalized Problem) is available in the users' guide [LSY98]. To use any of the modes listed below, the user is strongly urged to modify one of the driver routine **templates** that reside in the **EXAMPLES** directory.

When to Use a Spectral Transformation:

The first thing to decide is whether the problem will require a spectral transformation. If the problem is generalized ($M \neq I$), then a spectral transformation will be required. Such a transformation will most likely be needed for a standard problem if the desired eigenvalues are in the interior of the spectrum or if they are clustered at the desired part of the spectrum. Once this decision has been made and **OP** has been specified, an efficient means to implement the action of the operator **OP** on a vector must be devised. The expense of applying **OP** to a vector will of course have direct impact on performance.

Shift-invert spectral transformations may be implemented with or without the use of a weighted M -inner product. The relation between the eigenvalues of **OP** and the eigenvalues of the original problem must also be understood in order to make the appropriate specification of **which** in order to recover eigenvalues of interest for the original problem. The user must specify the number of eigenvalues to compute, which eigenvalues are of interest, the number of basis vectors to use, and whether or not the problem is standard or generalized.

Computational Modes for Real Nonsymmetric Problems:

The following subroutines are used to solve nonsymmetric generalized eigenvalue problems in real arithmetic. These routines are appropriate when A is a general nonsymmetric matrix and M is symmetric and positive semidefinite. The reverse communication interface routine for the nonsymmetric double precision eigenvalue problem is **dnaupd**. The routine is called as shown below. The specification of which **nev** eigenvalues is controlled by the **character*2** argument **which**. The most commonly used options are listed below. There are templates available as indicated for each of these.

```
call dnaupd (ido, bmat, n, which, nev, tol, resid, ncv, v,
&           ldv, iparam, ipntr, workd, worlk, lworkl, info)
```

There are three different shift-invert modes for nonsymmetric eigenvalue problems. These modes are specified by setting the parameter entry **iparam(7) = mode** where **mode** = 1, 2, 3, or 4.

In the following list, the specification of **OP** and B are given for the various modes. Also, the **iparam(7)** and **bmat** settings are listed along with the name of the sample driver for the given mode. Sample drivers

for the following modes may be found in the `EXAMPLES/NONSYM` subdirectory.

1. Regular mode (`iparam(7) = 1, bmat = 'I'`).
Use driver `dndrv1`.
 - (a) Solve $A\mathbf{x} = \mathbf{x}\lambda$ in regular mode.
 - (b) $OP = A$ and $B = I$.
2. Shift-invert mode (`iparam(7) = 3, bmat = 'I'`).
Use driver `dndrv2` with `sigma` a real shift.
 - (a) Solve $A\mathbf{x} = \mathbf{x}\lambda$ in shift-invert mode.
 - (b) $OP = (A - \sigma I)^{-1}$ and $B = I$.
3. Regular inverse mode (`iparam(7) = 2, bmat = 'G'`).
Use driver `dndrv3`.
 - (a) Solve $A\mathbf{x} = M\mathbf{x}\lambda$ in regular inverse mode.
 - (b) $OP = M^{-1}A$ and $B = M$.
4. Shift-invert mode (`iparam(7) = 3, bmat = 'G'`).
Use driver `dndrv4` with `sigma` a real shift.
 - (a) Solve $A\mathbf{x} = M\mathbf{x}\lambda$ in shift-invert mode.
 - (b) $OP = (A - \sigma M)^{-1}M$ and $B = M$.

76.9 MATLAB'S® EIGS

MATLAB has adopted ARPACK for the computation of a selected subset of the eigenvalues of a large (sparse) matrix. The MATLAB function for this is called `eigs` and it is a MATLAB driver to a mex-file compilation of ARPACK. In fact, a user can directly reference this mex-file as `arpackc`. Exactly the same sort of interfaces discussed above can be written in MATLAB to drive `arpackc`.

However, it is far more convenient just to use the provided `eigs` function. The command `D = eigs(A)` returns a vector of the 6 eigenvalues of A of largest magnitude, while `[V,D] = eigs(A)` returns eigenvectors in V and the corresponding (6 largest magnitude) eigenvalues on the diagonal of the matrix D .

The command `eigs(A,M)` solves the generalized eigenvalue problem $A^*V = M^*V^*D$. Here M must be symmetric (or Hermitian) positive definite and the same size as A . If M is nonsymmetric, this can also be handled directly, but the user will have to write and pass a function that will compute the action `w <- inv(A - sigma*M) * M*v` as needed. Of course, a sparse direct factorization of $(A - \sigma M)$ should be computed at the outset and reused to accomplish this action.

Other capabilities are easily utilized through `eigs`. The commands `eigs(A,k)` and `eigs(A,M,k)` return the k largest magnitude eigenvalues.

The commands `eigs(A,k,sigma)` and `eigs(A,M,k,sigma)` return k eigenvalues based on `sigma`, where `sigma` is either a scalar or one of the following strings: 'LM' or 'SM' - Largest or Smallest Magnitude

For real symmetric problems, `SIGMA` may also be:
 'LA' or 'SA' - Largest or Smallest Algebraic
 'BE' - Both Ends, one more from high end if K
 is odd

For nonsymmetric and complex problems, `SIGMA`
 may also be:
 'LR' or 'SR' - Largest or Smallest Real part
 'LI' or 'SI' - Largest or Smallest Imaginary
 part

If σ is a scalar including 0, `eigs` finds the eigenvalues in the complex plane closest to σ . The required sparse matrix factorizations are done automatically when a scalar is passed.

More control and flexibility can be obtained by using the commands `eigs(A,M,sigma,opts)` and `eigs(A,M,k,sigma,opts)` where `opts` is a struct. The fields of the struct `opts` will specify the following options:

```
opts.issym: symmetry of A or A-SIGMA*B
            represented by AFUN [{0} | 1]

opts.isreal: complexity of A or A-SIGMA*B
            represented by AFUN [0 | {1}]

opts.tol: convergence: Ritz estimate
          residual <= tol*NORM(A) [scalar | {eps}]

opts.maxit: maximum number of
            iterations [integer | {300}]

opts.p: number of Lanczos vectors:
        K+1<p<=N [integer | {2K}]

opts.v0: starting vector
        [N-by-1 vector | {randomly generated by ARPACK}]

opts.disp: diagnostic information display
          level [0 | {1} | 2]

opts.cholB: B is actually its Cholesky
            factor CHOL(B) [{0} | 1]

opts.permB: sparse B is actually
            CHOL(B(permB,permB)) [permB | {1:N}]

opts(AFUN,n) accepts the function AFUN
            instead of a matrix
```

In the above list, the items in square brackets denote the possible settings and the item given in curly brackets `{·}` is the default setting. For example, to set the convergence tolerance to 10^{-3} , one should give the command `opts.tol = .001`.

The function passing mechanism can be used to define a customized spectral transformation as in the example with a nonsymmetric M . Arguments may be passed to `AFUN(X,P1,P2,...)` by invoking the command `eigs (AFUN,n,k,sigma,opts,P1,P2,...)`.

Thus, all of the functionality of ARPACK is available through `eigs` and, in many cases, it is much easier to use. Moreover, the sparse matrix direct factorizations and reorderings available in the `sparfun` suite are easily used to implement various desirable spectral transformations. It is highly advisable to run sample problems with the characteristics of a desired very large problem in order to get an idea of the best spectral transformation to use and to get an indication of the expected convergence behavior.

References

- [LSY98] R. Lehoucq, D.C. Sorensen, and C. Yang, *ARPACK Users Guide: Solution of Large Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods*, SIAM Publications, Philadelphia, (1998). (Software available at: <http://www.caam.rice.edu/software/ARPACK>).
- [Sor92] D.C. Sorensen, Implicit application of polynomial filters in a k-step Arnoldi method, *SIAM J. Matrix Anal. Applic.*, 13: 357–385 (1992).
- [Sor02] D.C. Sorensen, Numerical methods for large eigenvalue problems, *Acta Numerica*, **11**, 519–584, (2002).

77

Summary of Software for Linear Algebra Freely Available on the Web

Jack Dongarra

University of Tennessee and Oakridge National Laboratory

Victor Eijkhout

University of Tennessee

Julien Langou

University of Tennessee

Table 77.1 to Table 77.5 present a list of freely available software for the solution of linear algebra problems. The interest is in software for high-performance computers that is available in “open source” form on the Web for solving problems in numerical linear algebra, specifically dense, sparse direct and iterative systems, and sparse iterative eigenvalue problems.

Additional pointers to software can be found at:

www.nhse.org/rib/repositories/nhse/catalog/#Numerical_Programs_and_Routines.

A survey of Iterative Linear System Solver Packages can be found at:

www.netlib.org/utk/papers/iterative-survey.

TABLE 77.1 Support Routines for Numerical Linear Algebra

Package	Support	Type		Language			Mode		Dense	Sparse
		Real	Complex	f77	c	c++	Seq	Dist		
ATLAS	yes	X	X	X	X		X		X	
BLAS	yes	X	X	X	X		X		X	
FLAME	yes	X	X	X	X		X		X	
LINALG	?									
MTL	yes	X				X	X		X	
NETMAT	yes	X				X	X		X	
NIST S-BLAS	yes	X	X	X	X		X			X
PSBLAS	yes	X	X	X	X		X	M		X
SparseLib++	yes	X	X		X	X	X			X
uBLAS	yes	X	X		X	X	X		X	

TABLE 77.2 Available Software for Dense Matrix

Package	Support	Type		Language			Mode	
		Real	Complex	f77	c	c++	Seq	Dist
LAPACK	yes	X	X	X	X		X	
LAPACK95	yes	X	X	95			X	
NAPACK	yes	X		X				
PLAPACK	yes	X	X	X	X			M
PRISM	yes	X		X			X	M
ScaLAPACK	yes	X	X	X	X			M/P

TABLE 77.3 Sparse Direct Solvers

Package	Support	Type		Language			Mode		SPD	Gen
		Real	Complex	f77	c	c++	Seq	Dist		
DSCPACK	yes	X			X		X	M	X	
HSL	yes	X	X	X			X		X	X
MFACT	yes	X			X		X		X	
MUMPS	yes	X	X	X	X		X	M	X	X
PSPASES	yes	X		X	X			M	X	
SPARSE	?	X	X		X		X		X	X
SPOOLES	?	X	X		X		X	M		X
SuperLU	yes	X	X	X	X		X	M		X
TAUCS	yes	X	X		X		X		X	X
UMFPACK	yes	X	X		X		X			X
Y12M	?	X		X			X		X	X

TABLE 77.4 Sparse Eigenvalue Solvers

Package	Support	Type		Language			Mode		Sym	Gen
		Real	Complex	f77	c	c++	Seq	Dist		
(B/H)LZPACK	yes	X	X	X			X	M/P	X	X
HYPRE	yes	X			X		X	M	X	
QMRPACK	?	X	X	X			X		X	X
LASO	?	X		X			X		X	
P_ARPACK	yes	X	X	X			X	M/P		X
PLANSO	yes	X		X			X	M	X	
SLEPc	yes	X	X		X		X	M	X	X
SPAM	yes	X		90			X		X	
TRLAN	yes	X		X			X	M	X	

TABLE 77.5 Sparse Iterative Solvers

Package	Support	Type		Language			Mode		Precond.		Iterative Solvers	
		Real	Comp.	f77	c	c++	Seq	Dist	SPD	Gen	SPD	Gen
AZTEC	no	X			X		X	M	X	X	X	X
BILUM	yes	X		X			X	M		X		X
BlockSolve95	?	X		X	X	X		M	X	X	X	X
BPKIT	yes	X		X	X	X	X	M	X	X		
CERFACS	yes	X	X	X			X	M			X	X
HYPRE	yes	X		X	X		X	M	X	X	X	X
IML++	?	X		X	X	X	X				X	X
ITL	yes	X				X	X				X	X
ITPACK	?	X		X			X				X	X
LASPack	yes	X			X		X				X	X
LSQR	yes	X		X	X		X					X
pARMS	yes	X		X	X		X	M			X	X
PARPRE	yes	X			X			M	X	X		
PETSc	yes	X	X	X	X		X	M	X	X	X	X
P-SparsLIB	yes	X		X				M		X		X
PSBLAS	yes	X	X	f90	X		X	M	X	X	X	X
QMRPACK	?	X	X	X			X				X	X
SLAP	?	X		X						X		
SPAI	yes	X			X		X	M			X	X
SPLIB	?	X		X			X				X	X
SPOOLES	?	X	X		X		X	M	X	X	X	X
SYMMLQ	yes	X	X	X			X				X	X
TAUCS	yes	X	X		X		X		X	X	X	X
Templates	yes	X		X	X		X				X	X
Trilinos	yes	X				X	X	M	X	X	X	X

Glossary

This glossary covers most of the terms defined in Chapters 1 to 49. It does not cover some terminology used in a single chapter (including most of the terminology that is specific to a particular application (Chapters 50 to 70)), nor does it cover most of the terminology used in computer software (Chapters 71 to 77). When two sections are listed, both define the term; the first listed chapter is the first instance, the second is the primary chapter dealing with the topic. Definitions in this glossary may not give all details; the reader is advised to consult the chapter/section following the term and definition).

A

abelian (group G): A commutative group, i.e., $ab = ba$ for all $a, b \in G$. **Preliminaries**

absolute (matrix norm): As a vector norm, each member of the family is absolute. **37.3**

absolute (vector norm $\|\cdot\|$): For all vectors $\mathbf{x}$, $\| |\mathbf{x}| \| = \|\mathbf{x}\|$. **37.1**

absolute algebraic connectivity (of simple graph $G = (V, E)$): $\max \alpha(G_w)$ where the maximum is taken over all nonnegative weights w of the edges of G such that $\sum_{e \in E} w(e) = |E|$. **36.5**

absolute error (in approximation $\hat{\mathbf{z}}$ to $\mathbf{z}$): $\|\mathbf{z} - \hat{\mathbf{z}}\|$ **37.4**

absolute value (of complex number $a + bi$): $\sqrt{a^2 + b^2}$. **Preliminaries**

absolute value (of real matrix A): The nonnegative matrix obtained by taking element-wise absolute values of A 's entries. **9.1**

access: Vertex u has access to vertex v in a digraph Γ if there is a walk in Γ from u to v ; also applied to sets of vertices. **9.1, 29.5**

access equivalence class (of a digraph): An equivalence class under the equivalence relation of access. **9.1, 29.5**

access equivalence class (of a matrix): An access equivalence class of its digraph. **9.1**

access equivalent: Two vertices in a digraph that have access to each other. **9.1, 29.5**

active branch (at a Type I characteristic vertex of a tree): For some Fiedler vector $\mathbf{y}$ the entries in $\mathbf{y}$ corresponding to the vertices in the branch are nonzero. **36.3**

acyclic (square matrix A): The graph of A has no cycles. **19.3**

additive (map $\phi : F^{m \times n} \rightarrow F^{m \times n}$): $\phi(A + B) = \phi(A) + \phi(B)$, for all $A, B \in F^{m \times n}$. **22.4**

additive coset (of subspace W): A subset of vectors the form $\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} : \mathbf{w} \in W\}$. **2.3**

additive D -stable (real square matrix A): $A + D$ is stable for every nonnegative diagonal matrix D . **19.4**

additive inverse eigenvalue problem (AIEP): Given $A \in \mathbb{C}^{n \times n}$ and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ find a diagonal matrix $D \in \mathbb{C}^{n \times n}$ such that $\sigma(A + D) = \{\lambda_1, \dots, \lambda_n\}$. **20.9**

additive preserver: An additive map preserving a certain property. **22.4**

adjacency matrix (of a digraph Γ of order n): The $n \times n$ 0, 1-matrix whose i, j th entry is 1 if there is an arc from the i th vertex to the j th vertex and 0 otherwise. **29.2**

adjacency matrix (of a graph G of order n): The symmetric $n \times n$ matrix whose i, j th entry is equal to the number of edges between the i th vertex and the j th vertex. **28.3**

adjacent (matrices A, B): $\text{rank}(A - B) = 1$. **22.4**

adjacent (vertices u and v in a graph): There exists an edge with endpoints u and v . **28.1**

adjoint (of a matrix): See *Hermitian adjoint*.

adjoint (of a linear operator T on an inner product space V): $\langle T(\mathbf{u}), \mathbf{v} \rangle = \langle \mathbf{u}, T^*(\mathbf{v}) \rangle$ for all $\mathbf{u}, \mathbf{v} \in V$. **5.3**

adjugate: The transpose of the matrix of cofactors. **4.2**

affine parameterized inverse eigenvalue problem: See *Section 20.9*.

AIEP: See *additive inverse eigenvalue problem*.

algebra (associative): A vector space A over a field F with a bilinear **multiplication** $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{xy}$ from $A \times A$ to A satisfying $(\mathbf{xy})\mathbf{z} = \mathbf{x}(\mathbf{yz})$. **Preliminaries, 69.1**

algebra (nonassociative): See *Section 69.1*.

algebraic connectivity (of simple a graph): The second least Laplacian eigenvalue. **28.4, 36.1**

algebraic multigrid preconditioner: A preconditioner that uses principles similar to those used for PDE problems on grids, when the “grid” for the problem is unknown or nonexistent and only the matrix is available. **41.4**

algebraic multiplicity (of an eigenvalue): The number of times the eigenvalue occurs as a root in the characteristic polynomial of the matrix. **4.3**

algorithm: A precise set of instructions to perform a task. **37.7**

allows: If P is a property referring to a real matrix, then a sign pattern A allows P if some real matrix in $Q(A)$ has property P . **33.1**

allows a properly signed nest ($n \times n$ sign pattern A): There exists $B \in Q(A)$ and a permutation matrix P such that $\text{sgn}(\det(P^T B P [[1, \dots, k]])) = (-1)^k$ for $k = 1, \dots, n$. **33.4**

Alt: the map $\text{Alt}(\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_k) = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) \mathbf{v}_{\pi(1)} \otimes \dots \otimes \mathbf{v}_{\pi(k)}$. **13.6**

alt multiplication: $(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_p) \wedge (\mathbf{v}_{p+1} \wedge \dots \wedge \mathbf{v}_{p+q}) = \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_{p+q}$. **13.7**

alternate path to a single arc (in a digraph): A path of length greater than 1 between vertices i and j such that the arc (i, j) is in the digraph. **35.7**

alternating (bilinear form f): $f(\mathbf{v}, \mathbf{v}) = 0$ for all $\mathbf{v} \in V$. **12.3**

alternating ($n \times n$ matrix A): $a_{ii} = 0$, $i = 1, 2, \dots, n$ and $a_{ji} = -a_{ij}$, $1 \leq i, j \leq n$. **12.3**

alternating (multilinear form): See *antisymmetric*.

alternating algebra: See *Grassman algebra*.

alternating product: See *exterior product*.

alternating space: See *Grassman space*.

alternator: See *Alt*.

angle (between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in a real inner product space): The real number θ , $0 \leq \theta \leq \pi$, such that $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$. **5.1**

annihilator: The set of linear functionals that annihilate every vector in the given set. **3.8**

antidiagonal (of $n \times n$ matrix A): The elements $a_{i,k-i}$, $i = 1, \dots, k-1$ with $2 \leq k \leq n+1$. **48.1**

anti-identity matrix: The $n \times n$ matrix with ones along the main antidiagonal, i.e., $a_{i,n+1-i} = 1$, $i = 1, \dots, n$ and zeros elsewhere. **48.1**

antisymmetric (bilinear form f): $f(\mathbf{u}, \mathbf{v}) = -f(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$. **12.3**

antisymmetric (multilinear form): A multilinear form that is an antisymmetric map. **13.4**

antisymmetric (map $\psi \in L^m(V; U)$): $\psi(\mathbf{v}_{\pi(1)}, \dots, \mathbf{v}_{\pi(m)}) = \text{sgn}(\pi) \psi(\mathbf{v}_1, \dots, \mathbf{v}_m)$ for all permutations π . **13.4**

aperiodic (matrix): An irreducible nonnegative matrix of period 1. **9.2**

appending G_2 at vertex v of G_1 : Constructing a simple graph from $G_1 \cup G_2$ by adding an edge between v and a vertex of G_2 . **36.2**

Arbitrary Precision Approximating (APA) algorithms: See *Section 47.4*.

arc: An ordered pair of vertices (part of a directed graph). **29.1**

argument (of a complex number): θ in the form $re^{i\theta}$ with $0 \leq \theta < 2\pi$. **Preliminaries**

Arnoldi factorization (k -step): $AV_k = V_k H_k + \mathbf{f}_k \mathbf{e}_k^*$ where $V_k \in \mathbb{C}^{n \times k}$ has orthonormal columns, $V_k^* \mathbf{f}_k = 0$ and $H_k = V_k^* A V_k$ is a $k \times k$ upper Hessenberg matrix with nonnegative subdiagonal elements. **41.3, 44.2**

associated undirected graph (of a digraph $\Gamma = (V, E)$): The undirected graph with vertex set V , having an edge between vertices u and v if and only if at least one of the arcs (u, v) and (v, u) is in Γ . **29.1**

associates (in a domain): a, b are associates if $a|b$ and $b|a$. **23.1**

association scheme: A set of graphs $G_0, \dots, G_d$ on a common vertex set satisfying certain axioms. **28.6**

associative algebra: See *algebra (associative)*.

asymmetric (digraph $\Gamma = (V, E)$): $(i, j) \in E$ implies $(j, i) \notin E$ for all distinct $i, j \in V$. **35.1**

augmented matrix (of a system of linear equations): Matrix obtained by adjoining the constant column to the coefficient matrix. **1.4**

B

backward error (for $\hat{f}(\mathbf{x})$): A vector $\mathbf{e} \in \mathbb{R}^n$ of smallest norm for which $f(\mathbf{x} + \mathbf{e}) = \hat{f}(\mathbf{x})$. **37.8**

backward stable (algorithm): The backward relative error $\mathbf{e}$ exists and is small for all valid input data $\mathbf{x}$ despite rounding and truncation errors in the algorithm. **37.8**

badly conditioned: See *ill-conditioned*.

balanced $((0, 1)$ -design matrix W): Every row of $m \times n$ matrix W has exactly $(n + 1)/2$ ones if n odd; exactly $n/2$ ones or exactly $(n + 2)/2$ ones if n even. **32.4**

balanced (real or sign pattern vector): It is a zero vector or it has both a positive entry and a negative entry. **33.3**

balanced column signing (of matrix A): A column signing of A in which all the rows are balanced. **33.3**

balanced row signing (of matrix A): A row signing of A in which all the columns are balanced. **33.3**

banded (family of Toeplitz matrices with symbol a): There exists some $m \geq 0$ such that $a_{\pm k} = 0$ for all $k \geq m$. **16.2**

barely L -matrix: An L -matrix that is not an L -matrix if any column is deleted. **33.3**

bases: Plural of *basis*.

basic class (of square nonnegative matrix P): An access equivalence class B of P with $\rho(P[B]) = \rho$. **9.3**

basic reduced digraph (of square nonnegative matrix P): The digraph whose vertex-set is the set of basic classes of P and whose arcs are the pairs (B, B') of distinct basic classes of P for which there exists a simple walk from B to B' in the reduced digraph of P . **9.3**

basic solution (to a least squares problem): A least squares solution with at least $n - \text{rank } A$ zero components. **39.1**

basic variable: A variable in a system of linear equations whose value is determined by the values of the free variables. **1.4**

basis: A set of vectors that is linearly independent and spans the vector space. **2.2**

BD: See *Bezout domain*.

Bezout domain (BD): A GCCD $\mathbb{D}$ such that for any two elements $a, b \in \mathbb{D}$, $(a, b) = pa + qb$, for some $p, q \in \mathbb{D}$. **23.1**

biacyclic (matrix): A matrix whose sparsity pattern has an acyclic bipartite graph. **46.3**

biadjacency matrix (of bipartite graph G): A matrix with rows indexed by one of the bipartition sets and columns indexed by the other, with the value of the entry being the number of edges between the vertices (or weight of the edge between the vertices). **30.1**

BiCG algorithm: Iterative method for solving a linear system using non-Hermitian Lanczos algorithm; the approximate solution $\mathbf{x}_k$ is chosen so that the residual $\mathbf{r}_k$ is orthogonal to $\text{Span}\{\mathbf{f}_0, A^*\mathbf{f}_0, \dots, A^{*k-1}\mathbf{f}_0\}$. **41.3**

BiCGSTAB algorithm: Iterative method for solving a linear system using non-Hermitian Lanczos algorithm with improved stability. See *Section 41.3*.

biclique (of a graph): A subgraph that is a complete bipartite graph. **30.3**

biclique cover (of graph $G = (V, E)$): A collection of bicliques of G such that each edge of E is in at least one biclique. **30.3**

biclique cover number (of graph G): The smallest k such that there is a cover of G by k bicliques. **30.3**

biclique partition (of $G = (V, E)$): A collection of bicliques of G whose edges partition E . **30.3**

biclique partition number (of graph G): The smallest k such that there is a partition of G into k bicliques. **30.3**

bidual space: The dual of the dual space. **3.8**

big-oh: function f is $O(g)$ if beyond a certain point $|f|$ is bounded by a multiple of $|g|$. **Preliminaries**

bigraph (of the $m \times n$ matrix A): The simple graph with vertex set $U \cup V$ where $U = \{1, 2, \dots, m\}$ and $V = \{1', 2', \dots, n'\}$, and edge set $\{\{i, j'\} : a_{ij} \neq 0\}$. **30.2**

bilinear form (on vector space V over field F): A map f from $V \times V$ into F that satisfies $f(a\mathbf{u}_1 + b\mathbf{u}_2, \mathbf{v}) = af(\mathbf{u}_1, \mathbf{v}) + bf(\mathbf{u}_2, \mathbf{v})$ and $f(\mathbf{u}, a\mathbf{v}_1 + b\mathbf{v}_2) = af(\mathbf{u}, \mathbf{v}_1) + bf(\mathbf{u}, \mathbf{v}_2)$. **12.1**

bilinear noncommutative algorithms: See *Section 47.2*.

binary matrix: A $(0, 1)$ -matrix, i.e., a matrix in which each entry is either 0 or 1. **31.3**

biorthogonal (sets of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ in an inner product space): $\langle \mathbf{v}_i, \mathbf{w}_j \rangle = 0$ if $i \neq j$. **41.3**

bipartite (graph G): The vertices of G can be partitioned into disjoint subsets U and V such that each edge of G has the form $\{u, v\}$ where $u \in U$ and $v \in V$. **28.1, 30.1**

bipartite fill-graph of $n \times n$ matrix A with each diagonal entry nonzero: The simple bipartite graph with vertex set $\{1, 2, \dots, n\} \cup \{1', 2', \dots, n'\}$ with an edge joining i and j' if and only if there exists a path from i to j in the directed graph, $\Gamma(A)$, of A each of whose intermediate vertices has label less than $\min\{i, j\}$. **30.2**

bipartite graph (of sparsity pattern S): The graph with vertices partitioned into row vertices $r_1, \dots, r_m$ and column vertices $c_1, \dots, c_n$, where r_k and r_l are connected if and only if $(k, l) \in S$. **46.3**

bipartite sign pattern: A sign pattern whose digraph is bipartite. **33.5**

bipartition: The sets into which the vertices of a bipartite graph are partitioned. **30.1**

block (in a 2-design): A subset B_i of X (see 2-design). **32.2**

block (of a graph or digraph): A maximal nonseparable sub(di)graph. **35.1**

block (of a matrix): An entry of a block matrix, which is a submatrix of the original matrix. **10.1**

block-clique (graph or digraph): Every block is a clique. **35.1**

block diagonal (for a particular block matrix structure): A square block matrix A with all off-diagonal blocks 0. **10.2**

block matrix: A matrix that is partitioned into submatrices with the row indices and column indices partitioned into consecutive subsets sequentially. **10.1**

block lower triangular (matrix A): A^T is block upper triangular. **10.2**

block-Toeplitz matrix: A block matrix $A = [A_{ij}]$ such that $A_{i+k,i} = A^{(k)}$, **48.1**

block-Toeplitz-Toeplitz-block (BTTB) matrices: a block-Toeplitz matrix in which the blocks $A^{(k)}$ are themselves Toeplitz matrices. **48.1**

block upper triangular (for a particular block matrix structure): A block matrix having every subdiagonal block 0. **10.2**

(0,1)-Boolean algebra: $\{0, 1\}$, with $0 + 0 = 0$, $0 + 1 = 1 = 1 + 0$, $1 + 1 = 1$, $0 * 1 = 0 = 1 * 0$, $0 * 0 = 0$, and $1 * 1 = 1$. **30.3**

Boolean matrix: A matrix whose entries belong to the Boolean algebra. **30.3**

Boolean rank (of $m \times n$ Boolean matrix A): The minimum k such that there exists an $m \times k$ Boolean matrix B and a $k \times n$ Boolean matrix C such that $A = BC$. **30.3**

Bose-Mesner algebra: The algebra generated by the adjacency matrices of the graphs of an association scheme. **28.6**

bottleneck matrix (of a branch of tree T at vertex v): The inverse of the principal submatrix of the Laplacian matrix corresponding to the vertices of that branch. **36.3**

boundary (of a subset S of $\mathbb{R}$ or $\mathbb{C}$): The intersection of the closure of S and the closure of the complement of S . **Preliminaries**

branch (of tree T at vertex v): A component of $T - v$. **34.1**

Bunch-Parlett factorization (of symmetric real matrix H): The factorization $P^T H P = L B L^T$, where P is a permutation matrix, B is a block-diagonal matrix with diagonal blocks of sizes 1×1 or 2×2 , and L is a full column rank unit lower triangular matrix, where the diagonal blocks in L which correspond to 2×2 blocks in B are 2×2 identity matrices. **15.5**

Businger-Golub pivoting: a particular pivoting strategy for QR-factorization. **46.2**

C

canonical angles between the column spaces of $X, Y \in \mathbb{C}^{n \times k}$: $\theta_i = \arccos \sigma_i$, where $\{\sigma_i\}_{i=1}^k$ are the singular values of $(Y^* Y)^{-1/2} Y^* X (X^* X)^{-1/2}$ **15.1** (equivalent to principal angles **17.1**)

canonical angle matrix: $\text{diag}(\theta_1, \theta_2, \dots, \theta_k)$, where $\theta_1, \theta_2, \dots, \theta_k$ are the canonical angles. **15.1**

canonical correlations (between subspaces X and Y of $\mathbb{C}^r$): Cosines of principal (canonical) angles **17.7**

Cauchy matrix: Given vectors $\mathbf{x} = [x_1, \dots, x_n]^T$ and $\mathbf{y} = [y_1, \dots, y_n]^T$, the Cauchy matrix $C(\mathbf{x}, \mathbf{y})$ has i, j -entry equal to $\frac{1}{x_i + y_j}$. **48.1**

center See *central vertex*.

central (real matrix B): The zero vector is in the convex hull of the columns of B . 33.11

central vertex (of a generalized star): a vertex such that each of its neighbours are pendant vertices of their branches, and each branch is a path. 34.1

CG: See *conjugate gradient algorithm*.

CGS algorithm: Iterative method for solving a linear system using non-Hermitian Lanczos algorithm. See Section 41.3.

change-of-basis matrix (from $\mathcal{B}$ to $\mathcal{C}$): The matrix consisting of the coordinate vectors with respect to basis $\mathcal{C}$ of the basis vectors in $\mathcal{B}$. 2.6

characteristic equation (of the pencil $A - xB$): $\det(xB - A) = 0$. 43.1

characteristic polynomial (of matrix A): $\det(xI - A)$. 4.3

characteristic polynomial (of the pencil $A - xB$): $\det(xB - A)$. 43.1

characteristic polynomial (of a graph): The characteristic polynomial of its adjacency matrix. 28.3

characteristic vector (of a subset of m vertices of a graph): The m -vector whose i th entry is 1 if $i \in X$, and 0 otherwise. 30.3

characteristic vertex (of tree T): See Section 36.3.

Cholesky decomposition (of positive-definite matrix A): $A = G G^*$ with $G \in \mathbb{C}^{n \times n}$ lower triangular and having positive diagonal entries. 38.5

Cholesky factorization: See *Cholesky decomposition*.

chord (of a cycle in a graph): An edge joining two nonconsecutive vertices on the cycle. 30.1

chordal bipartite graph: Every cycle of length 6 or more has a chord. 30.1

chordal graph: Every cycle of length 4 or more has a chord. 30.1

1-chordal: See *block-clique*.

chromatic index (of graph G): The smallest number of classes in a partition of edges of G so that no two edges in the same class meet. 27.6

chromatic number (of graph G): The smallest number of color classes of any vertex coloring of G (not defined if G has loops). 28.5

circulant matrix: A Toeplitz matrix in which every row is obtained by a single cyclic shift of the previous row. 20.3, 48.1

clique (of a digraph): Every vertex has a loop and for any two distinct vertices u, v , both arcs $(u, v), (v, u)$ are present. 35.1

clique (of a graph that allows loops but not multiple edges): Every vertex has a loop and for any two distinct vertices u, v , the edge $\{u, v\}$ is present. 35.1

clique (of a simple graph): A subgraph isomorphic to a complete graph. 28.5

clique number (of graph G): The largest order of a clique in G . 28.5

closed cone: A cone that is a closed subset of the vector space. 8.5, 26.1

closed under matrix direct sums (class of matrices X): Whenever $A_1, A_2, \dots, A_k$ are X -matrices, then $A_1 \oplus A_2 \oplus \dots \oplus A_k$ is an X -matrix. 35.1

closed under permutation similarity (class of matrices X): Whenever A is an X -matrix and P is a permutation matrix, then $P^T A P$ is an X -matrix. 35.1

closed under taking principal submatrices: See *hereditary*.

closed walk (in a digraph): A walk in which the first vertex equals the last vertex. 54.2

4-cockade: A bipartite graph created by adding 4 cycles through edge identification to a 4-cycle. 30.2

cocktail party graph: The graph obtained by deleting a disjoint edges from the complete graph K_{2a} . 28.2

coclique (of a graph): An induced subgraph with no edges. 28.5

co-cover (of a $(0, 1)$ -matrix): A collection of 1s such that each line of A contains at least one of the 1s. 27.1

co-index (of square nonnegative matrix P): $\max\{v_P(\lambda) : \lambda \in \sigma(P), |\lambda| = \rho \text{ and } \lambda \neq \rho\}$. 9.3

codomain (of $T: V \rightarrow W$): The vector space W . 3.1

coefficient matrix: The matrix of coefficients of a system of linear equations. 1.4

coefficients (of a linear equation): The scalars that occur multiplied by variables. 1.4

i, j -cofactor: $(-1)^{i+j}$ times the i, j -minor. 4.1

Colin de Verdière parameter: A parameter of a simple graph. **28.5**

color class: A coclique in a vertex coloring partition of the vertices of a graph. **28.5**

column: The entries of a matrix lying in a vertical line in the matrix. **1.2**

column equivalent (matrices $A, B \in \mathbb{D}^{n \times n}$): $B = AP$ for some $\mathbb{D}$ -invertible P . **23.2**

column signing (of (real or sign pattern) matrix A): AD where D is a signing. **33.3**

column space: See *range*.

column-stochastic (matrix): A square nonnegative matrix having all column sums equal to 1. **9.4**

column sum vector (of a matrix): The vector of column sums. **27.4**

combinatorially orthogonal (sign pattern vectors $\mathbf{x}, \mathbf{y}$): $|\{i : x_i y_i \neq 0\}| \neq 1$. **33.10**

combinatorially symmetric (partial matrix B): b_{ij} specified implies b_{ji} specified. **35.1**

combinatorially symmetric sign pattern A : $a_{ij} \neq 0$ if and only if $a_{ji} \neq 0$. **33.5**

communicate: See *access equivalent*.

commute: Matrices or linear operators A, B such that $AB = BA$. **1.2, 3.2**

companion matrix (of a monic polynomial): Square matrix with ones on the subdiagonal and last column consisting of negatives of the polynomial coefficients **4.3, 6.4**

comparison matrix (of real square matrix A): Matrix having i, j -entry $-|a_{ij}|$ for $i \neq j$ and i, i -entry $|a_{ii}|$. **19.5**

compatible (generalized sign patterns): The intersection of their qualitative classes is nonempty. **33.1**

complement (of set X in universe S): The elements of S not in X . **Preliminaries**

complement (of binary matrix M): $J - M$ (where J is the all 1s matrix). **31.3**

complement (of simple graph $G = (V, E)$): The simple graph having vertex set V and as edges all unordered pairs from V that are not in E . **28.1**

complement (orthogonal): See *orthogonal complement*.

complete (bipartite graph): A simple bipartite graph with bipartition $\{U, V\}$ such that each $\{u, v\}$ is an edge (for all $u \in U, v \in V$). **28.1, 30.1**

complete (simple graph): The edge set consists of all unordered pairs of distinct vertices. **28.1**

complete orthonormal set: An orthonormal set of vectors whose orthogonal complement is 0. **5.2**

completed max-plus semiring: The set $\mathbb{R} \cup \{\pm\infty\}$ equipped with the addition $(a, b) \mapsto \max(a, b)$ and the multiplication $(a, b) \mapsto a + b$, with the convention that $-\infty + (+\infty) = +\infty + (-\infty) = -\infty$. **25.1**

completely positive (matrix A): $A = CC^T$ for some nonnegative $n \times m$ matrix C . **35.4**

completely reducible (matrix A): There is a permutation matrix P such that $PAP^T = A_1 \oplus A_2 \oplus \cdots \oplus A_k$ where $A_1, A_2, \dots, A_k$ are irreducible and $k \geq 2$. **27.3**

completion (of a partial matrix): A choice of values for the unspecified entries. **35.1**

X -completion property (where X is a type of matrix): Digraph (respectively, graph) G has this property if every partial X -matrix B such that $\mathcal{D}(B) = G$ (respectively, $\mathcal{G}(B) = G$) can be completed to an X -matrix. **35.1**

complex conjugate (of $a + bi \in \mathbb{C}$): $a - bi$. **Preliminaries**

complex sign pattern matrix: A matrix of the form $A = A_1 + iA_2$ for sign patterns A_1 and A_2 . **33.8**

complex vector space: A vector space over the field of complex numbers. **1.1**

component-wise relative backward error of the linear system: See *Section 38.1*.

component-wise relative distance (between matrices H and $\tilde{H}$): $\text{reldist}(H, \tilde{H}) = \max_{i,j} \frac{|h_{ij} - \tilde{h}_{ij}|}{|h_{ij}|}$, where $0/0 = 0$. **15.4**

composite cycle (in sign pattern A): A product of disjoint simple cycles. **33.1**

$k \times k$ -compound matrix: A matrix formed from the $k \times k$ minors of a given matrix. **4.2**

condensation digraph: See *reduced digraph*.

condition number (of $\mathbf{z}$ with respect to problem P):

$$\text{cond}_P(\mathbf{z}) = \lim_{\varepsilon \rightarrow 0} \sup \left\{ \left(\frac{\|P(\mathbf{z} + \delta\mathbf{z}) - P(\mathbf{z})\|}{\|P(\mathbf{z})\|} \right) \left(\frac{\|\mathbf{z}\|}{\|\delta\mathbf{z}\|} \right) \mid \|\delta\mathbf{z}\| \leq \varepsilon \right\}. \quad \mathbf{37.4}$$

condition number (of an eigenvalue): See *individual condition number*.

condition number (of matrix A for linear systems): $\kappa_v(A) = \|A^{-1}\|_v \|A\|_v$. **37.5, 38.1**

condition number of the linear system: $A\hat{\mathbf{x}} = \mathbf{b}$: $\kappa(A, \hat{\mathbf{x}}) = \|A^{-1}\| \frac{\|\mathbf{b}\|}{\|\hat{\mathbf{x}}\|}$. **38.1**

condition number (of least squares problem $A\mathbf{x} = \mathbf{b}$): $\kappa(A) = \sigma_1/\sigma_p$, where $\text{rank } A = p$. **39.6**

condition numbers for polar factors ($A = UP$) in the Frobenius norm:

$\text{cond}_F(X) = \lim_{\delta \rightarrow 0} \sup_{\|\Delta A\|_F \leq \delta} \frac{\|\Delta X\|_F}{\delta}$, for $X = P$ or U . **15.3**

conductance: A parameter of a simple graph. **28.5**

cone: A subset K of a real or complex vector space such that for each $\mathbf{x}, \mathbf{y} \in K, c \geq 0, \mathbf{x} + \mathbf{y} \in K$ and $c\mathbf{x} \in K$; in Section 26.1 cone is used to mean proper cone. **8.5, 26.1**

conformable: See *conformal*.

conformal (partitions of matrices A, B): Partitions of block matrices that allow multiplication via the block structure. **10.1**

congruent (square matrices A, B over field F): There is an invertible matrix C such that $B = C^T AC$. **12.1**

***congruent** (square matrices A, B over $\mathbb{C}$): There is an invertible matrix C such that $B = C^* AC$. **8.3**

φ -congruent (square matrices A, B over field F with automorphism φ): There is an invertible matrix C such that $B = C^T A\varphi(C)$. **12.4**

conjugate: See *complex conjugate*.

conjugate gradient (CG) algorithm (to solve $H\mathbf{x} = \mathbf{b}$ for preconditioned Hermitian positive definite matrix H): At each step k , the approximation $\mathbf{x}_k$ of the form $\mathbf{x}_k \in \mathbf{x}_0 + \text{Span}\{\mathbf{r}_0, H\mathbf{r}_0, \dots, H^{k-1}\mathbf{r}_0\}$ for which the $\sqrt{H}$ -norm of the error, $\|\mathbf{e}_k\|_{\sqrt{H}} \equiv \langle \mathbf{e}_k, H\mathbf{e}_k \rangle^{1/2}$, is minimal. **41.2**

conjugate partition (of a sequence of positive integers $(u_1, u_2, \dots, u_n)$): The i th element of the conjugate partition is the number of j s such that $u_j \geq i$. **Preliminaries**

connected (graph): A graph with nonempty vertex set such that there exists a walk between any two distinct vertices. **28.1**

connected (digraph): A digraph whose associated undirected graph is connected. **29.1**

connected component (of a graph): A connected (induced) subgraph not properly contained in a connected subgraph. **28.1**

consecutive ones property (of a $(0, 1)$ -matrix A): There exists a permutation matrix P such that PA is a Petrie matrix. **30.2**

consistent: A system of linear equations that has one or more solutions. **1.4**

k -consistent sign pattern A : Every matrix $B \in Q(A)$ has exactly k real eigenvalues. **33.5**

constant (of a linear equation): The scalar not multiplied by a variable. **1.4**

constant vector: The vector of constants of a system of linear equations. **1.4**

contraction (matrix): A matrix $A \in \mathbb{C}^{n \times n}$ such that $\|A\|_2 \leq 1$. **18.6**

contraction (of edge e in graph $G = (V, E)$): The operation that merges the endpoints of e in V , and deletes e from E . **28.2**

convergent (square nonnegative matrix P): $\lim_{m \rightarrow \infty} P^m = 0$. **9.3**

convergent regular splitting: Square real matrix A has a convergent regular splitting if A has a representation $A = M - N$ such that $N \geq 0$, M invertible with $M^{-1} \geq 0$ and $M^{-1}N$ is convergent. **9.5**

converges geometrically to a with (geometric) rate β (complex sequence $\{a_m\}_{m=0,1,\dots}$): $\{\frac{a_m - a}{\gamma^m} : m = 0, 1, \dots\}$ is bounded for each $\beta < \gamma < 1$. **9.1**

convex (set of vectors): Closed under convex combinations. **Preliminaries**

convex combination (of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$): A vector of the form $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_k\mathbf{v}_k$ with a_i nonnegative and $\sum a_i = 1$ (vector space is real or complex). **Preliminaries**

convex cone: See *cone*.

convex function: A function $f : S \rightarrow \mathbb{R}$ where S is a convex set and for all $a \in \mathbb{R}$ such that $0 < a < 1$ and $\mathbf{x}, \mathbf{y} \in S$, $f(a\mathbf{x} + (1-a)\mathbf{y}) \leq af(\mathbf{x}) + (1-a)f(\mathbf{y})$. **Preliminaries**

convex hull (of a set of vectors): The set of all convex combinations of the vectors. **Preliminaries**

convex polytope: The convex hull of a finite set of vectors in $\mathbb{R}^n$. **Preliminaries**

coordinate (part of a vector): In the vector space F^n , one of the entries of a vector. **1.1**

coordinate mapping: The function that maps a vector to its coordinate vector. **2.6**

coordinate vector: The vector of coordinates of a vector relative to an ordered basis. **2.6**

coordinates (of a vector relative to a basis): The scalars that occur when the vector is expressed as a linear combination of the basis vectors. **2.6**

copositive (matrix A): $\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \geq 0$. **35.5**

coprime (elements in a domain): 1 is a greatest common divisor of the elements. **23.1**

corner minor: The determinant of a submatrix in the upper right or lower left corner of a matrix. **21.3**

correlation matrix: A positive semidefinite matrix in which every main diagonal entry is 1. **8.4**

cosine (of a complex square matrix A): The matrix defined by the cosine power series,

$$\cos(A) = I - \frac{A^2}{2!} + \cdots + \frac{(-1)^k}{(2k)!} A^{2k} + \cdots. \quad \mathbf{11.5}$$

cospectral (graphs): Having the same spectrum. **28.3**

cover (of a $(0, 1)$ -matrix): A collection of lines that contain all the 1s of the matrix. **27.1**

curve segment (from $\mathbf{x}$ to $\mathbf{y}$): The range of a continuous map ϕ from $[0, 1]$ to $\mathbb{R}^n$ with $\phi(0) = \mathbf{x}$ and $\phi(1) = \mathbf{y}$. **28.2**

cut-vertex (of connected simple graph G): A vertex v of G such that $G - v$ is disconnected. **36.2**

(k -)cycle (in a graph or digraph): A walk (of length k) with all vertices distinct except the first vertex equals the last vertex. **28.1, 29.1**

cycle (permutation): A permutation τ that maps a subset to itself cyclically. **Preliminaries**

k -cycle (in sign pattern A): A formal product of the form $a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_k i_1}$, where each of the elements is nonzero and the index set $\{i_1, i_2, \dots, i_k\}$ consists of distinct indices. **33.1**

n -cycle matrix: The $n \times n$ matrix with ones along the first subdiagonal and in the $1, n$ -entry, and zeros elsewhere. **48.1**

n -cycle pattern: A sign pattern A where the digraph of A is an n -cycle. **33.5**

cycle-clique (digraph): The induced subdigraph of every cycle is a clique. **35.7**

cycle product: A walk product where the walk is a cycle. **29.3**

cyclic normal form (of matrix A): A matrix in specific form that is permutation similar to A . **29.7**

cyclically real (square ray pattern A): The product of every cycle in A is real. **33.8**

D

$\Delta(G)$: $\max(p - q)$ over all ways in which q vertices may be deleted from graph G leaving p paths. **34.2**

$\mathbb{D}$ -invertible, $\mathbb{D}$ -module, $\mathbb{D}$ -submodule: Alphabetized under *invertible, module, submodule*.

D-optimal (design matrix W): $\det W^T W$ is maximal over all (± 1) - (or $(0, 1)$ -) matrices of the given size. **32.1**

damped least squares solution: The solution to the problem $(A^T A + \alpha I)\mathbf{x} = A^T \mathbf{b}$, where $\alpha > 0$. **39.8**

data fitting: See *Section 39.2*.

data perturbations (for linear system $A\mathbf{x} = \mathbf{b}$): Perturbations of A and/or $\mathbf{b}$. **38.1**

decomposable tensor: A tensor of the form $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_m$. **13.2**

defective (matrix $A \in F^{n \times n}$): There is an eigenvalue of A (over algebraic closure of F) having geometric multiplicity less than its algebraic multiplicity. **4.3**

deflation: A process of reducing the size of the matrix whose eigenvalue decomposition is to be determined, given that one eigenvector is known. **42.1**

degenerate (bilinear form f on vector space V): Not nondegenerate, i.e., the rank of f less than $\dim V$, **12.1**; also applied to φ -sesquilinear form **12.4**.

degree (of polynomial $p(x_1, \dots, x_n) \neq 0$): The largest integer m such that there is a term $cx_1^{m_1} \cdots x_n^{m_n}$ with $c \neq 0$ in p having degree m . **23.1**

degree (of term $cx_1^{m_1} \cdots x_n^{m_n}$ in polynomial $p(x_1, \dots, x_n)$): The sum of the degrees of the x_i , i.e., $\sum_i m_i$.

degree (of a vertex v): The number of times that v occurs as an endpoint of an edge. **28.1**

deletion (of edge e from graph $G = (V, E)$): The operation that deletes e from E . **28.2**

deletion (of vertex v from graph $G = (V, E)$): The operation that deletes v from V and all edges with endpoint v from E . **28.2**

depth (of eigenvector $\mathbf{x}$ for eigenvalue λ of A): The natural number h such that $\mathbf{x} \in \text{range}(A - \lambda I)^h - \text{range}(A - \lambda I)^{h+1}$. **6.2**

derogatory (matrix $A \in F^{n \times n}$): Some eigenvalue of A (over algebraic closure of F) has geometric multiplicity greater than 1. **4.3**

2-design with parameters (v, k, λ) : A collection of k -element subsets (blocks) B_i of a finite set X with $|X| = v$, such that each 2-element subset of X is contained in exactly λ blocks. **32.1**

(± 1) -design matrix: A matrix whose entries are 1 or -1 . **32.1**

$(0, 1)$ -design matrix: A matrix whose entries are 1 or 0. **32.1**

det: See *determinant*.

determinant (det): A scalar-valued function of a square matrix or linear operator, defined inductively, and also given by $\det(A) = \sum_{\sigma \in S_n} \text{sgn } \sigma \prod_{i=1}^n a_{i\sigma(i)}$. **4.1**

determinantal region (of complex sign pattern or ray pattern A): $S_A = \{\det(B) : B \in Q(A)\}$. **33.8**

determined by its spectrum (graph G): Every graph cospectral with G is isomorphic to G . **28.3**

diagonal (of a matrix): The diagonal means the *main diagonal*, i.e., the set of diagonal entries of a matrix, **1.2**. A diagonal is a collection of n nonzero entries in A no two on the same line. **27.2**

diagonal entry: An entry that lies in row k and column k for some k . **1.2**

diagonal matrix: A matrix all of whose off-diagonal entries are zero. **1.2, 10.2**

diagonal pattern: A square sign pattern all of whose off-diagonal entries are zero. **33.1**

diagonal product (of a matrix): The product of the entries on a diagonal of the matrix. **27.6**

diagonalizable (matrix $A \in F^{n \times n}$): There exist nonsingular matrix $B \in F^{n \times n}$ and diagonal matrix $D \in F^{n \times n}$ such that $A = BDB^{-1}$. **4.3**

diagonally dominant ($n \times n$ complex matrix A): $|a_{ii}| \geq \sum_{j=1, j \neq i}^n |a_{ij}|$ for $i = 1, \dots, n$. **9.5**

diagonally scaled representation (of symmetric matrix H with positive diagonal entries): A factored representation $H = DAD$ with $D = \text{diag}(\sqrt{h_{11}}, \dots, \sqrt{h_{nn}})$, $a_{ij} = \frac{h_{ij}}{\sqrt{h_{ii}h_{jj}}}$. **15.4**

diagonally scaled totally unimodular (DSTU) (real matrix A): There exist diagonal matrices D_1, D_2 and totally unimodular Z such that $A = D_1 Z D_2$. **46.3**

diagonally stable: See *Lyapunov diagonally stable*.

diameter (of a connected graph): The largest distance that occurs between two vertices. **28.1**

digraph: A directed graph having at most one arc between each ordered pair of vertices (loops permitted). **29.1**

digraph (of $n \times n$ matrix A): The digraph having vertices $\{1, \dots, n\}$ and having arc (i, j) exactly when $a_{ij} \neq 0$. **9.1, 29.2**

digraph (of $n \times n$ partial matrix B): The digraph having vertices $V = \{1, \dots, n\}$ and for each $i, j \in V$, (i, j) is an arc exactly when b_{ij} is specified. **35.1**

dilation: Matrix $A \in \mathbb{C}^{n \times n}$ has a dilation $B \in \mathbb{C}^{m \times m}$ if there is $X \in \mathbb{C}^{m \times n}$ such that $X^*X = I_n$ and $X^*BX = A$. **18.6**

dim: See *dimension*.

dimension (dim) (of a vector space): The number of vectors in a basis for the vector space. **2.2**

dimension (dim) (of a convex polytope): The smallest dimension of an affine space containing it. **27.6**

direct sum (of matrices): A block diagonal matrix with these matrices as the diagonal blocks. **2.4, 10.2**

direct sum (of subspaces): $W_1 + \dots + W_k$ is direct if for all $i = 1, \dots, k$, $W_i \cap \sum_{j \neq i} W_j = \{0\}$. **2.3**

direct sum (of vector spaces): See *external direct sum*.

directed bigraph (of the real $m \times n$ matrix A): The directed graph with vertices $1, 2, \dots, m, 1', 2', \dots, n'$, with arc (i, j') if and only if $a_{ij} > 0$, and with arc (j', i) if and only if $a_{ij} < 0$. **30.2**

directed edge: See *arc*.

directed graph: A finite nonempty set of vertices and a finite multiset of arcs, where each arc is an ordered pair of vertices. **29.1**

directed multigraph: A directed graph having more than one arc between at least one pair of vertices. **29.1**

disjoint (graphs): The vertex sets are disjoint sets. **28.1**

dispersion: See *Section 21.3*.

distance (between two vectors): The norm of their difference. **5.1**

distance (between two vertices in a graph): The length of a shortest path between the vertices (infinite if no path). **28.1**

distance of $W(A)$ to the origin: $\tilde{w}(A) = \min\{|\mu| : \mu \in W(A)\}$. **18.1**

distance-regular graph: A connected graph with the property that for every pair of vertices u, v , the number of vertices that have distance i from u and distance j from v depends only on i, j , and the distance between u and v . **28.6**

distinguished eigenvalue (of nonnegative square matrix P): An eigenvalue of P that has a semipositive eigenvector. **9.1**

d **divides** a (in a domain $\mathbb{D}$): $a = db$ for some $b \in \mathbb{D}$. **23.1**

divisor (i.e., d is a divisor of a): See *divides*.

domain (ring): An integral domain, i.e., a commutative ring without zero divisors and containing identity 1. **23.1**

domain (of $T: V \rightarrow W$): The vector space V . **3.1**

dominant eigenvalue: λ_1 where $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$ are the eigenvalues. **42.1**

dot product: See *standard inner product*.

double echelon form: A special type of real matrix, see **21.1**

double generalized star: A tree constructed by joining the centers of two generalized stars by an edge. **34.7**

double path: A tree constructed by joining two degree two vertices of two paths by an edge. **34.7**

double precision: Typically, floating-point numbers have machine epsilon roughly 10^{-16} and precision roughly 16 decimal digits. **37.6**

doubly directed tree: A digraph Γ whose associated undirected graph is a tree and whenever (i, j) is an arc in Γ , (j, i) is also an arc in Γ . **29.1**

doubly nonnegative (matrix): A positive semidefinite matrix having every entry nonnegative. **35.4**

doubly stochastic (matrix): A square nonnegative matrix having all row and column sums equal to 1. **9.4**

downdating (QR factorization): See *Section 39.7*.

DSTU: See *diagonally scaled totally unimodular*.

Δ TP: See *triangular totally positive*.

dual basis: A specific basis for the dual space defined from a given basis for the vector space. **3.8**

dual cone (of cone K): See *dual space of cone*.

dual norm (of vector norm $\|\cdot\|$): $\|\mathbf{y}\|^D = \max_{\mathbf{x} \neq 0} \frac{|\mathbf{y}^* \mathbf{x}|}{\|\mathbf{x}\|}$. **37.1**

dual space: The vector space of linear functionals on a given vector space. **3.8**

dual space (of cone K): $K^* = \{\mathbf{y} \in V \mid \langle \mathbf{x}, \mathbf{y} \rangle \geq 0 \quad \forall \mathbf{x} \in K\}$ **8.5, 26.1**

E

ED: See *Euclidean domain*.

ED-RCF basis (for a linear operator): See *Section 6.4*.

ED-RCF matrix (over F): A block diagonal matrix of the form $C(h_1^{m_1}) \oplus \dots \oplus C(h_t^{m_t})$ where each $h_i(x)$ is a monic polynomial that is irreducible over F . **6.4**

EDD: See *elementary divisor domain*.

edge: A pair of vertices, part of a graph. **28.1**

edge cut: Let $G = (V, E)$ be a graph, and let $X, V \setminus X$ be a nontrivial partitioning of V . The set E_X of edges of G that have one end point in X and the other end point in $V \setminus X$ is an edge cut. **36.2**

efficacy (of a processor): See *speed*.

efficient: One algorithm is more efficient than another, if it accomplishes the same task with a lower cost of computation. **37.7**

eigenpair (of $A \in \mathbb{C}^{n \times n}$): A (eigenvalue, eigenvector) pair. **15.1**

eigenspace (of eigenvalue λ of A): $\ker(A - \lambda I)$. **4.3**

k -**eigenspace** (of matrix or linear operator A at λ): $\ker(A - \lambda I)^k$ **6.1**

eigentriplet (of $A \in \mathbb{C}^{n \times n}$): A vector-scalar-vector triplet $(\mathbf{y}, \lambda, \mathbf{x})$ such that $\mathbf{x} \neq \mathbf{0}$, $\mathbf{y} \neq \mathbf{0}$, and $A\mathbf{x} = \lambda\mathbf{x}$, $\mathbf{y}^*A = \lambda\mathbf{y}^*$. **15.1**

eigenvalue (of matrix or linear operator A): A scalar λ such that there exists a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$. **4.3**

eigenvalue (of the pencil $A - xB$): $\lambda = \mu/\nu$ where $\nu A\mathbf{v} = \mu B\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$ (if $\nu = 0, \lambda = \infty$). **43.1**

eigenvalue (of a graph): An eigenvalue of its adjacency matrix. **28.3**

eigenvalue decomposition (EVD) (of real symmetric matrix A): $A = U\Lambda U^T$, where $U^T U = U U^T = I_n$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a real diagonal matrix of eigenvalues, and the columns of U are eigenvectors. **42.1**

eigenvector (of matrix or linear operator A): A nonzero vector $\mathbf{x}$ such that there exists a scalar λ such that $A\mathbf{x} = \lambda\mathbf{x}$. **4.3**

eigenvector (of the pencil $A - xB$): Nonzero vector $\mathbf{v} \in \mathbb{C}^n$ such that there are scalars $\mu, \nu \in \mathbb{C}$, not both zero, such that $\nu A\mathbf{v} = \mu B\mathbf{v}$. **43.1**

elementary bidiagonal matrix: An $n \times n$ real matrix whose main diagonal entries are all equal to one, and there is at most one nonzero off-diagonal entry and this entry must occur on the super- or sub-diagonal. **21.2**

elementary divisor domain (EDD): A GCDD such that for any three elements $a, b, c \in \mathbb{D}$ there exists $p, q, x, y \in \mathbb{D}$ such that $(a, b, c) = (px)a + (py)b + (qy)c$. **23.1**

elementary divisors (of an ED-RCF matrix): The polynomials whose companion matrices are the diagonal blocks. **6.4**

elementary divisors (of matrix or linear operator A): The elementary divisors of $\text{RCF}_{ED}(A)$. **6.4**

elementary divisors rational canonical form (of matrix $A \in F^{n \times n}$): An ED-RCF matrix that is similar to A . **6.4**

elementary divisors rational canonical form (of a linear operator): See *Section 6.4*.

elementary divisors rational canonical form matrix: See *ED-RCF matrix*.

elementary matrix: The result of doing one elementary row operation on the identity matrix. **1.5**

elementary column operation: Column analog of an elementary row operation. **6.5, 23.2**

elementary row operation (on a matrix over a domain): Add a multiple of one row to another, interchange two rows, or multiply a row by an invertible domain element. **6.5, 23.2**

elementary row operation (on a matrix over a field): Add a multiple of one row to another, interchange two rows, or multiply a row by a nonzero field element. **1.3**

elementary symmetric function: The sum of all products of n variables taken k at a time. **Preliminaries**

elimination graph: A graph, digraph or bigraph associated with the principal submatrix that remains to be factored during Gaussian elimination. **40.4**

elimination ordering (for the rows of a matrix): A bijection that specifies the order in which the rows of the matrix are eliminated during Gaussian elimination. **40.5**

elimination sequence: See *elimination ordering*.

embedding: A representation of a graph in $\mathbb{R}^n$ where edges intersect only at vertices. **28.2**

empty graph: A graph with no edges. **28.1**

endpoint (of an edge in a graph): One of the two vertices on the edge. **28.1**

energy norm: See *M-norm* (alphabetized under norm).

i, j -entry: The scalar in row i and column j in a matrix. **1.2**

entry sign symmetric P -matrix A : A P -matrix such that for all i, j , either $a_{ij}a_{ji} > 0$ or $a_{ij} = a_{ji} = 0$. **35.10**

entry sign symmetric P_0 -matrix A : A P_0 -matrix such that for all i, j , either $a_{ij}a_{ji} > 0$ or $a_{ij} = a_{ji} = 0$. **35.10**

entry sign symmetric $P_{0,1}$ -matrix A : A $P_{0,1}$ -matrix such that for all i, j , either $a_{ij}a_{ji} > 0$ or $a_{ij} = a_{ji} = 0$. **35.10**

entry weakly sign symmetric P -matrix A : A P -matrix such that for all i, j , $a_{ij}a_{ji} \geq 0$. **35.10**

entry weakly sign symmetric P_0 -matrix A : A P_0 -matrix such that for all i, j , $a_{ij}a_{ji} \geq 0$. **35.10**

entry weakly sign symmetric $P_{0,1}$ -matrix A : A $P_{0,1}$ -matrix such that for all i, j , $a_{ij}a_{ji} \geq 0$. **35.10**

envelope (of sparse matrix A): The set of indices of the elements between the first and last nonzero elements of every row. **40.5**

equivalence relation: A relation that is reflexive, symmetric, and transitive. **Preliminaries**

equivalent (bilinear forms f, g): There exists an ordered basis $\mathcal{B}$ such that the matrix of g relative to $\mathcal{B}$ is congruent to the matrix of f relative to $\mathcal{B}$. **12.1**

φ -equivalent (bilinear forms f, g): There exists an ordered basis $\mathcal{B}$ such that the matrix of g relative to $\mathcal{B}$ is φ -congruent to the matrix of f relative to $\mathcal{B}$. **12.4**

***equivalent** (bilinear forms f, g on a complex vector space): φ -equivalent with $\varphi =$ complex conjugation. **12.5**

equivalent (matrices $A, B \in F^{m \times n}$): $B = QAP$ for some invertible matrices Q, P . **2.4**

equivalent (matrices $A, B \in \mathbb{D}^{m \times n}$): $B = QAP$ for some $\mathbb{D}$ -invertible matrices Q, P . **23.2**

equivalent: Systems of linear equations that have the same solution set. **1.4**

ergodic class (of stochastic matrix P): A basic class of P . **9.4**

ergodic state: A state in an ergodic class. **9.4**

ergodicity coefficient (of complex matrix A): $\max\{|\lambda| : \lambda \in \sigma(P) \text{ and } |\lambda| \neq \rho(A)\}$. **9.1**

error at step k of an iterative method for solving $A\mathbf{x} = \mathbf{b}$: The difference between the true solution $A^{-1}\mathbf{b}$ and the approximate solution $\mathbf{x}_k$: $\mathbf{e}_k \equiv A^{-1}\mathbf{b} - \mathbf{x}_k$. **41.1**

Euclid's Algorithm: See *Section 23.1*.

Euclidean distance (matrix A): There exist vectors $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ (for some $d \geq 1$) such that $a_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ for all $i, j = 1, \dots, n$. **35.3**

Euclidean domain (ED): A domain $\mathbb{D}$ with a function $d : \mathbb{D} \setminus \{0\} \rightarrow \mathbb{Z}^+$ such that for all nonzero $a, b \in \mathbb{D}$, $d(a) \leq d(ab)$ and there exist $t, r \in \mathbb{D}$ such that $a = tb + r$, where either $r = 0$ or $d(r) < d(b)$. **23.1**

Eulidean norm (of a matrix): See *Frobenius norm*.

Eulidean norm (of a vector): See *2-norm*.

Euclidean space: Finite dimensional real vector space with standard inner product. **5.1**

EVD: See *eigenvalue decomposition*.

even (permutation): Can be written as the product of an even number of transpositions. **Preliminaries**

even (cycle in a sign pattern): The length of the simple or composite cycle is even. **33.1**

expanders (or family of expanders): An infinite family of graphs with constant degree and isoperimetric number bounded from below. **28.5**

explicit restarting (of the Arnoldi algorithm): A straightforward but inferior way to implement polynomial restarting by explicitly constructing the starting vector $\psi(A)\mathbf{v}$ by applying $\psi(A)$ through a sequence of matrix-vector products. **44.5**

exponent (of floating point number x): e in floating point number $x = \pm \left(\frac{m}{b^{p-1}}\right) b^e$. **37.6**

exponent (of a primitive digraph): Period; equivalently, the smallest value of k that works in the definition of primitivity. **29.6**

exponent (of a primitive matrix A): The exponent of the digraph of A . **29.6**

exponent of matrix multiplication complexity: The infimum of the real numbers w such that there exists an algorithm for multiplying $n \times n$ matrices with $O(n^w)$ arithmetic operations. **47.2**

exponential (of a square complex matrix A): The matrix defined by the exponential power series, $e^A = I + A + \frac{A^2}{2!} + \dots + \frac{A^k}{k!} + \dots$. **11.3**

extended precision: Floating point numbers that have greater than double precision. **37.6**

extension: Signing $D' = \text{diag}(d'_1, d'_2, \dots, d'_k)$ is an extension of signing D if $D' \neq D$ and $d'_i = d_i$ whenever $d_i \neq 0$. **33.3**

external direct sum (of vector spaces): The cartesian product of the vector spaces with component-wise operations. **2.3**

exterior power (of a vector space): See *Grassman space*.

exterior product (of vectors $\mathbf{v}_1, \dots, \mathbf{v}_m$): $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_m = m! \text{Alt}(\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_k)$. **13.6**

extreme point (of a closed convex set S): A point in S that is not a nontrivial convex combination of other points in S . **Preliminaries**

extreme vector $\mathbf{v} \in K$: Either $\mathbf{v}$ is the zero vector or $\mathbf{v}$ is nonzero and $\Phi(\mathbf{v}) = \{\lambda \mathbf{v} : \lambda \geq 0\}$. **26.1**

F

face F (of cone K): A subcone of K such that $\mathbf{v} \in F$, $\mathbf{v} \geq^K \mathbf{w} \geq^K \mathbf{0} \implies \mathbf{w} \in F$. **26.1**

face of K generated by $S \subseteq K$: The intersection of all faces of K containing S . **26.1**

family of Toeplitz matrices (defined by symbol $a = \sum_{k=-\infty}^{\infty} a_k z^k$): The set $\{T_n\}_{n \geq 1}$ where T_n has constants $a_{1-n}, \dots, a_{n-1}$. **16.2**

Fiedler vector (of simple graph G): An eigenvector of the Laplacian of G corresponding to the algebraic connectivity. **36.1**

field: A nonempty set with two binary operations, addition and multiplication, satisfying commutativity, associativity, distributivity and existence of identities and inverses. **Preliminaries**

field of rational functions: The quotient field of the field of polynomials over F . **23.1**

field of values: See *numerical range*.

fill element (in a sparse matrix A): An element that is zero in A but becomes nonzero during Gaussian elimination. **40.3**

fill graph (of sparse matrix A): The (graph, digraph, or bigraph) of A , together with all the additional edges corresponding to the fill elements that occur during Gaussian elimination. **30.2, 40.5**

fill matrix (of Gaussian elimination on a matrix): The matrix $M_1 + M_2 + \dots + M_{n-1} - (n-1)I + U$, where the M_i are Gauss transformations used in the elimination and U is the resulting upper triangular matrix. **40.3**

final (subset α of vertices): no vertex in α has access to a vertex not in α . **9.1**

finite dimensional (vector space V): V has a basis containing a finite number of vectors. **2.2**

finitely generated (ideal I of a domain $\mathbb{D}$): An ideal of $\mathbb{D}$ that is finitely generated as a $\mathbb{D}$ -module. **23.1**

finitely generated (module $\mathbf{M}$ over a domain $\mathbb{D}$): There exist k elements (*generators*) $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbf{M}$ so that every $\mathbf{v} \in \mathbf{M}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_k$, over $\mathbb{D}$. **23.3**

fixed space: the set of vectors fixed by a linear transformation. **3.6**

flip map: See *Section 22.2*.

floating point addition, subtraction multiplication, division: Operations on floating point numbers, See *Section 37.6*.

floating point number: A real number of the form $x = \pm \left(\frac{m}{b^{p-1}}\right) b^e$. **37.6**

floating point operation: One of floating point addition, subtraction, multiplication, division. **37.6**

flop: A floating point operation. **37.7**

FOM: See *full orthogonalization method*.

forest: A graph with no cycles. **28.1**

forward error (in $\hat{f}(\mathbf{x})$): $f(\mathbf{x}) - \hat{f}(\mathbf{x})$, the difference between the mathematically exact function evaluation and the perturbed function evaluation. **37.8**

forward stable (algorithm): The forward relative error is small for all valid input data $\mathbf{x}$ despite rounding and truncation errors in the algorithm. **37.8**

free variable: A variable x_i in the solution to a system of linear equations that is a free parameter. **1.4**

Frobenius norm (of $m \times n$ complex matrix A): $\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$. **7.1, 37.3**

Frobenius normal form (of matrix A): A block upper triangular matrix with irreducible diagonal blocks that is permutation similar to A . **27.3**

full (cone): Has a nonempty interior. **8.5, 26.1**

fully indecomposable ($(0, 1)$ -matrix): Not partly decomposable. **27.2**

full orthogonalization method (FOM): Generates, at each step k , the approximation $\mathbf{x}_k$ of the form $\mathbf{x}_k \in \mathbf{x}_0 + \text{Span}\{\mathbf{r}_0, A\mathbf{r}_0, \dots, A^{k-1}\mathbf{r}_0\}$ for which the residual is orthogonal to the Krylov space. **41.3**

fundamental subspaces (of the least squares problem $A\mathbf{x} = \mathbf{b}$): $\text{range } A, \ker A^T, \ker A$ and $\text{range } A^T$. **39.3**

G

Galerkin condition: The defining condition for a Ritz vector and Ritz value. **44.1**

Gauss multipliers: The nonzero entries in a Gauss vector. **38.3**

Gauss transformation: The matrix $M_k = I - \ell_k \mathbf{e}_k^T$, where ℓ_k is a Gauss vector. **38.3**

Gauss vector: A vector $\ell_k \in \mathbb{C}^n$ with the leading k entries equal to zero. **38.3**

Gauss–Jordan elimination: A process used to find the reduced row echelon form of a matrix. **1.3**

Gaussian elimination: A process used to find a row echelon form of a matrix. **1.3, 38.3**

g.c.d.: See *greatest common divisor*.

GCDD: See *greatest common divisor domain*.

general linear group (of order n over F): The multiplicative group consisting of all invertible $n \times n$ matrices over the field F . **67.1**

general solution: A formula that describes every vector in the solution set to a system of linear equations. **1.4**

generalized cycle (in a digraph): A disjoint union of one or more cycles such that every vertex lies on exactly one cycle. **29.1**

generalized cycle product: A walk product where the walk is a generalized cycle. **29.4**

generalized diagonal (of square matrix A): Entries of A corresponding to a generalized cycle in the digraph of A . **29.4**

generalized eigenspace (of matrix or linear operator A at λ): $\ker(A - \lambda I)^v$, where v is the index of A at λ . **6.1**

generalized eigenvector (of a matrix or linear operator): A nonzero vector in the generalized eigenspace. **6.1**

generalized Laplacian (of simple graph G): A matrix M such that for $i \neq j$, $m_{ij} < 0$ if the i th and j th vertices are adjacent and $m_{ij} = 0$ otherwise (no restriction for $i = j$). **28.4**

generalized line graph: See *Section 28.2*.

generalized minimal residual (GMRES): Algorithm generates, at each step k , the approximation $\mathbf{x}_k$ of the form $\mathbf{x}_k \in \mathbf{x}_0 + \text{Span}\{\mathbf{r}_0, A\mathbf{r}_0, \dots, A^{k-1}\mathbf{r}_0\}$ for which the 2-norm of the residual is minimal. **41.3**

generalized sign pattern: A matrix whose entries are from the set $\{+, -, 0, \#\}$, where $\#$ indicates an ambiguous sum (the result of adding $+$ with $-$). **33.1**

generalized star: A tree in which at most one vertex has degree greater than two. **34.6**

generators (of an ideal): See *finitely generated*.

generic matrix: A matrix whose nonzero elements are independent indeterminates over the field F . **30.2**

geometric multiplicity: The dimension of the eigenspace. **4.3**

Geršgorin discs (of $A \in \mathbb{C}^{n \times n}$): $\{z \in \mathbb{C} : |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|\}$ for $i = 1, \dots, n$. **14.2**

Givens matrix: See *Givens transformation*.

Givens rotation: See *Givens transformation*.

Givens transformation: An identity matrix modified so that the (i, i) and (j, j) entries are replaced by $c = \cos(\theta)$, the (i, j) entry is replaced by $s = e^{i\vartheta} \sin(\theta)$, and the (j, i) entry is replaced by $-\bar{s} = -e^{-i\vartheta} \sin(\theta)$. **38.4**

GMRES: See *generalized minimal residual, restarted GMRES algorithm*.

graded (associative algebra A): There exist $(A_k)_{k \in \mathbb{N}}$ vector subspaces of A such that $A = \bigoplus_{k \in \mathbb{N}} A_k$ and $A_i A_j \subseteq A_{i+j}$ for every $i, j \in \mathbb{N}$. **13.9**

graded matrix $A \in \mathbb{C}^{m \times n}$: A can be scaled as $A = GS$ (or $A = S^*GS$, depending on situation) such that G is “well-behaved” (i.e., $\kappa_2(G)$ is of modest magnitude), where S is a scaling matrix (often diagonal). **15.3**

Gram matrix (of vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in a complex inner product space): the matrix whose i, j -entry is $\langle \mathbf{v}_i, \mathbf{v}_j \rangle$. **8.1**

graph: A finite set of vertices and a finite multiset of edges, where each edge is an unordered pair of vertices (not necessarily distinct). **28.1**

graph (of $n \times n$ matrix A): The simple graph whose vertex set is $\{1, \dots, n\}$ and having an edge between vertices i and j ($i \neq j$) if and only if $a_{ij} \neq 0$ or $a_{ji} \neq 0$. **19.3, 34.1**

graph (of a convex polytope $\mathcal{P}$): Vertices are the extreme points of $\mathcal{P}$ and edges are the pairs of extreme points of 1-dimensional faces of $\mathcal{P}$. **27.6**

graph (of $n \times n$ combinatorially symmetric partial matrix B): The graph having vertices $V = \{1, \dots, n\}$ and for each $i, j \in V$, $\{i, j\}$ is an edge exactly when b_{ij} is specified (loops permitted, no multiple edges). **35.1**

graph of a combinatorially symmetric $n \times n$ sign pattern $A = [a_{ij}]$: The graph with vertex set $\{1, 2, \dots, n\}$ where $\{i, j\}$ is an edge iff $a_{ij} \neq 0$ (note: the graph may have loops). **33.5**

Grassman algebra (on vector space V): $\bigwedge V = \bigoplus_{k \in \mathbb{N}} \left(\bigwedge^k V \right)$ with alt multiplication. **13.9**

Grassmann space (of vector space V): $\bigwedge^m V = \text{Alt}(\bigotimes^m V)$. **13.6**

greatest common divisor (g.c.d.) (of $a_1, \dots, a_n \in \mathbb{D}$): $d \in \mathbb{D}$ such that $d|a_i$ for $i = 1, \dots, n$, and if $d'|a_i, i = 1, \dots, n$, then $d'|d$. **23.1**

greatest common divisor domain (GCDD): A domain in which any two elements have a g.c.d. **23.1**

greatest integer (of a real number x): the greatest integer less than or equal to x . **Preliminaries**

group: A set with one binary operation, satisfying associativity, existence of identity and inverses. **Preliminaries**

group inverse (of square complex matrix A): A matrix X satisfying $AXA = A$, $XAX = X$ and $AX = XA$. **9.1**

H

H-matrix (real square matrix A): The comparison matrix of A is an M -matrix. **19.5**

Hadamard matrix: A ± 1 -matrix H_n with $H_n H_n^T = nI_n$. **32.2**

Hadamard product (of $A, B \in F^{n \times n}$): The $n \times n$ matrix whose i, j -entry is $a_{ij}b_{ij}$. **8.5**

Hall matrix: An $n \times n$ $(0, 1)$ -matrix that does not have a $p \times q$ zero submatrix for positive integers p, q with $p + q > n$. **27.2**

Hamilton cycle: A cycle in a graph that includes all vertices. **28.1**

Hamiltonian cycle: See *Hamilton cycle*.

Hankel matrix: A matrix with constant elements along its antidiagonals. **48.1**

height (of basic class B): The largest number of vertices on a simple walk that ends at B in the basic reduced digraph. **9.3**

hereditary (class of matrices X): Whenever A is an X -matrix and $\alpha \subseteq \{1, \dots, n\}$, then $A[\alpha]$ is an X -matrix. **35.1**

Hermite normal form: A generalization of reduced row echelon form used in domains. **23.2**

Hermitian (linear operator on an inner product space): An operator that is equal to its adjoint. **5.3**

Hermitian (matrix): A real or complex matrix equal to its conjugate transpose. **1.2, 7.2, 8.1**

Hermitian adjoint (of a matrix): The conjugate transpose of a real or complex matrix. **1.2**

Hermitian form f (on complex vector space V): A sesquilinear form such that $f(\mathbf{v}, \mathbf{u}) = \overline{f(\mathbf{u}, \mathbf{v})}$ for all $\mathbf{u}, \mathbf{v} \in V$. **12.5**

Hoffman polynomial (of graph G): A polynomial $h(x)$ of minimum degree such that $h(A_G) = J$. **28.3**

Hölder norm: See *p-norm*.

homogeneous (digraph): Either symmetric or asymmetric. **35.7**

homogeneous (element of graded algebra $A = \bigoplus_{k \in \mathbb{N}} A_k$): An element of some A_k . **13.9**

homogeneous (pencil associated with $A(x) = A_0 + xA_1 \in \mathbb{D}[x]^{m \times n}$): $A(x_0, x_1) = x_0A_0 + x_1A_1 \in \mathbb{D}[x_0, x_1]^{m \times n}$. **23.4**

homogeneous (polynomial): All terms have the same degree. **23.1**

homogeneous (system of linear equations): A system in which all the constants are zero. **1.4**

Householder matrix: See *Householder transformation*.

Householder reflector: See *Householder transformation*.

Householder transformation (defined by $\mathbf{v} \in \mathbb{C}^n$): The matrix $I - \frac{2}{\|\mathbf{v}\|_2^2} \mathbf{v}\mathbf{v}^*$. **38.4**

Householder vector: The vector $\mathbf{v}$ used to define a Householder transformation. **38.4**

hyperbolic SVD decomposition (of matrix pair $(G, \mathcal{J})$): A decomposition of G , $G = W\Sigma B^{-1}$, where W is orthogonal, Σ is diagonal, and B is $\mathcal{J}$ -orthogonal. **46.5**

I

IAP: See *inertially arbitrary pattern*.

ideal (of a domain $\mathbb{D}$): A nonempty subset I of $\mathbb{D}$ such that if $a, b \in I$ and $p, q \in \mathbb{D}$, then $pa + qb \in I$. **23.1**

idempotent: A matrix or linear transformation that squares to itself. **2.7, 3.6**

identity matrix: A diagonal matrix having all diagonal elements equal to one. **1.2**

identity pattern (of order n): The $n \times n$ diagonal pattern with + diagonal entries. **33.1**

identity transformation: A linear operator that maps each vector to itself. **3.1**

IEP: See *inverse eigenvalue problem*.

IF-RCF matrix: A block diagonal matrix of the form $C(a_1) \oplus \cdots \oplus C(a_s)$, where $a_i(x)$ divides $a_{i+1}(x)$ for $i = 1, \dots, s-1$. **6.6**

ill-conditioned (input $\mathbf{z}$ for P): Some small relative perturbation of $\mathbf{z}$ causes a large relative perturbation of $P(\mathbf{z})$. **37.4**

image: See *range*.

imaginary part (of complex number $a + bi$): b . **Preliminaries**

immanant: A function f_λ from complex square matrices to complex numbers defined by the irreducible character of a partition λ of n , specifically, $f_\lambda(M) = \sum_{\sigma \in S_n} \chi_\lambda(\sigma) \prod_{i=1}^n m_{i\sigma(i)}$. **31.10**

implicit restarting (of the Arnoldi algorithm): Apply a sequence of implicitly shifted QR steps to an m -step Arnoldi or Lanczos factorization to obtain a truncated form of the implicitly shifted QR-iteration. **44.5**

imprimitive (digraph): Not primitive. **29.6**

imprimitive (matrix A): The digraph $\Gamma(A)$ is imprimitive. **29.6**

improper (divisor of a in a domain): An associate of a or a unit. **23.1**

incidence matrix (of a graph): A matrix with rows indexed by the vertices and columns indexed by the edges, having i, j entry 1 if vertex v_i is an endpoint of edge e_j and 0 otherwise. **28.4**

incidence matrix (of subsets $S_1, S_2, \dots, S_m$ of a finite set $\{x_1, x_2, \dots, x_n\}$): The $m \times n$ (0,1)-matrix $M = [m_{ij}]$ in which $m_{ij} = 1$ iff $x_j \in S_i$. **31.3**

(± 1)-incidence matrix (of a 2-design): A matrix W whose rows are indexed by the elements x_i of X and whose columns are indexed by the blocks B_j having $w_{ij} = -1$ if $x_i \in B_j$ and $w_{ij} = +1$ otherwise. **32.2**

incomplete Cholesky decomposition: A preconditioner for a Hermitian positive definite matrix A of the form $M = LL^*$, where L is a sparse lower triangular matrix. **41.4**

incomplete LU decomposition: A preconditioner for a general matrix A of the form $M = LU$, where L and U are sparse lower and upper triangular matrices. **41.4**

inconsistent: A system of linear equations that has no solution. **1.4**

indefinite (Hermitian matrix A): Neither A nor $-A$ is positive semidefinite. **8.4**

independent (subspaces W_i): $\mathbf{w}_1 + \cdots + \mathbf{w}_k = \mathbf{0}$ and $\mathbf{w}_i \in W_i, i = 1, \dots, k$ implies $\mathbf{w}_i = \mathbf{0}$ for all $i = 1, \dots, k$. **2.3**

independent set of vertices: See *clique*.

index (of an eigenvalue): The smallest integer k such that the k -eigenspace equals the $k+1$ -eigenspace. **6.1**

index (of square nonnegative matrix P): The index of the spectral radius of P . **9.3**

index (of $A \in H_0$): See *Section 23.3*.

index of imprimitivity (of an irreducible matrix A): The greatest common divisor of the lengths of all cycles in the digraph $\Gamma(A)$; same as period. **29.7**

index of primitivity: See *exponent*.

individual condition number for eigenvalue λ : $\frac{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}{|\mathbf{y}^* \mathbf{x}|}$ where $(\mathbf{y}, \lambda, \mathbf{x})$ is an eigentriplet. **15.1**

induced basis: The nonzero images in the quotient V/W of vectors in a basis that contains a basis for W . **2.3**

induced norm: The matrix norm induced by the family of vector norms $\|\cdot\|$ is $\|A\| = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|A\mathbf{x}\|}{\|\mathbf{x}\|}$. **37.3**

induced subdigraph (of $\Gamma = (V, E)$): A subdigraph $\Gamma = (V', E')$ containing all arcs from E with endpoints in V' . **29.1**

induced subgraph (of $G = (V, E)$): A subgraph $G = (V', E')$ containing all edges from E with endpoints in V' . **28.1**

inertia (of complex square matrix A): The ordered triple $\text{in}(A) = (\pi(A), \nu(A), \delta(A))$ where $\pi(A)$ is the number of eigenvalues of A with positive real part, $\nu(A)$ is the number of eigenvalues of A with negative real part, and $\delta(A)$ is the number of eigenvalues of A on the imaginary axis. **8.3, 19.1**

inertia preserving (real square matrix A): The inertia of AD is equal to the inertia of D for every nonsingular real diagonal matrix D . **19.3**

inertially arbitrary pattern (IAP): A sign pattern A of size n such that every possible ordered triple (p, q, z) of nonnegative integers p, q , and z with $p + q + z = n$ can be achieved as the inertia of some $B \in Q(A)$. **33.6**

infinite dimensional (vector space): A vector space that is not finite dimensional. **2.2**

initial (minor): See *Section 21.3*

injective: See *one-to-one*.

inner distribution: Vector parameter of an association scheme. **28.6**

inner product (on a vector space V over $F = \mathbb{R}$ or $\mathbb{C}$): A function $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ satisfying certain conditions. For $F = \mathbb{R}$, $\langle \cdot, \cdot \rangle$ is symmetric, bilinear and positive definite. **5.1**

inner product space: A real or complex vector space with an inner product. **5.1**

integral domain: A commutative ring without zero divisors and containing identity 1. **23.1**

iterative method (for solving a linear system $Ax = b$): Any algorithm that starts with an initial guess x_0 for the solution and successively modifies that guess in an attempt to obtain improved approximate solutions $x_1, x_2, \dots$. **41.1**

internal direct sum: See *direct sum*.

intersection (of two graphs $G = (V, E)$ and $G' = (V', E')$): The graph with vertex set $V \cap V'$, and edge (multi)set $E \cap E'$. **28.1**

intersection numbers: Parameters of an association scheme. **28.6**

P -invariant face (of K -nonnegative matrix P): $PF \subseteq F$. **26.1**

invariant factors (of IF-RCF matrix $C(a_1) \oplus \dots \oplus C(a_s)$): The polynomials $a_i(x), i = 1, \dots, s$. **6.6**

invariant factors (of matrix or linear operator $A \in F^{n \times n}$): The invariant factors of the invariant factors rational canonical form of A . **6.6**

invariant factors (of a matrix over a domain): See *Section 23.1*.

invariant factors rational canonical form (of matrix $A \in F^{n \times n}$): The IF-RCF matrix similar to A . **6.6**

invariant factors rational canonical form (of a linear operator): See *Section 6.6*.

invariant factors rational canonical form matrix: See *IF-RCF matrix*.

invariant polynomials: See *invariant factors*.

invariant subspace (of linear transformation $T : V \rightarrow W$): A subspace U of V such that for all $u \in U$, $T(u) \in U$. **3.6**

inverse (of square matrix A): A matrix B such that $AB = BA = I$. **1.5**

inverse (of linear transformation $T : V \rightarrow W$): A linear transformation $S : W \rightarrow V$ such that $TS = I_W$ and $ST = I_V$. **3.7**

inverse eigenvalue problem: The problem of constructing a matrix with prescribed structural and spectral constraints. **20.1**

inverse eigenvalue problem with prescribed entries: Construct a matrix with given eigenvalues subject to given entries in given positions. **20.1**

Inverse Eigenvalue Problem of tree T : Determine all possible spectra that occur among matrices in $\mathcal{S}(T)$. **34.5**

inverse iteration: The power method applied to the inverse of a shifted matrix. **42.1**

inverse M -matrix: An invertible matrix whose inverse is an M -matrix. **9.5**

inverse nonnegative: A square sign pattern that allows an entrywise nonnegative inverse. **33.7**

inverse-positive (real square matrix A): A is nonsingular and $A^{-1} \geq 0$. **9.5**

inverse positive (square sign pattern): Allows an entrywise positive inverse. **33.7**

invertible (matrix or linear transformation): Has an inverse. **1.5, 3.7**

$\mathbb{D}$ -invertible (matrix in $\mathbb{D}^{n \times n}$): Has an inverse within $\mathbb{D}^{n \times n}$. **23.2**

irreducible (element a of a domain): a is not a unit and every divisor of a is an associate of a or a unit. **23.2**

irreducible (matrix): Not reducible. **9.2, 27.3**

K -irreducible (K -nonnegative matrix P): The only P -invariant faces are the trivial faces. **26.1**

irreducible components (of matrix A): The diagonal blocks in the Frobenius normal form of A . **27.3**

isolated vertex: A vertex of a graph having no incident edges. **28.1**

isomorphic (graphs $G = (V, E)$ and $G' = (V', E')$): There exist bijections $\phi : V \rightarrow V'$ and $\psi : E \rightarrow E'$, such that $v \in V$ is an endpoint of $e \in E$ if and only if $\phi(v)$ is an endpoint of $\psi(e)$. **28.1**

isomorphic (vector spaces): There is an isomorphism from one vector space onto the other. **3.7**

isomorphism (of vector spaces): An invertible linear transformation. **3.7**

isoperimetric number: A parameter of a simple graph. **28.5**

J

$\mathcal{J}$ -orthogonal: Alphabetized under *orthogonal*.

Jacobi method (for computing the EVD of A): A sequence of matrices, $A_0 = A$, $A_{k+1} = G(i_k, j_k, c, s) A_k G(i_k, j_k, c, s)^T$, $k = 1, 2, \dots$, where $G(i_k, j_k, c, s)$ is a Givens rotation matrix. **42.7**

Jacobi rotation: A Givens rotation used in the Jacobi method. **42.7**

join (of disjoint graphs $G = (V, E)$ and $G' = (V', E')$): The union of $G \cup G'$ and the complete bipartite graph with vertex set $V \cup V'$ and partition $\{V, V'\}$. **28.1**

join (of faces F, G of cone K): Face generated by $F \cup G$. **26.1**

Jordan basis (for matrix A): The ordered set of columns of C where $\text{JCF}(A) = C^{-1}AC$. **6.2**

Jordan basis (of a linear operator): See *Section 6.2*.

Jordan block (of size k with eigenvalue λ): The $k \times k$ matrix having every diagonal entry equal to λ , every first superdiagonal entry equal to 1, and every other entry equal to 0. **6.2**

Jordan canonical form (of matrix A): A Jordan matrix that is similar to A . **6.2**

Jordan canonical form (of a linear operator T): ${}_B[T]_B$ that is a Jordan matrix. **6.2**

Jordan chain (above eigenvector $\mathbf{x}$ for eigenvalue λ of A): A sequence of vectors $\mathbf{x}_0 = \mathbf{x}, \mathbf{x}_1, \dots, \mathbf{x}_h$ satisfying $\mathbf{x}_i = (A - \lambda I)\mathbf{x}_{i+1}$ for $i = 0, \dots, h - 1$ (where h is the depth of $\mathbf{x}$). **6.2**

Jordan invariants (of matrix or linear operator A): The set of distinct eigenvalues of A and for each eigenvalue λ , the number b_λ and sizes $p_1, \dots, p_{b_\lambda}$ of the Jordan blocks with eigenvalue λ in a Jordan canonical form of A . **6.2**

Jordan matrix: A block diagonal matrix having Jordan blocks as the diagonal blocks. **6.2**

K

K -irreducible, K -nonnegative, K -positive, K -semipositive: Alphabetized under *irreducible, nonnegative, positive, semipositive*.

ker: See *kernel*.

kernel (ker) (of a linear transformation): The set of vectors mapped to zero by the transformation. **3.5**

kernel (ker) (of a matrix A): The set of solutions to $A\mathbf{x} = \mathbf{0}$. **2.4**

Kronecker product (of matrices A and B): The block matrix whose i, j block is $a_{ij}B$. **10.4**

Krylov matrix (of matrix A , vector $\mathbf{x}$, and positive integer k): $[\mathbf{x}, A\mathbf{x}, A^2\mathbf{x}, \dots, A^{k-1}\mathbf{x}]$. **42.8**

Krylov space: A subspace of the form $\text{Span}\{\mathbf{q}, A\mathbf{q}, A^2\mathbf{q}, \dots, A^{k-1}\mathbf{q}\}$, where A is an n by n matrix and $\mathbf{q}$ is an n -vector. **41.1, 44.1**

Ky Fan k norms (of $A \in \mathbb{C}^{m \times n}$): $\|A\|_{K,k} = \sum_{i=1}^k \sigma_i(A)$. **17.3**

L

L -matrix (sign pattern or real matrix A): For every $B \in Q(A)$, the rows of B are linearly independent. **33.3**

Lanczos algorithm (for Hermitian matrices): A short recurrence for constructing an orthonormal basis for a Krylov space. **41.2**

Laplacian: See *Laplacian matrix*.

Laplacian eigenvalues (of simple graph G): The eigenvalues of the Laplacian matrix of G . **28.4**

Laplacian matrix (of simple graph G): The matrix $D - A_G$, where D is the diagonal matrix of vertex degrees and A_G is the adjacency matrix of G . **28.4**

Laplacian matrix (of a weighted graph): The matrix $L = [\ell_{ij}]$ such that $\ell_{ij} = 0$ if vertices i and j are distinct and not adjacent, $\ell_{ij} = -w(e)$ if $e = \{i, j\}$ is an edge, and $\ell_{i,i} = \sum w(e)$, where the sum is taken over all edges e incident with vertex i . **36.4**

largest size of a zero submatrix: The maximum of $r + s$ such that the matrix has an $r \times s$ (possibly vacuous) zero submatrix. **27.1**

L DU factorization: A factorization of a matrix as the product of a unit lower triangular matrix, a diagonal matrix, and a unit upper triangular matrix. **1.6**

leading principal minor: The determinant of a leading principal submatrix. **4.2**

leading principal submatrix: A principal submatrix lying in rows and columns 1 to k . **1.2**

leading entry: The first nonzero entry in a row of a matrix in REF. **1.3**

least squares data fitting: See *Section 39.2*.

least squares problem ($A\mathbf{x} = \mathbf{b}$): Find a vector $\mathbf{x} \in \mathbb{R}^n$ such that $\|\mathbf{b} - A\mathbf{x}\|_2$ is minimized. **5.8, 39.1**

least squares solution: A solution to a least squares problem. **5.8, 39.1**

left singular space: See *singular spaces*.

left singular vector: See *singular vectors*.

left eigenvector (of matrix A): A nonzero row vector $\mathbf{y}$ such that there exists a scalar λ such that $\mathbf{y}A = \lambda\mathbf{y}$. **4.3**

length (of a walk, path, cycle in a graph or digraph): The number of edges (arcs) in the walk. **28.1, 29.1**

length (of a permutation cycle): The number of elements in the cycle. **Preliminaries**

length (of a vector): See *norm*.

($C, 0$)-limit (of complex sequence $\{a_m\}_{m=0,1,\dots}$): Ordinary limit $\lim_{m \rightarrow \infty} a_m$. **9.1**

($C, 1$)-limit (of complex sequence $\{a_m\}_{m=0,1,\dots}$): $\lim_{m \rightarrow \infty} m^{-1} \sum_{s=0}^{m-1} a_s$. **9.1**

(C, k)-limit (of complex sequence $\{a_m\}_{m=0,1,\dots}$): Defined inductively from $(C, k-1)$ -limit. See *Section 9.1*.

line (of a matrix): A row or column. **27.1**

line graph (of simple graph G): The simple graph that has as vertices the edges of G , and vertices are adjacent if the corresponding edges of G have an endpoint in common. **28.2**

linear combination: A (finite) sum of scalar multiples of vectors. **2.1**

linear equation: An equation of the form $a_1x_1 + \dots + a_nx_n = b$. **1.4**

linear form: See *linear functional*.

linear functional: A linear transformation from a vector space to the field. **3.8**

m -linear map: See *multilinear map*.

linear mapping: See *linear transformation*.

linear operator: A linear transformation from a vector space to itself. **3.1**

linear preserver (of function f): A linear operator $\phi : \mathcal{V} \rightarrow \mathcal{V}$ such that $f(\phi(A)) = f(A)$ for every $A \in \mathcal{V}$, where $\mathcal{V}$ is a subspace of $F^{m \times n}$. **22.1**

linear preserver problem (corresponding to the property P): The problem of characterizing all linear (bijective) maps on a subspace of matrices satisfying P . **22.2**

linear transformation: A function from a vector space to a vector space that preserves addition and scalar multiplication. **3.1**

linear transformation associated to matrix A : The transformation that multiplies each vector in F^n by A . **3.3**

linearly dependent (set of vectors): There is a linear combination of the vectors with at least one nonzero coefficient, that is equal to the zero vector. **2.1**

linearly independent (set of vectors): Not linearly dependent. **2.1**

linked: Two disjoint Jordan curves in $\mathbb{R}^3$ such that there is no topological 2-sphere in $\mathbb{R}^3$ separating them. **28.2**

linklessly embeddable: There is an embedding of graph G in $\mathbb{R}^3$ such that no two disjoint cycles of G are linked. **28.2**

little-oh: Function f is $o(g)$ if the limit of $\frac{|f|}{|g|}$ is 0. **Preliminaries**

local similarity: A map $\phi : F^{n \times n} \rightarrow F^{n \times n}$ such that for every $A \in F^{n \times n}$ there exists an invertible $R_A \in F^{n \times n}$ such that $\phi(A) = R_A A R_A^{-1}$. **22.4**

Loewner (partial) ordering (on Hermitian matrices A, B): $A \succ B$ if $A - B$ is positive definite and $A \succeq B$ if $A - B$ is positive semidefinite. **8.5**

logarithm (of a square complex matrix A): Any matrix B such that $e^B = A$. **11.4**

look-ahead strategy: A technique to overcome breakdown in some algorithms. **41.3**

loop: An edge (arc) of a graph (directed graph) having both endpoints equal. **29.1**

Lovász parameter: A parameter of a simple graph. **28.5**

lower Hessenberg (matrix A): A^T is upper Hessenberg. **10.2**

lower shift matrix: A square matrix with ones on the first subdiagonal and zero elsewhere. **48.1**

lower triangular matrix: A matrix such that every entry having row number less than column number is zero. **1.2, 10.2**

Löwner (partial) ordering: See *Loewner (partial) ordering*

LU factorization: A factorization of a matrix as the product of a unit lower triangular matrix and an upper triangular matrix. **1.6, 38.3**

Lyapunov diagonally semistable (real square matrix A): There exists a positive diagonal matrix D such that $AD + DA^T$ is positive semidefinite. **19.5**

Lyapunov diagonally stable (real square matrix A): There exists a positive diagonal matrix D such that $AD + DA^T$ is positive definite. **19.5**

Lyapunov scaling factor (of real square matrix A): A positive diagonal matrix D such that $AD + DA^T$ is positive (semi)definite. **19.5**

M

M-matrix (real square matrix A): A can be written as $A = sI - P$ with P a nonnegative matrix and s a scalar satisfying $s > \rho(P)$. **9.5**

M_0 -matrix: (real square matrix A): A can be written as $A = sI - P$ with P a nonnegative matrix and s a scalar satisfying $s \geq \rho(P)$. **9.5**

machine epsilon: The distance between the number one and the next larger floating point number. **37.6**

main angles (of a graph): The cosines of the angles between the eigenspaces of the adjacency matrix and the all-ones vector. **28.3**

main diagonal (of a matrix): The set of diagonal entries of a matrix. **1.2**

majorizes: Weakly majorizes and the sums of all entries are equal. **Preliminaries**

(k -)matching (of a graph): A set of k mutually disjoint edges. **30.1**

matrices: Plural of *matrix*.

matrix: a rectangular array of elements of a field. **1.2**

Or if specifically stated, a rectangular array of elements of a domain, ring, or algebra. The following standard matrix terms are applied to the matrices over a domain $\mathbb{D}$ viewed as in the quotient field of $\mathbb{D}$: determinant, minor, rank, *Section 23.2*. The following standard matrix terms are applied to matrices over a Bezout domain within the domain: kernel, range, system of linear equations, coefficient matrix, augmented matrix, *Section 23.3*

matrix condition number: See *condition number (of matrix A for linear systems)*.

matrix cosine: See *cosine*.

matrix direct sum: A block diagonal matrix. **2.4, 10.2**

matrix exponential: See *exponential*.

matrix function: A function of a square complex matrix; can be defined using power series, Jordan Canonical Form, polynomial interpretation, Cauchy integral, etc. **11.1**

matrix logarithm: See *logarithm*.

matrix of a transformation T (with respect to bases $\mathcal{B}$ and $\mathcal{C}$): The matrix consisting of the coordinate vectors with respect to $\mathcal{C}$ of the images under T of the vectors in $\mathcal{B}$. **3.3**

matrix norm $\|\cdot\|$: A family of real-valued functions defined on $m \times n$ real or complex matrices for all positive integers m and n , such that for all matrices A and B (where A and B are compatible for the given operation) and all scalars, (1) $\|A\| \geq 0$, and $\|A\| = 0$ implies $A = 0$; (2) $\|\alpha A\| = |\alpha| \|A\|$; (3) $\|A + B\| \leq \|A\| + \|B\|$; (4) $\|AB\| \leq \|A\| \|B\|$. **37.3**

matrix pencil: See *pencil*.

matrix product: The result of multiplying two matrices. **1.2**

matrix representing f relative to basis $\mathcal{B} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$: The matrix whose i, j -entry is $f(\mathbf{w}_i, \mathbf{w}_j)$, *Section 12.1*; also applied to φ -sesquilinear form. **12.4**

matrix sine: See *sine*.

matrix sign function: See *sign (of a matrix)*.

matrix square root: See *square root, principal square root*.

matrix–vector product: The result of multiplying a matrix and a vector. 1.2

maximal (ideal I of a domain $\mathbb{D}$): The only ideals that contain I are I and $\mathbb{D}$, and $I \neq \mathbb{D}$. 23.1

maximal (sign-nonsingular-sign pattern matrix): If a zero entry is set nonzero, then the resulting pattern is not SNS. 33.2

maximal rank (of sign-pattern A): $\max\{\text{rank } B : B \in Q(A)\}$. 33.6

maximum column sum norm (of matrix $A \in \mathbb{C}^{m \times n}$): $\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|$. 37.3

maximum multiplicity (of simple graph G): The maximum multiplicity of any eigenvalue among matrices in $\mathcal{S}(G)$. 34.2

maximum row sum norm (of matrix $A \in \mathbb{C}^{m \times n}$): $\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$. 37.3

max-plus semiring: The set $\mathbb{R} \cup \{-\infty\}$, equipped with the addition $(a, b) \mapsto \max(a, b)$ and the multiplication $(a, b) \mapsto a + b$. The identity element for the addition, zero, is $-\infty$, and the identity element for the multiplication, unit, is 0. 25.1

measure of relative separation (of positive real numbers a, b): $|\sqrt{a/b} - \sqrt{b/a}|$. 17.4

meet (of faces F, G of cone K): $F \cap G$ 26.1

metric: A distance function. Preliminaries

MIEP: See *multiplicative inverse eigenvalue problem*.

minimal (matrix norm $\|\cdot\|$): For any matrix norm $\|\cdot\|_v$, $\|A\|_v \leq \|A\|$ for all A implies $\|\cdot\|_v = \|\cdot\|$. 37.3

minimal polynomial (of matrix A): The unique monic polynomial of least degree for which $q_A(A) = 0$. 4.3

minimal rank (of sign pattern A): $\min\{\text{rank } B : B \in Q(A)\}$. 33.6

minimal residual (MINRES) algorithm (to solve $Hx = b$ with H preconditioned Hermitian): At each step k , the approximation x_k is of the form $x_k \in x_0 + \text{Span}\{r_0, Hr_0, \dots, H^{k-1}r_0\}$ for which the 2-norm of the residual, $\|r_k\|_2$, is minimal. 41.2

minimal sign-central: A sign-central pattern that is not sign-central if any column of A is deleted. 33.11

minimally connected (digraph): Strongly connected and the deletion of any arc produces a subdigraph that is not strongly connected. 29.8

minimally potentially stable (sign pattern A): A is a potentially stable, irreducible sign pattern such that replacing any nonzero entry by zero results in a pattern that is not potentially stable. 33.4

minimum co-cover (of a $(0, 1)$ -matrix): A co-cover with the smallest number of 1s. 27.1

minimum cover (of a $(0, 1)$ -matrix): A cover with the smallest number of lines. 27.1

minimum-norm least squares solution: The least squares solution of minimum Euclidean norm. 39.1

minimum rank (of simple graph G): The minimum of $\text{rank } A$ among matrices $A \in \mathcal{S}(G)$. 34.2

minor (of a graph G): Any graph that can be obtained from G by a sequence of edge deletions, vertex deletions, and contractions. 28.2

minor (of a matrix): The determinant of a submatrix; for the i, j -minor, the submatrix is obtained by deleting row i and column j . 4.1

MINRES: See *minimal residual*.

Möbius function: $\mu : \mathbb{N} \mapsto \{-1, 0, 1\}$ is defined by $\mu(1) = 1$, $\mu(m) = (-1)^e$ if m is a product of e distinct primes, and $\mu(m) = 0$ otherwise. 20.5

$\mathbb{D}$ -module: A generalization of vector space over a field; an additive group with a (scalar) multiplication by elements $a \in \mathbb{D}$ that satisfies the standard distribution properties and $1v = v$ for all $v \in \mathbb{D}$. The following standard terms are also applied to modules over a Bezout domain—linear combination, basis (may not exist), standard basis for $\mathbb{D}^n$, dimension (if there is a basis). 23.3

monic (polynomial $p(x)$): The coefficient of the highest power of x in $p(x)$ is 1. 23.1

monotone (real vector $v = [v_i]$): $v_1 \geq v_2 \geq \dots \geq v_n$. 27.4

monotone (vector norm $\|\cdot\|$): For all x, y , $|x| \leq |y|$ implies $\|x\| \leq \|y\|$. 37.1

Moore–Penrose pseudo-inverse (of matrix $A \in \mathbb{C}^{m \times n}$): A matrix $A^\dagger \in \mathbb{C}^{n \times m}$ satisfying: $AA^\dagger A = A$; $A^\dagger AA^\dagger = A^\dagger$; $(AA^\dagger)^* = AA^\dagger$; $(A^\dagger A)^* = A^\dagger A$. 5.7

Moore–Penrose inverse (of a sign pattern): Analog of the Moore–Penrose inverse of a real matrix. 33.7

multibigraph (of a nonnegative integer matrix A): Like the bigraph of A , except that edge $\{i, j\}$ has multiplicity a_{ij} . 30.2

multigrid preconditioner: A preconditioner designed for problems arising from partial differential equations discretized on grids. **41.4**

multilinear form: A multilinear map into the field F . **13.1**

multilinear map (m -linear map): A map φ from $V_1 \times \cdots \times V_m$ into U that is linear on each coordinate. **13.1**

multiple (eigenvalue λ of real symmetric matrix B): $\alpha_B(\lambda) > 1$. **34.1**

multiplicative (map $\phi : F^{n \times n} \rightarrow F^{n \times n}$): $\phi(AB) = \phi(A)\phi(B)$, for all $A, B \in F^{n \times n}$. **22.4**

multiplicative D -stable (real square matrix A): DA is stable for every positive diagonal matrix D . **19.3**

multiplicative inverse eigenvalue problem (MIEP): Given $B \in \mathbb{C}^{n \times n}$ and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ find a diagonal matrix $D \in \mathbb{C}^{n \times n}$ such that $\sigma(BD) = \{\lambda_1, \dots, \lambda_n\}$. **20.9**

multiplicative perturbation (of $A \in \mathbb{C}^{m \times n}$): $D_L^* A D_R$ for some $D_L \in \mathbb{C}^{m \times m}$, $D_R \in \mathbb{C}^{n \times n}$. **15.3**

multiplicative preserver: A multiplicative map preserving a certain property. **22.4**

multiplicity (of an eigenvalue): See *algebraic multiplicity*, *geometric multiplicity*.

multiplicity (of a singular value): The dimension of the right or left singular space. **45.1**

multiset: An unordered list of elements that allows repetition. **Preliminaries**

N

NaN: Result of an operation that is not a number in a standard-conforming floating point system. **37.6**

nearly decomposable ($(0, 1)$ -matrix): Fully indecomposable and each matrix obtained by replacing a 1 with a 0 is partly decomposable. **27.2**

nearly reducible (matrix A): Irreducible and each matrix obtained from A by replacing a nonzero entry with a zero is reducible. **27.3**

nearly sign-central: A sign pattern that is not sign-central, but can be augmented to a sign-central pattern by adjoining a column. **33.11**

nearly sign nonsingular (NSNS) (sign pattern): A square pattern having at least two nonzero terms in the expansion of its determinant, with precisely one nonzero term having opposite sign to the others. **33.2**

negative definite (real symmetric or Hermitian bilinear form f): $-f$ is positive definite. **12.2, 12.5**

negative semidefinite (real symmetric or Hermitian bilinear form f): $-f$ is positive semidefinite. **12.2, 12.5**

negative set of edges α (in a signed bipartite graph): $\text{sgn}(\alpha)$ is -1 . **30.1**

negative stable (complex polynomial): Its roots lie in the open left half plane. **19.2**

negative stable (complex square matrix): Its eigenvalues lie in the open left half plane. **19.2**

neighbor (of vertex v): Any vertex adjacent to v . **28.1**

NIEP: See *nonnegative inverse eigenvalue problem*.

nilpotent: A matrix or linear transformation that can be multiplied by itself a finite number of times to obtain the zero matrix or transformation. **2.7, 3.6**

node: See *vertex*.

noncommutative algorithms: Algorithms that do not use commutativity of multiplication. **47.2**

nondefective (matrix $A \in F^{n \times n}$): For each eigenvalue of A (over algebraic closure of F), the geometric multiplicity equals the algebraic multiplicity. **4.3**

nondegenerate (bilinear form f on vector space V): The rank of f is equal to $\dim V$, *Section 12.1*; also applied to φ -sesquilinear form. **12.4**

nonderogatory (matrix $A \in F^{n \times n}$): The geometric multiplicity of each eigenvalue of A (over algebraic closure of F) is 1. **4.3**

nondifferentiable (boundary point μ of convex set S): There is more than one support line of S passing through μ . **18.2**

non-Hermitian Lanczos algorithm See *Section 41.3*

nonnegative (real matrix A): All of A 's elements are nonnegative. **9.1**

nonnegative (sign pattern): All of its entries are nonnegative. **33.7**

K -nonnegative (matrix $A \in \mathbb{R}^{n \times n}$): $AK \subseteq K$. **26.1**

K -nonnegative (vector $\mathbf{v} \in \mathbb{R}^n$): $\mathbf{v} \in K$. **26.1**

nonnegative inverse eigenvalue problem: Construct a nonnegative matrix with given eigenvalues. **20.3**

nonnegative integer rank (of nonnegative integer matrix A): The minimum k such that there exists an $m \times k$ nonnegative integer matrix B and a $k \times n$ nonnegative integer matrix C with $A = BC$. **30.3**

nonnegative P -matrix: A P -matrix in which every entry is nonnegative. **35.9**

nonnegative P_0 -matrix: A P_0 -matrix in which every entry is nonnegative. **35.9**

nonnegative $P_{0,1}$ -matrix: A $P_{0,1}$ -matrix in which every entry is nonnegative. **35.9**

nonnegative stable (complex square matrix A): The real part of each eigenvalue of A is nonnegative. **9.5**

nonprimary matrix function: Function of a matrix defined using the Jordan Canonical Form using different branches for f and its derivatives for two Jordan blocks for the same eigenvalue. **11.1**

nonseparable (graph or digraph): Connected and does not have a cut-vertex. **35.1**

nonsingular (linear transformation): A linear transformation whose kernel is zero. **3.7**

nonsingular (matrix): An invertible matrix. **1.5**

norm (of a vector $\mathbf{v}$, in an inner product space): $\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$. **5.1**

norm: See *vector norm*, *matrix norm* or *specific norm*.

1-norm (of vector $\mathbf{x} \in \mathbb{C}^n$): $\|\mathbf{x}\|_1 = |x_1| + |x_2| + \cdots + |x_n|$. **37.1**

1-norm (of matrix $A \in \mathbb{C}^{m \times n}$): See *maximum column sum norm*.

2-norm (of vector $\mathbf{x} \in \mathbb{C}^n$): $\|\mathbf{x}\|_2 = \sqrt{|x_1|^2 + |x_2|^2 + \cdots + |x_n|^2}$. **37.1**

2-norm (of matrix $A \in \mathbb{C}^{m \times n}$): see *maximum column sum norm*.

∞ -norm (of vector $\mathbf{x} \in \mathbb{C}^n$): $\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$. **37.1**

∞ -norm (of matrix A): See *maximum row sum norm*.

M -norm (where $\|\cdot\|$ is a vector norm and M is a nonsingular matrix): $\|\mathbf{x}\|_M \equiv \|M\mathbf{x}\|$. **37.1**

$\mathcal{M}$ -norm (matrix norm): Norm induced by a family of M_n -norms for $\mathcal{M} = \{M_n : n \geq 1\}$ a family of nonsingular $n \times n$ matrices. **37.3**

p -norm (of matrix $A \in \mathbb{C}^{m \times n}$, with $p \geq 1$): The matrix norm induced by the (vector) p -norm. **37.1**

p -norm (of vector $\mathbf{x} \in \mathbb{C}^n$, with $p \geq 1$): $\|\mathbf{x}\|_p = (|x_1|^p + \cdots + |x_n|^p)^{\frac{1}{p}}$. **37.1**

normal (complex square matrix or linear operator): Commutes with its Hermitian adjoint. **7.2**

normal equations (for the least squares problem $A\mathbf{x} = \mathbf{b}$): The system $A^*A\mathbf{x} = A^*\mathbf{b}$. **5.8**

normalized (representation $\pm(m/b^{p-1})b^e$ of a floating point number): $b^{p-1} \leq m < b^p$. **37.6**

normalized immanant: Function defined by the irreducible character of a partition λ of n , specifically, $f_\lambda(M) = \frac{1}{\chi_\lambda(\mathbf{e})} \sum_{\sigma \in S_n} \chi_\lambda(\sigma) \prod_{i=1}^n m_{i\sigma(i)}$. **31.10**

normalized scaling (of rectangular matrix A): A scaling DAE of A with $\det(D) = \det(E) = 1$. **9.6**

NSNS: See *nearly sign nonsingular*.

null graph: A graph with no vertices. **28.1**

null space: See *kernel*.

nullity: The dimension of the kernel. **2.4, 3.5**

numerical radius (of $A \in \mathbb{C}^{n \times n}$): $w(A) = \max\{|\mu| : \mu \in W(A)\}$. **18.1**

numerical range (of $n \times n$ complex matrix A): $W(A) = \{\mathbf{v}^*A\mathbf{v} | \mathbf{v}^*\mathbf{v} = 1, \mathbf{v} \in \mathbb{C}^n\}$. **7.1, 18.1**

numerical rank (of matrix A , with respect to the threshold τ): $\min\{\text{rank}(A + E) : \|E\|_2 \leq \tau\}$. **39.9**

numerically orthogonal (vectors $\mathbf{x}, \mathbf{y}$): $|\mathbf{x}^T \mathbf{y}| \leq \varepsilon \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$. **46.1**

numerically orthogonal matrix: Each pair of its columns are numerically orthogonal. **46.1**

numerically stable (algorithm): Produces results that are roughly as accurate as the errors in the input data allow. **37.8**

numerically unstable (algorithm): Allows rounding and truncation errors to produce results that are substantially less accurate than the errors in the input data allow. **37.8**

O

odd (permutation): Can be written as the product of an odd number of transpositions. **Preliminaries**

odd (cycle in a sign pattern): The length of the simple or composite cycle is odd. **33.1**

off-diagonal entry: An entry in a matrix that is not a diagonal entry. **1.2**

off-norm (of A): The Frobenius norm of the matrix consisting of all off-diagonal elements of A and a zero diagonal. **42.7**

oh: See *big-oh*, *little-oh*.

one-to-one (linear transformation T): $\mathbf{v}_1 \neq \mathbf{v}_2$ implies $T(\mathbf{v}_1) \neq T(\mathbf{v}_2)$. 3.5

onto (linear transformation): The codomain equals the range. 3.5

open left half plane: $\{z \in \mathbb{C} : \operatorname{re}(z) < 0\}$. Preliminaries

open right half plane: $\{z \in \mathbb{C} : \operatorname{re}(z) > 0\}$. Preliminaries

open sector (from ray $e^{i\alpha}$ to ray $e^{i\beta}$): The set of rays $\{re^{i\theta} : r > 0, \alpha < \theta < \beta\}$. 33.8

operator norm: See *induced norm*.

order (of a graph): The number of vertices in the graph. 28.1

order (of a square matrix): See *size*.

ordered multiplicity list (of real symmetric matrix B): $(m_1, \dots, m_r)$ where the distinct eigenvalues of B are $\beta_1 < \dots < \beta_r$ with multiplicities $m_1, \dots, m_r$. 34.5

oriented (vertex-edge) incidence matrix (of graph G): A matrix obtained from the incidence matrix of G by replacing a 1 in each column by a -1 . 28.4

orthogonal (vectors): Two vectors whose inner product is 0. 5.2

orthogonal (set of vectors): Any two distinct vectors in the set are orthogonal. 5.2

$\mathcal{J}$ -orthogonal (real matrix B) $B^T \mathcal{J} B = \mathcal{J}$ (where $\mathcal{J}$ is a nonsingular real symmetric matrix). 46.5

orthogonal basis: A basis that is orthogonal as a set. 5.2

orthogonal complement (of subset S): Subspace of vectors orthogonal to every vector in S . 5.2

orthogonal iteration: Method for computing the EVD of real symmetric A : given starting $n \times p$ matrix X_0 with orthonormal columns, the sequence of matrices $Y_{k+1} = AX_k$, $Y_{k+1} = X_{k+1}R_{k+1}$ $k = 0, 1, 2, \dots$, where $X_{k+1}R_{k+1}$ is the reduced QR factorization of Y_{k+1} . 42.1

orthogonal matrix: A real or complex matrix Q such that $Q^T Q = I$. 5.2, 7.1

orthogonal projection: Projection onto a subspace along its orthogonal complement. 5.4

orthogonal with respect to symmetric bilinear form f (vectors $\mathbf{v}, \mathbf{u}$): $f(\mathbf{u}, \mathbf{v}) = 0$. 12.2

orthonormal (set of vectors): An orthogonal set of unit vectors. 5.2

orthonormal basis: A basis that is orthonormal as a set. 5.2

oscillatory (real square matrix A): A is totally nonnegative and A^k is totally positive for some integer $k \geq 1$. 21.1

outerplanar (graph G): There is an embedding of G into $\mathbb{R}^2$ with the vertices on the unit circle and the edges contained in the unit disc. 28.2

overflow: $|x|$ equals or exceeds a threshold at or near the largest floating point number. 37.6

P

P -invariant face: Alphabetized under *invariant*.

P -matrix: every principal minor is positive. 19.2

P_0 -matrix: Every principal minor is nonnegative. 35.8

P_0^+ -matrix: Every principal minor is nonnegative and at least one principal minor of each order is positive. 19.2

$P_{0,1}$ -matrix: Every principal minor is nonnegative and every diagonal element is positive. 35.8

pairwise orthogonal (orthogonal projections P and Q): $PQ = QP = 0$. 7.2

Parter vertex: See *Parter–Wiener vertex*.

Parter–Wiener vertex (for eigenvalue λ of $n \times n$ Hermitian matrix A): j such that $\alpha_{A(j)}(\lambda) = \alpha_A(\lambda) + 1$. 34.1

partial completely positive matrix B : Every fully specified principal submatrix of B is a completely positive matrix, whenever b_{ij} is specified, then so is b_{ji} and $b_{ji} = b_{ij}$, and all specified off-diagonal entries are nonnegative. 35.4

partial copositive matrix B : Every fully specified principal submatrix of B is a copositive matrix and whenever b_{ij} is specified, then so is b_{ji} and $b_{ji} = b_{ij}$. 35.5

partial doubly nonnegative matrix B : Every fully specified principal submatrix of B is a doubly nonnegative matrix, whenever b_{ij} is specified, then so is b_{ji} and $b_{ji} = b_{ij}$, and all specified off-diagonal entries are nonnegative. 35.4

partial entry sign symmetric P -matrix B : Every fully specified principal submatrix of B is an entry sign symmetric P -matrix and if both b_{ij} and b_{ji} are specified, then $b_{ij}b_{ji} > 0$ or $b_{ij} = b_{ji} = 0$. **35.10**

partial entry sign symmetric P_0 -matrix B : Every fully specified principal submatrix of B is an entry sign symmetric P_0 -matrix and if both b_{ij} and b_{ji} are specified, then $b_{ij}b_{ji} > 0$ or $b_{ij} = b_{ji} = 0$. **35.10**

partial entry sign symmetric $P_{0,1}$ -matrix B : Every fully specified principal submatrix of B is an entry sign symmetric $P_{0,1}$ -matrix and if both b_{ij} and b_{ji} are specified, then $b_{ij}b_{ji} > 0$ or $b_{ij} = b_{ji} = 0$. **35.10**

partial entry weakly sign symmetric P -matrix B : Every fully specified principal submatrix of B is an entry weakly sign symmetric P -matrix and if both b_{ij} and b_{ji} are specified, then $b_{ij}b_{ji} \geq 0$. **35.10**

partial entry weakly sign symmetric P_0 -matrix B : Every fully specified principal submatrix of B is an entry weakly sign symmetric P_0 -matrix and if both b_{ij} and b_{ji} are specified, then $b_{ij}b_{ji} \geq 0$. **35.10**

partial entry weakly sign symmetric $P_{0,1}$ -matrix B : Every fully specified principal submatrix of B is an entry weakly sign symmetric $P_{0,1}$ -matrix and if both b_{ij} and b_{ji} are specified, then $b_{ij}b_{ji} \geq 0$. **35.10**

partial Euclidean distance matrix B : Every diagonal entry is specified and equal to 0, every fully specified principal submatrix of B is a Euclidean distance matrix, and whenever b_{ij} is specified, then so is b_{ji} and $b_{ji} = b_{ij}$. **35.3**

partial inverse M -matrix B : Every fully specified principal submatrix of B is an inverse M -matrix and every specified entry of B is nonnegative. **35.7**

partial M -matrix B : Every fully specified principal submatrix of B is an M -matrix and every specified off-diagonal entry of B is nonpositive. **35.6**

partial M_0 -matrix B : Every fully specified principal submatrix of B is an M_0 -matrix and every specified off-diagonal entry of B is nonpositive. **35.6**

partial matrix: A square array in which some entries are specified and others are not. **35.1**

partial nonnegative P -matrix: Every fully specified principal submatrix is a nonnegative P -matrix and all specified entries are nonnegative. **35.9**

partial nonnegative P_0 -matrix: Every fully specified principal submatrix is a nonnegative P_0 -matrix and all specified entries are nonnegative. **35.9**

partial nonnegative $P_{0,1}$ -matrix: Every fully specified principal submatrix is a nonnegative $P_{0,1}$ -matrix and all specified entries are nonnegative. **35.9**

partial P -matrix: Every fully specified principal submatrix is a P -matrix. **35.8**

partial P_0 -matrix: Every fully specified principal submatrix is a P_0 -matrix. **35.8**

partial $P_{0,1}$ -matrix: Every fully specified principal submatrix is a $P_{0,1}$ -matrix. **35.8**

partial positive P -matrix: Every fully specified principal submatrix is a positive P -matrix and all specified entries are positive. **35.9**

partial semidefinite ordering: See *Loewner ordering*.

partial strictly copositive matrix B : Every fully specified principal submatrix of B is a strictly copositive matrix and whenever b_{ij} is specified, then so is b_{ji} and $b_{ji} = b_{ij}$. **35.5**

partitioned (matrix): A matrix partitioned into submatrices by partitions of the row and column indices. **10.1**

partly decomposable ($n \times n$ (0,1)-matrix): Has a $p \times q$ zero submatrix for positive integers p, q with $p + q = n$. **27.2**

path (in a graph): A walk with all vertices distinct. **28.1**

path (in sign pattern A): A formal product of the form $\gamma = a_{i_1 i_2} a_{i_2 i_3} \dots a_{i_k i_{k+1}}$, where each of the elements is nonzero and the index set $\{i_1, i_2, \dots, i_{k+1}\}$ consists of distinct indices. **33.1**

path-clique (digraph): The induced subdigraph of every alternate path to a single arc is a clique. **35.7**

path-connected (subset S of complex numbers): There exists a path in S (i.e., a continuous function from $[0,1]$ to S) from any point in S to any other point in S . **Preliminaries**

path cover number (of simple graph G): The minimum number of vertex disjoint induced paths of G that cover all vertices of G . **34.2**

pattern (of matrix A): The (0,1)-matrix obtained from A by replacing each nonzero entry with a 1. **27.1**

pattern block triangular form (partial matrix B): The adjacency matrix of $\mathcal{D}(B)$ is in block triangular form. **35.1**